Order Chiroptera

With at least 17 families, 175 genera, and over 900 species, bats are second only to rodents in phylogenetic diversity. This taxonomic diversity is paralleled by an equally varied reproductive biology. Bats are extraordinarily flexible with respect to the timing of the major events of the reproductive cycle. Sperm storage in both males and females dissociates the timing of copulation from that of conception. Similarly, embryonic diapause and delayed implantation release the timing of birth from dependence on the timing of conception.

In general, females produce one offspring each breeding season. Testes are often abdominal during non-reproductive seasons.

Ordinal comparisons of anatomy (419, 2071, 6720, 6966, 9178), reproductive asymmetry (11804), ovum development (11095), physiology (9350), sperm storage (4449, 8830, 8834, 11018, 11792), implantation (8947), gestational delays (1460, 8827), birth (11801), placentation (1228, 3926, 11799), endocrinology (6139), male reproductive cycles (4160, 6065), reproductive energetics/ecology (6110, 8176, 8835), and methodology (6110a) are available. One journal, Myotis, focuses on bats.

Order Chiroptera, Family Pteropodidae

Flying foxes and fruit bats (*Acerodon* 450-565 g; *Aethalops* 19 g; *Alionycteris* 64 mm; *Aproteles*, *Balionycteris* 9-15 g; *Boneia* 150-194 g; *Casinycteris* 40 g; *Chironax* 12-22 g; *Cynopterus* 26-100 g; *Dobsonia* 164-827 g; *Dyacopterus* 70-85 g; *Eidolon* 215-390 g; *Eonycteris* 35-80 g; *Epomophorus* 40-310 g; *Epomops* 60-220 g; *Haplonycteris* 16-20 g; *Harpyionycteris* 83-142 g; *Hypsignathus* 218-450 g; *Latidens* FA 67.5 mm, n1; *Macroglossus* 10-23 g; *Megaerops* 20-38 g; *Megaloglossus* 8-20 g; *Melonycteris* 77-106 mm; *Micropteropus* 10-35 g; *Myonycteris* 23-61 g; *Nanomycteris* 16-35 g; *Neopteryx* 190-250 g; *Notopteris* 50-60 g; *Nyctimene* 25-50 g; *Otopteropus* 17-21 g; *Paranyctimene* 22 g; *Penthetor* 35-65 g; *Plerotes* 87 mm; *Ptenochirus* 86-105 mm; *Pteralopex* 255-280 mm; *Pteropus* 45-1600 g; *Rousettus* 55-910 g; *Scotonycteris* 20-64 g; *Sphaerias* 26-30 g; *Styloctenium* 175-220 g; *Syconycteris* 11-25 g; *Thoopterus* 67-99 g) are primarily frugivorous or nectivorous inhabitants of the old-world tropics and subtropics. Males are often larger than females (191, 696, 1185, 3215, 5226, 6845, 7717, 8494, 8817, 9175, 10780, 11150, 11270; cf 3903, 5759, 8494, 9297). *Hypsignathus* mates in leks (1185).

The hemochorial or endotheliochorial, labyrinthine, discoid or diffuse, deciduate placenta has a mesometrial or circumpherential implantation of intermediate depth into a duplex or bicornuate uterus (181, 2198, 2948, 3914, 3917, 3924, 3926, 4573, 5527, 5539, 5617, 5924, 6627, 6846, 6847, 7416, 7417, 7563, 7677, 7679, 9178, 9746, 10310, 11799, 11898; cf 6627) and is eaten after birth (6093, 6094, 9365). A postpartum estrus occurs (252, 567, 696, 3914, 3916, 3921, 4572, 6031, 8081, 8082, 8871, 8874, 8875, 10780) as do gestational delays (*Eidolon* 7676, 7677; *Haplonycteris* 4571, 4572, 4573).

Dobsonia may be born with open eyes (2835), but most pteropids open their eyes 1-10 d after birth (334, 898, 3903, 5759, 6093, 6094, 7679, 7859, 9175, 9297, 10158; cf 9365). Neonates usually have some fur at birth (331, 334, 2835, 2948, 5759, 6093, 7679, 9297, 10158; cf 898, 1342, 9175). Females have 2 mammae (331, 6720, 9175, 9297, 11898) and an anatomical bridge linking the ovary and the uterus (3937, 3940, 6847). Males have seminal vesicles, ampullary glands, urethral glands, and a baculum (6065, 7441; *Cynopterus* 7439, 7440, 11071; *Harpyia* 9178; *Pteropus* 4097, 7665, 8863; *Rousettus* 3938, 6721).

Details of female anatomy (419, 3937, 5617, 6720, 6846, 6847, 9178, 9490, 10878, 11898), male anatomy (568, 5599), spermatogenesis (7064), and endocrinology (252, 567, 5128, 5526, 7065, 10878) are available.

NEONATAL MASS

Cynopterus sphinx	10.1 g; n1 stillborn	8873
	11.0 g; r10.8-11.2; n2	8873
	11 g	9471
	12-14 g	3915
Dobsonia moluccensis	50-60 g	2835
Dobsonia peroni	48 g; n1 emb	3903
Eidolon helvum	min 39 g	3566
	40-50 g	5759
	40-53 g; term emb	7677
	45-55 g; term emb	3184
	m: ~50 g; n1	2948
	f: 58.5 g	3566

	m: 58.7 g	3566
Eonycteris spelaea	9.0 g; r8.2-9.8; n2 smallest yg	898
	14.6 g; term emb	898
Epomophorus labiatus	11 g	8082
Epomops buettikoferi	24 g; n1 term emb	11273
Epomops franqueti	~19 g	8081
	20 g; near term	5127
Hypsignathus monstrosus	~40 g	1185
Myonycteris torquata	11 g; n1 term emb	11273
Pteropus spp	45 g; n1	9365
Pteropus poliocephalus	~70 g; r46-92	7833a
	90 g	10879
Rousettus aegyptiacus	17.9 g; n1	567
	18.7 g	7679
	22.5 g; n1; 2 d	6093, 6094
	22.7 g; se2.5; n11	7951
	23-24 g	5226
Rousettus leschenaulti	12 g	3915, 3921
NEONATAL SIZE		
Dobsonia beauforti	FA: 42.7 mm; n1	807b
Epomophorus crypturus		
or *Epomophorus gambianus*	HB: ~70 mm; FA: 32.5 mm	334
Myonycteris torquata	greatest length: 24 mm; n1 largest emb	807c
	greatest length: ~34 mm; n1 smallest yg	807c
Pteropus spp	130 mm; n1	9365
Pteropus giganteus	FA: 72 mm; n1	7013b
Pteropus poliocephalus	FA: 60 mm; n1	7833a
Rousettus aegyptiacus	56 mm; n1	6093
WEANING MASS		
Cynopterus sphinx	21.2 g; r19.8-22.3; n3 juveniles	10286
	~35 g	9471
Eidolon helvum	120-150 g	3566
Rousettus aegyptiacus	63.1 g; n1; 3 mo	567
	84.5 g; r67-98; 3 mo	5226
LITTER SIZE		
Acerodon celebensis	1 emb; n1 f	809
Acerodon mackloti	1 emb ea	3903
Boneia bidens	1 emb ea; n2 f	809
Casinycteris argynnis	1 emb	252
Chironax melanocephalus	1 emb ea; n2 f	809
Cynopterus brachyotis	1	7161
	1	8494
	1 yg; n1 lt	6394
	1 emb ea; n5 f	809
Cynopterus sphinx	1	9471
	1 cl	8874
	1 emb	5617
	1 emb	10286
	1 yg	3915
	1; n1 lt	12128
	1 emb; n1 f	3903
	1 emb ea; n165 f	3927
	1-2	1340
	2 cl; n1 f	3903
	2 yg; n1 lt	3903
Dobsonia beauforti	1 emb ea; n6 f	807b
Dobsonia inermis	1 emb; n1 f	7073
Dobsonia minor	1 emb ea; n3 f	7073
Dobsonia moluccensis	1 emb; n1 f	7073
	1 yg ea; n9 f	2835
Dobsonia peroni	1 emb; n1 f	3903
Eidolon helvum	1	7677
	1	3566
	1 yg	972

	1; 2 rare	252
	1.2 emb; r1-2; n5 f	191
	1; 2 rare, 1 pr twins in > 100 pregancies	5759
Eonycteris spelaea	1 mode; 2 rare	696
	1.13 emb; mode 1; r1-2; n23 f	898
Epomophorus angolensis	1	9945
Epomophorus crypturus	1	252
	1; twins reported	10158
	1 emb; n1 f	331, 337
Epomophorus gambianus	1 emb ea; n2 f	71
	1 emb ea; n5 f	8611
Epomophorus labiatus	1	252
	1 emb; n1 f	191
	1 emb; n1 f	185
	1 emb ea; n2 f	11272
Epomophorus wahlbergi	1	10158
	1 emb; n1 f	331, 337
	mode 1; r1-2	252
Epomops buettikoferi	1 emb; n1 f	11273
	1 emb ea; n8 f	1948
Epomops franqueti	1	252
	1	1346
	1	191
	1 yg	972
	1 ea; n2 lt	71
Haplonycteris fischeri	1	4572, 4573
Harpionycteris celebensis	1 emb ea; n2 f	809
Hypsignathus monstrosus	1	252
	1	191
	1; twins recorded	5759
	2 emb; n1 f	1347
Macroglossus minimus	1	10698
	1; n1 lt	7075
	1 emb; n1 f	6447
	1 emb ea; n3 f	809
	1 emb ea; n12 f	7073
Megaloglossus woermanni	1 emb ea; n2 f	8084
	1 emb ea; n3 f	5387
	2 emb, different sizes; n1 f	2198
Melonycteris woodfordi	1 emb ea; n2 f	7073
Micropteropus pusillus	1 emb; n1 f	4747
Myonycteris torquata	1	1346
	1 emb ea; n2 f; case of twins	1948
	1 emb ea; n2 f	11881
	1 emb ea; n3 f	5387
	2 emb, different sizes; n1 f	11273
Nanonycteris veldkampi	1 emb ea; n9 f	1948
Neopteryx frosti	1 emb; n1 f	809
Notopteris macdonaldi	1 yg	9457
	1 yg ea; n4 f	7835
	1 emb ea; n11 f	7835
Nyctimene aello	1 emb ea; n4 f	7073
Nyctimene albiventer	1 emb ea; n6 f	7073
Nyctimene cephalotes	1 emb ea; n1 f	809
Nyctimene robinsoni	1 emb ea; n2 f	7075, 7076
Paranyctimene raptor	1 emb ea; n7 f	7073
Penthetor lucasi	1 emb; n1 f	10314
Pteropus alecto	1; n many	7075
	2 yg; n1 lactating f	2214
Pteropus anetianus	1 emb ea; n8 f	517
Pteropus conspicillatus	1; n many	7075
Pteropus giganteus	1	3915, 3927
	1	1340
	1	8494, 8501

	1	6845
	1; n1 lt	12128
	1 emb ea; n3 f	7013b
	1 ea; n3 lt	3370
	1 emb ea; n70 f	3927
	1 emb ea; n~100	7416
Pteropus howensis	1 emb ea; n2 f	9457
Pteropus ornatus	1 emb ea; n14 f	9457
Pteropus poliocephalus	1	10878, 10879
	1	7075
	1; n1 lt	12128
Pteropus rayneri	1 emb; n1 f	9457
Pteropus rodricensis	1	1625
	1 yg ea; n3 f	8652
Pteropus rufus	2 emb ea; n3 f	8906
Pteropus scapulatus	1	7075
	1 yg; n2 lt	7833a
Pteropus tokudae	1 yg; n1 f	8371
Pteropus tonganus	1 yg	517
	1 yg	9457
Pteropus vampyrus	1	7161
	1 yg	6369
	1 emb; n1 f	2313
Rousettus spp	1; n1 lt	6093
Rousettus aegyptiacus	1	5226
	1 emb	7679
	1 emb; n1 f	11272
	1 emb; n1 f	11273
	1 emb; n6 f	7717
	1 emb ea; n11 f	1948
	1 emb; n12 f	331
	1+; r1-2; occasionally 2	4698
	1; r1-2	6094
	1 yg; 1.3% twins	8084
	1.25; n73 lt	12128
Rousettus angolensis	1	252
	1 emb ea; n2 f	11273
	1 emb ea; n2 f	11881
Rousettus amplexicaudatus	1 emb ea; n2 f	809
	1 emb ea; n2 f	7073
Rousettus celebensis	1 emb or yg ea; n5 f	809
Rousettus leschenaulti	1	8494
	1	9175
	1 yg	1340, 1342
	1 emb; n1 f	10872
	1 emb ea; n636 f	3914, 3915, 3916, 3921 3922, 3927, 3931
	1-2; twins rare	5537
Scotonycteris ophiodon	1 emb	252
	1 emb; n1 f	1948
	1 emb; n1 f	2955
Scotonycteris zenkeri	1	252
	1 emb; n1 f	1948
Syconycteris australis	1 yg; n3 lt	7833
	1 emb ea; n12 f	7073
Thoopterus nigrescens	1 emb ea; n2 f	809
SEXUAL MATURITY		
Cynopterus sphinx	f: ~5 mo; ~50 g	9471
	m: 15-20 mo	9471
Eidolon helvum	f: 200 g	2430
	m: testes large: 200-249 g	7677
Eonycteris spelaea	f: 1 yr	696
	m: 1 yr	696

Epomops buettikoferi	f: 1st mate: 6 mo	10780
	m: 11 mo	10780
Haplonycteris fischeri	f: 1st conceive: 3-5 mo	4571, 4573
Hypsignathus monstrosus	f: 1st preg: 6-7 mo	1185
	m: epididymal sperm: 12-18 mo	1185
Micropteropus pusillus	f: 1st mate: 6 mo	10780
	m: 7 mo	10780
Pteropus spp	f: 1st birth: 26 mo; n1	9365
Pteropus giganteus	f/m: 1 yr	8798
	f/m: 2 yr	1346
Pteropus poliocephalus	m/f: 18 mo	7833a
	m: spermatogenesis: 16 mo; 500-600 g	10879
Pteropus scapulatus	m/f: 18 mo	7833a
Pteropus tonganus	m: testes large: 600 g	517
	f: 1st preg: 450 g	517
Rousettus aegyptiacus	f/m: ~1 yr; ~90 g	7679
Rousettus leschenaulti	f: 7-8 mo; 55 g	3921
	m: 73 g	3921
GESTATION		
Cynopterus sphinx	3 mo	10286
	115-125 d	7417
	~120 d	9471
	~5 mo	8871
	150 d 1st pregnancy; 120 d 2nd pregnancy	6031
Eidolon helvum	implant: 3 mo	7676, 7677
	implant: 4-6 mo; field data	3566
	postimplantation growth: ~4 mo	7676, 7677
	~4 mo	3184
	~4 mo; field data	252, 254
	7 mo	7676, 7677
	9-12 mo with delay; field data	3566
Eonycteris spelaea	> 6 mo; poss 200 d; n35 recaptured f	696
Epomophorus crypturus	est ~3 mo; field data	252
Epomophorus gambianus	3? mo; field data	252
	5-6 mo	10780
Epomops buettikoferi	5-6 mo	10780
Epomops franqueti	5-6 mo	8081
Haplonycteris fischeri	delay: 2-6 mo primiparous f; max 8 mo	4573
	delay: 3-8 mo	4572
	slow growth: 3-9 mo; rapid growth: 2.5-3 mo	4571
	11.5 mo	4573
	1 yr	4571
Micropteropus pusillus	5-6 mo	10780
Pteropus spp	6.5 mo	9365
Pteropus giganteus	~5 mo; field data	7416
	140-150 d	1340, 1342, 3915
	6 mo; field data	6845
	199 d; r198-200; n3	3370
Pteropus poliocephalus	6 mo	10878, 10879
Pteropus tonganus	5-6 mo	517
Rousettus aegyptiacus	4 mo; field data	7679
	est min 4 mo	6093
	105-107 d	5759
Rousettus angolensis	4 mo; field data	255
Rousettus leschenaulti	15 wk	8494
	3.5 mo	1640
	4 mo; field data	3916
	125 d	3915, 3921

LACTATION

Cynopterus brachyotis	1st flight: 3 mo; n1	6394
Cynopterus sphinx	wean: 40-45 d	9471
Dobsonia moluccensis	independent: 6 mo	10698
Eidolon helvum	wean: ~1 mo	7677
	wean: 2 mo	254
	wean: 3-4 mo	3566
Eonycteris spelaea	teat attachment: 4-6 wk	696
	nurse: est 3 mo	696
Epomophorus crypturus	wean: est ~2 mo	252
Epomops buettikoferi	wean: 7-8 wk	10780
Haplonycteris fischeri	wean: 2.5 mo	4572
Micropteropus pusillus	wean: 7-8 wk	10780
Pteropus giganteus	solid food: ~4 mo	8798
	wean: 3 mo	7859
	wean: 5 mo	1346
Pteropus rodricensis	solid food: 11 wk	8652
Rousettus aegyptiacus	den emergence: 70 d	6093
	nurse: 6 wk	7679
	independent: 3 mo	5226
	independent: by 4 mo	567
	wean: min 4 mo	6093

LITTERS/YEAR

Cynopterus sphinx	2	10286
	2	8871
	2	3915
Eidolon helvum	1	7677
	1	3566
Eonycteris spelaea	2	696
Epomophorus crypturus	est 2	252
Epomophorus gambianus	1	252
	2	10780
Epomops buettikoferi	2	10780
Micropteropus pusillus	2	10780
Pteropus giganteus	1	1340, 1342
Rousettus aegyptiacus	1	331
	1	9175
	2	567
	2	7679
Rousettus leschenaulti	1	9175
	2	3914, 3915, 3916, 3921, 3922

SEASONALITY

Acerodon celebensis	preg f: Feb?; n1; Sulawesi, Indonesia	809
Acerodon mackloti	preg f: Mar, May; Timor, Indonesia	3903
Aethalops alecto	preg f: Feb-Mar; Malaysia	7161
	preg f: Feb, Apr, June; n7; Malaysia	6446
Balionycteris maculata	preg f: Jan-Mar; Malaysia	7161
	preg f: Feb, Apr; n2; Malaysia	6446
	preg f: Aug; Malaysia	7161
Boneia bidens	preg f: June-July; n2; n Sulawesi, Indonesia	809
Casinycteris argynnis	birth: May; central Africa	252
	lact f: May-July; central Africa	252
Chironax melanocephalus	preg f: Jan-Mar; Malaysia	7161
	preg f: Feb, Apr; n2; Malaysia	6446
	preg f: Mar-Apr; n2; Sulawesi, Indonesia	809
Cynopterus brachyotis	preg f: Jan-Feb, Apr-May, Sept; n117; Malaysia	6445a, 6446
	preg f: Jan, Mar; n5; Sulawesi, Indonesia	809
	preg f: May, Aug; Sri Lanka	8494
	preg f: yr rd; Malaysia	7161
	preg/lact f: peak Feb; Philippines	135
	preg/lact f: May, Aug, Oct; Philippines	7586
	lact f: June; n1; Sarawak, Malaysia	6447
Cynopterus horsfieldi	preg f: yr rd; Selangor, Malaysia	7161

Cynopterus sphinx	breeding condition: Mar-May; no data May-Jan; Timor, Indonesia	3903
	reproductive tract: enlarged Dec; India	11071
	sperm: yr rd; India	10286
	anestrus July-Oct; India	9471
	mate: Mar; India	8874
	mate: early Oct; India	3937
	mate: yr rd; Sri Lanka	8501
	conceive: Mar; India	8874
	preg f: Feb-July; India	10286
	preg f: Oct-July; India	3915
	birth: Feb-Apr, June-July; India	9471
	birth: Mar-Aug; India	1340
Dobsonia inermis	preg f: Sept; n1; Bougainville, Solomon I	7073
Dobsonia minor	preg f: Jan, May-June; n3; Papua, New Guinea	7073
Dobsonia moluccensis	preg f: Mar; n1; Papua, New Guinea	7073
	preg/lact f: June-Aug; New Guinea	10114
	birth: peak mid Aug-mid Nov; New Guinea	2835
	birth: end winter-spring; Australia	10698
Dobsonia peroni	testes: large Mar-Apr; no data June-Jan; Timor, Indonesia	3903
Eidolon helvum	spermatogenesis: Apr-June; e Africa	5759
	testes: large Apr-June, regress July-Mar; Uganda	7676, 7677
	testes: large May-June; W Africa	5388
	mate: Apr-June; Uganda	7676, 7677
	mate: May-July; Nigeria	3566
	mate: June-July, Sept-Oct; central Africa	254
	implant: Oct; e Africa	5759
	implant: Oct-Nov; Uganda	7676, 7677
	implant: Nov-Dec; Nigeria	3566
	preg f: Feb; n1; Liberia	11273
	preg f: Oct-Feb; Uganda	7677, 7678, 7680
	preg f: Nov; n5/12; Zaire	191
	preg/lact f: Jan-Apr; n4; Cameroun	2948
	birth: Feb-Mar; Uganda	7676, 7677, 7678
	birth: mid Mar-May, peak Apr-May; Nigeria	3566
	birth: Apr; e Africa	7684
	birth: Oct-Dec, Feb-Mar; central Africa	254
	birth: Dec-Feb; e Africa	5759
Eonycteris spelaea	preg f: Feb, Apr; n5; Malaysia	6446
	preg f: yr rd, peaks May-June, Oct-Nov; Malaysia	696
	preg f: yr rd; Selangor caves, Malaysia	7161
Epomophorus crypturus	preg f: end July-Mar; S Africa	10158
	birth: peak Sept; S Africa	10158
	lact f: Mar; n1; n Zimbabwe	331
Epomophorus gambianus	mate: Sept-Oct; few data; central Africa	252
	preg f: Feb-Mar, May; Niger	8611
	preg f: Mar, May, Sept; Burkina Faso	1139
	birth: Jan-Feb; few data; central Africa	252
	birth: Apr, Oct; Sudan	10780
	birth: Dec-Jan; few data; W Africa	5903
Epomophorus labiatus	testes: large Sept-Oct,; Uganda	8082
	mate: end Mar-early Apr?, end Sept-early Oct?; Uganda	8082
	mate: peak Oct-June; central Africa	252
	preg f: Feb; n1; Kenya	185
	preg f: May, Dec; n2; Zaire	11272
	birth: Feb-Mar; central Africa	11270
	birth: Feb-Apr; Sept-Oct; Uganda	8082
	birth: Mar; Kenya, Tanzania	252
	birth: Oct; Rwanda, Tanzania	252
	birth: Nov; Zambia	252
	young found: Nov-Jan; Zaire	191
Epomophorus wahlbergi	preg f: Feb; n1; Zambia	331
	birth: Mar; Kenya	252
	birth: Mar, Nov; Zambia	252

	birth: Oct; Tanzania	252
Epomops buettikoferi	preg f: Dec; n6; Mt Nimba, Liberia	11273
	birth: peaks Mar-Apr, Sept-Oct; W Africa	10780
	birth: peaks Aug, Dec-Jan; Mt Nimba, Liberia	1948, 11881
Epomops franqueti	spermatogenesis: Jan-Apr; June-Sept; Gabon	252
	mate: yr rd; Uganda; e Africa	5759
	preg f: Feb-Mar; n9; Cameroun	2955
	preg f: Mar-July, Sept-Jan; Kisangani, Zaire	5127
	preg f: peak Apr-Sept, Sept-Feb; Uganda	8081
	preg/lact f: yr rd; central Africa	252
	embryo weight: peaks Jan, July; Kisangani, Zaire	5127
	birth: early rainy season; Uganda	8081
	lact f: Mar-Apr; Rio Muni, W Africa	5387, 5388
	young found: yr rd; Zaire	191, 1346
Haplonycteris fischeri	testes: large May-July; Phillippines	4573
	mate: June; multiparous; Sept-Dec; nulliparous; Philippines	4571, 4573
	mate: yr rd; Philippines	4572
	implant: Sept-Dec; Philippines	4571
	preg/lact f: May, Aug, Oct-Nov; Philippines	7586
	birth: May-June; Philippines	4571, 4572
	wean: end Sept-Oct; Philippines	4572
Harpionycteris celebensis	preg f: Jan, Sept; n2; Sulawesi, Indonesia	809
	lact f: Jan; n1; Sulawesi, Indonesia	809
Hypsignathus monstrosus	m calling: mid Dec-early Feb, June-Aug; Gabon	1185
	preg f: May, Dec; Zaire	191
	preg f: July, Oct-Sept; n4; Mt Nimba, Liberia	11881
	preg f: Nov; n1; Cameroun	2948
	birth: Jan-Mar; June-July; Cameroun?, central Africa	252
	birth: est peaks Feb, July; e Africa	5759
	birth: Dec-Mar; Gabon	1346
Macroglossus minimus	preg f: Jan-Mar, July-Sept; n12; Papau, Newguinea, Bougainville, Solomon I	7073
	preg f: Feb, Mar, May; n3; Indonesia	809
	preg f: Feb, Apr, June; n4; Malaysia	6446, 6447
	preg f: Aug; n1; Australia	10698
	preg f: yr rd; New Guinea, Solomon I	7073
	preg/lact f: peak Jan-Aug; Philippines	135
	preg/lact f: June-Aug; New Guinea	10114
	preg/lact f: Aug, Oct; Philippines	7586
	lact f: Dec-Jan, Mar; n6; Solomon I	8481
	repro: yr rd; Bismarck Archipelago	10114
Magaerops ecaudatus	preg f: Feb-Mar, June; Malaysia	7161
	preg f: Feb, Apr, June; n10; Malaysia	6446
Megaloglossus woermanni	preg f: Mar; Zaire	808
	preg f: Aug; n2; Gabon	5387
	preg f: Aug; n4; Mt Nimba, Liberia	11881
	preg f: Sept; n1; Rio Muni, W Africa	5387
	preg/lact f: July-Aug; Liberia	11881
	lact f: Jan, July; Gabon	252
	lact f: Jan, Mar; Cameroun	252
	lact f: Feb, May; Zaire	808
	lact f: Apr; n1; Rio Muni, W Africa	5387
	lact f: July-Aug; n2; Mt Nimba, Liberia	11881
	lact f: Oct; Femando Po, central Africa	252
Melonycteris aurantius	lact f: Oct; Solomon I	8481
Melonycteris woodfordi	preg f: July, Sept; n2; Bougainville, Solomon I	7073
Micropteropus pusillus	conceive: Oct-early Nov; Zaire	808
	preg f: May, Nov; n2; W Africa	5387
	preg f: Aug; n1; Ethiopia	4747
	birth: Mar-Apr, Sept-Oct; Sudan	6844, 10780
	lact f: Jan; n2; s Burkina Faso	5903
	lact f: May; n1; W Africa	5387
	young found: Mar-Apr; W Africa	5387, 5388
	repro: Dec-Jan; tropical Africa	1346

Myonycteris torquata	preg f: Jan-Mar; n21; Mt Nimba, Liberia	11273
	preg f: Oct; n3; W Africa	5387
	birth: Aug-Sept, Feb-Mar; Ivory Coast	10778
	birth: peak Sept; Mt Nimba, Liberia	11881
	birth: Dec-Feb; n16; Gabon	1346
Nanonycteris veldkampi	preg f: mid Nov-mid Dec; n12; Mt Nimba, Liberia	11881
	preg f: Dec-Mar; n7; Mt Nimba, Liberia	11273
	birth: May-June; Mole Nat Park, w Africa	6844
	lact f: Jan; n1; Mt Nimba, Liberia	11273
Neopteryx frosti	preg f: Mar; n1; Sulawesi, Indonesia	809
Notopteris macdonaldi	preg/lact f: Dec-Jan; New Caledonia	7835
	preg f: July; New Caledonia	9457
	lact f: Aug; New Caledonia	9457
Nyctimene aello	preg f: Jan-Feb; n4; Papua, New Guinea	7073
Nyctimene albiventer	preg f: Jan, July-Aug; n6; Papua, New Guinea	7073
	lact f: Feb, May, Aug; n8; Papua, New Guinea	7073
	lact f: July; n1; Bougainville, Solomon I	7073
Nyctimene cephalotes	preg f: Jan, Oct; n2; Sulawesi, Indonesia	809
Nyctimene robinsoni	preg f: Aug; n2; QLD, Australia	7076
Otopteropus cartilagonodus	preg f: Apr; n1; Luzon, Philippines	7586
Paranyctimene raptor	preg f: Jan-Feb, May; n7; New Guinea	7073
Penthetor lucasi	preg f: Jan; n1; Sarawak, Malaysia	10314
	preg f: Feb, Apr, June; n4; Malaysia	6446
	lact f: May; n1; Sarawak, Malaysia	10314
	preg f: June, Sept; none Jan-Mar, July; Malaysia	7161
Ptenochirus jagori	preg/lact f: peak Jan-Mar; Philippines	135
	preg/lact f: May, Aug, Oct; Luzon, Philippines	7586
Pteralopex anceps	lact f: July; n1; Solomon I	8481
Pteropus alecto	mate: Mar-Apr; n Australia	2214
	preg f: Jan; n1; Sulawesi, Indonesia	809
Pteropus anetianus	testes large: Oct-Jan; New Hebrides	517
	preg f: May; n2; New Hebrides	9457
Pteropus giganteus	mate: July-Aug; Madras, India	7859
	mate: early Sept; India	7416
	mate: early Oct; India	3937
	mate: mid Oct; zoo	3370
	mate: Dec-Jan; India, Burma, Sri Lanka	6845, 6847
	preg f: Nov-Apr; India	3915
	birth: Jan-Mar; w India	1340, 1342, 1346
	birth: May-June; Sri Lanka	1340, 1342, 1346
	birth: end Jan-early Feb; India	7416
	birth: Mar; Madras, India	7859
	birth: end Apr-early May; zoo	3370
	birth: May-June, Dec-Mar; zoo	517
	birth: end May (perhaps early June); Sri Lanka	6845
	birth: end May-early June; Sri Lanka	8494, 8501
	wean: May; Madras, India	7859
	young found: yr rd; Africa	8798
Pteropus howensis	preg f: Aug; Solomon I	8481
	preg f: Aug; n2; Solomon I	9457
Pteropus neohibernicus	preg/lact f: June-Aug; New Guinea	10114
Pteropus ornatus	preg f: Apr, June-Oct; n14; New Caledonia	9457
Pteropus poliocephalus	testes large: Feb-Apr; QLD, Australia	7065
	mate: peak Mar-Apr; QLD, Australia	10878, 10879
	birth: Sept-Oct; e Australia	7833a
Pteropus rayneri	preg f: July; n1; Solomon I	9457
	preg f: July or Nov; Solomon I	8481
Pteropus rodricensis	mate: June, Nov-Dec; Rodriques I, UK	1625
	birth: Nov-Dec, Mar; Rodriques I, UK	1625
Pteropus rufus	preg f: Oct; n3; Madagascar	8906
Pteropus scapulatus	mate: Oct; n1; QLD, Australia	8954a
	preg f: Feb; NSW, Australia	8954a
	birth: Apr-May; n21; e Australia	7833a
	birth: May; Australia	623

Pteropus temmincki	f with yg: June, July, or Aug; n1; New Guinea	10114
Pteropus tonganus	testes: large Nov-Feb; New Hebrides	517
	Leydig cells: large Oct-Feb; New Hebrides	4097
	mate: Feb; New Hebrides	4097
	preg f: Feb-Oct; New Hebrides	517
	preg f: May; New Hebrides	9457
	preg f: June; New Caledonia	9457
	birth: peak end Aug-early Sept; New Hebrides	517
Pteropus vampyrus	preg f: Apr; n1; Philippines	7586
	preg f: Nov-Jan; few data; Selangor, Malaysia	7161
	birth: end Mar-early Apr; Thailand	6369
Rousettus aegyptiacus	sperm: yr rd, peak Feb-Apr; Egypt	6721
	testes: peak Apr, Aug, min Nov, Feb, June-Aug; Uganda	7679
	mate: end Apr; not seasonal; Egypt	6093
	mate: June-mid Sept; Cape Province, S Africa	6549
	mate: est Nov-Jan; Egypt	8817
	mate: yr rd; Egypt	6721
	preg f: Mar; n1; Mt Nimba, Liberia	11273
	preg f: Aug; n1; Mt Nimba, Liberia	11881
	preg f: Oct 90%; Transvaal, S Africa	5226
	preg f: end Oct-early Dec; n11; Mt Nimba, Liberia	1948
	preg f: Dec-Mar, July-Sept; Uganda	7678
	birth: Mar, Sept, before peak rain; Uganda	7678, 7679
	birth: peak Mar, Sept; e Africa	8084
	birth: Oct-June; S Africa	4698
	birth: Nov-Dec; Transvaal, S Africa	5226
	birth: yr rd; se Zaire, Rwanda	255
	birth: yr rd; n73; zoo	12128
	birth: yr rd; zoo	517
	lact f: Feb; W Africa	5387
	lact f: Mar-May; Egypt	8817
	lact f: Aug; n1; Mt Nimba, Liberia	
Rousettus amplexicaudatus	mate: May-July; Java	5924
	preg f: Feb, Apr, June, Dec; n10; Malaysia	6446
	preg f: Mar; n2; Sulawesi, Indonesia	809
	preg f: July, Sept; n2; Bougainville, Solomon I	7073
	preg/lact f: peak Feb-Apr; Philippines	135
	preg/lact f: June-Aug; New Guinea	10114
	birth: Mar-May; no data June-Jan; Timor, Indonesia	3903
	lact f: Jan; n1; New Guinea	7073
	lact f: Dec; n1; Sulawesi, Indonesia	809
	lact f: Dec-Jan; n11; Solomon I	8481
Rousettus angolensis	preg f: Feb, Dec; n2; Mt Nimba, Liberia	11273
	preg f: July-Sept; Mt Nimba, Liberia	11881
	birth: June, Nov-Feb; se Zaire, Rwanda	255
	birth: Dec-Jan; Cameroun	2948
Rousettus celebensis	preg f: Mar; n1; Sulawesi, Indonesia	809
	birth: Mar; n1; Sulawesi, Indonesia	809
	lact f: Apr; n3; Sulawesi, Indonesia	809
Rousettus lanosus	mate: yr rd; e Africa	5759
Rousettus leschenaulti	anestrus: July-Nov; India	3916
	seminal vesicles enlarged: Oct-Apr; India	3938
	spermatogenesis: Nov-Mar; India	1342
	testes: abdominal May-Oct; India	3938
	mate: Nov-early Dec; Pakistan	9175
	conceive: mid Nov-mid Mar; India	3916
	preg f: Nov-July; India	3915, 3921
	birth: mid Feb-early Mar; Pakistan	9175
	birth: Mar, July; India	3916
	birth: Mar, July-Aug; India	1340, 1342
Scotonycteris ophiodon	preg f: end Aug-early Sept, Nov; n3; Mt Nimba, Liberia	11881
	birth: end Dec-early Jan; central Africa	252
	lact: Feb-Apr; central Africa	252

Scotonycteris zenkeri	mate: Aug-Sept; central Africa	252
	preg f: Aug; n1; Mt Nimba, Liberia	11881
	preg f: Nov; n1; Mt Nimba, Liberia	1948
	birth: Dec-Jan; central Africa	252
	lact: Dec-Mar/Apr; central Africa	252
	lact f: Dec; n1; Mt Nimba, Liberia	11273
Syconycteris australis	preg f: Jan, Mar-May, July, Sept; n12; Papua, New Guinea	7073
	preg f: Feb; n4; Beerwah, QLD, Australia	2841
	lact f: Jan; n2; Papua, New Guinea	7073
	lact f: Apr; n1; New Guinea	7073
Thoopterus nigrescens	preg f: Jan; n2; Sulawesi, Indonesia	809

Order Chiroptera, Family Rhinopomatidae, Genus *Rhinopoma*

The insectivorous, mouse-tailed bats (*Rhinopoma* 10-45 g) range from western Africa to southeast Asia. The zygote enters the uterus in the morula stage (5536). The labyrinthine, endotheliochorial, discoidal placenta has a superficial, mesometrial, circumferential, circumpuntial implantation into a symmetrical bicornuate uterus (3913, 3917, 5532, 5534, 5535, 5536, 10288, 11799) and is not normally eaten (248). Birth lasts 20-25 min and the naked, blind neonate is born head first (248, 5534, 9175). Females have 2 pectoral mammary glands, 2 pubic false teats (248, 5534, 5759, 6720), and are monoestrous (248, 5534). Males have ampullary glands, which store sperm, abdominal testes (248, 5533, 6065, 9999), urethral glands (7440), and seminal vesicles (9178). Details of female anatomy (553, 3937, 5534, 6161, 10288) and endocrinology (553) are available.

NEONATAL MASS		
Rhinopoma hardwickei	2-3 g; n6	5534
WEANING MASS		
Rhinopoma hardwickei	7-8 g; 20 d	5534
	12-13 g; 2 mo	5534
LITTER SIZE		
Rhinopoma hardwickei	1 emb	553, 5534
Rhinopoma microphyllum	1	248
	1 yg	8720
	1 emb; r1-2; 1 set twins; n237 (table), n273 (text) f	6161
SEXUAL MATURITY		
Rhinopoma hardwickei	f: 8.5-9 mo	5534
	f: 9-9.5 mo	553
	m: 16-17 mo	5534
Rhinopoma microphyllum	f/m: 2nd yr	248
GESTATION		
Rhinopoma hardwickei	95-100 d	553, 5534
Rhinopoma microphyllum	70-80 d	1340
	4 mo	6161
	123 d	3915
	min 123 d	248
LACTATION		
Rhinopoma hardwickei	teat attachment: 20 d	5534
	wean: 2 mo	553
Rhinopoma microphyllum	wean: 4-6 wk	248
INTERLITTER INTERVAL		
Rhinopoma hardwickei	1 yr	553, 5534
Rhinopoma microphyllum	1 yr	248
SEASONALITY		
Rhinopoma hardwickei	testes: regress Mar; India	5533
	spermatogenesis: Nov-Dec; India	5533
	mate: mid Feb-early Mar; India	553
	mate: mid Feb-mid Apr; India	5533
	mate: end Feb-mid Apr; India	5534
	ovulate: mid Mar; India	553
	ovulate: mid Mar-Apr; India	5534
	conceive: Apr; India	5533
	preg f: Apr; Egypt	8818

	birth: June; India	1340
	birth: June; Pakistan	9175
	birth: mid June-early July; Agra, India	553
	birth: mid June-July; India	5534
	birth: est July; Sudan	5759
	lact f: July; n8; India	9062
	half grown juveniles: mid July; n4; W Africa	5903
Rhinopoma microphyllum	torpor: end Oct-Feb/early Mar; India	248
	testes size: increase Oct, large end Jan, small end Apr; India	248
	mate: Feb-Mar; India	6161
	mate: Mar; India	248
	mate: Mar; India	3937
	mate: Apr-early July; Bengal, India	10288
	birth: June; India	1340
	birth: end June; India	6161
	birth: end July-early Aug; India	248
	half grown yg: Aug; W Africa	5903

Order Chiroptera, Family Emballonuridae

Sheath-tailed, sac-winged, and ghost bats (*Balantiopteryx* 5-9 g; *Centronycteris* 5 g; *Coleura* 9-11 g; *Cormura* 10 g; *Cyttarops* 50-55 mm; *Diclidurus* 19 g; *Emballonura* 4-40 g; *Peropteryx* 6-19 g; *Rhynchonycteris* 2-4 g; *Saccolaimus* 55-105 g; *Saccopteryx* 3-11 g; *Taphozous* 10-30 g) are insectivores from the tropics and subtropics. Females may be larger than males (1186, 1187, 6568, 7122, 8494, 10114; cf 5759). *Taphozous* exhibits reproductive asymmetry as the right ovary and uterus are used preferentially (4399, 5261, 5262, 5715, 5716, 5829, 9491; cf 3911, 3912, 6030, 9490). *Balantiopteryx* and *Peropteryx* use both ovaries (6568, 11804). The labyrinthine, endotheliochorial, discoidal placenta has a placental hematoma and a superficial, circumferential or lateral implantation into a bipartite or bicornuate uterus (3912, 3913, 3917, 5715, 5716, 6568, 6627, 6966, 8948, 9178, 9491, 11799, 11808). Development may be delayed (*Taphozous* 6029). Placentophagia is reported (9175), as is estrus postpartum (1340, 1346, 4399, 6029, 6030, 8948, 9491; cf 5831), but monoestry also occurs (*Balantiopteryx* 383).

Neonates are 25-30% of adult mass and most are born blind and naked (5759, 7161, 8501, 9175, 9491, 10669; cf 5903); however, *Taphozous* is born furred with open eyes (5380). *Taphozous* females have 2 mammae (6720). Males have penial spines and ampullary glands or seminal vesicles (5829, 6065, 6966, 7440, 7441, 7662, 9178). Details of anatomy (419, 6721, 9490), endocrinology (5383), spermatogenesis (5382, 7439), and social organization (1186, 1187, 1188, 1189) are available.

NEONATAL MASS		
Balantiopteryx plicata	~2 g	2341
	2 g	6568
Saccopteryx bilineata	3.5 g; r3.2-4.0	10669
Taphozous georgianus	7.8 g; r7.3-8.1; n4	5380
Taphozous melanopogon	7.5 g; r7-8; n2 neonates; 7-8 g full term	9491
NEONATAL SIZE		
Taphozous georgianus	HB: 50.2 mm; r46-56; n4; FA: 35.5 mm; r34-36; n4	5380
Taphozous nudiventris	47-51 mm	111
WEANING MASS		
Taphozous georgianus	13-16 g; 18-33 d	5380
Taphozous melanopogon	~20 g	9491
Saccopteryx bilineata	m: 6.6 g; se0.41; n13	10669
	f: 7.0 g; se0.31; n8	10669
WEANING SIZE		
Taphozous georgianus	HB: 62-68 mm; FA: 54-61 mm; 18-33 d	5380
LITTER SIZE		
Balantiopteryx io	1 yg	384
Balantiopteryx plicata	1 yg	2341
	1 emb; n1 f	2350, 2351
	1 emb ea; n3 f	3204
	1 emb ea; n10 f	1937
	1 emb ea; n21 f	5409

Centronycteris maximiliani	1 emb; n1 f	6260
Coleura afra	1	10158
	1	5759
	1	252
Diclidurus albus	1 yg	1671
Emballonura nigrescens	1 emb ea; n9 f	7073
Emballonura monticola	1	7161
Peropteryx macrotis	1 emb; n1 f	9113
	1 emb ea; n3 f	10974
	1 emb, yg ea; n14 f	7624
	1 emb ea; n20 f	3999a
Rhynchonycteris naso	1 emb; n1	11526
	1 yg; n1	452
	1 emb ea; n2 f	10974
	1 emb ea; n3 f	1475
	1 emb ea; n3 f	5405e
	1 emb ea; n4 f	1635
	1 emb ea; n6 f	2230, 4228
	1 yg ea; n9 lt	3999a
	1 emb ea; n15 f	7624
	1 emb, yg ea; n17 f	7624
Saccolaimus peli	1	191
Saccolaimus saccolaimus	1	8494, 8501
	1 yg; n1 f	7161
Saccopteryx bilineata	1 emb; n1 f	1162
	1 yg; n1 f	3024
	1 emb ea; n2 f	4228
	1 emb ea; n2 f	11471a
	1 emb ea; n4 f	10974
	1 emb, yg ea; n4 f	9113
	1; n5	452
	1 emb ea; n7 f	7624
	1 emb ea; n38 f	3999a
	1 ea; n41 f	1186
Saccopteryx canescens	1 ea; n2 lt	452
Saccopteryx leptura	1 emb; n1 f	3999a
	1 emb; n1 f	5425
	1 ea; n2 lt	452
	1 emb ea; n2 f	5405c
	1 emb ea; n3 f	2345
Taphozous georgianus	1 ea; n3 lt	7075
	1 yg; n282	5380
Taphozous longimanus	1	1340, 1346
	1	3931
	1	6030
	1 yg	3912, 3927
	1 emb; n1 f	8501
	1 emb ea; n129 f	3927
Taphozous mauritianus	1 yg	5759
	1	252
	1	10158
	1	1346
	1	191
Taphozous melanopogon	1	3927
	1	1340, 1342, 1346
	1	7161
	1 cl	5261, 5262
	1 emb ea; n57 f	3927
	1 emb ea; n89 f	9491
	1 emb, yg ea; n146 lt	5714, 5716
Taphozous nudicluniatus?	1; n1 lt	7075

Taphozous nudiventris	est 1	10006
	1	1340
	1	9175
	1	111
Taphozous perforatus	1	1340
	1 emb ea, n2 f	8611
	1 emb ea; n2 f	10158
SEXUAL MATURITY		
Balantiopteryx plicata	f: 1st birth: 1 yr	1187
Coleura afra	f: 1st birth: 1 yr	7122
Rhynchonycteris naso	f: 1st birth: max 18 mo	1187
Saccopteryx bilineata	f: 1st birth: 1 yr	1187
Saccopteryx leptura	f: 1st birth: max 18 mo	1187
Taphozous georgianus	f: 1st mate: 9 mo	5380
	m: 1st sperm: 8-9 mo	5380
	m: 1st mate: 21 mo	5380
Taphozous melanopogon	f: 1st conceive: 8 mo	5716
	f/m: not w/in 1st yr	9491
GESTATION		
Balantiopteryx plicata	~4.5 mo; r4.5-8 mo	6568
Peropteryx kappleri	~5 mo; field data	8948
Peropteryx macrotis	4-4.5 mo	6568
Saccopteryx leptura	4-4.5 mo	6568
Taphozous georgianus	~3 mo	5379, 5380
	est 4 mo	5829
Taphozous longimanus	105 d; 1st preg; 86 d; 2nd preg; indirect	6029, 6030
Taphozous mauritianus	~6 wk; field data; not many	5573
Taphozous melanopogon	120-125 d	9491
	~4 mo	5714
Taphozous perforatus	3 mo	255
LACTATION		
Balantiopteryx plicata	1st fly: 2 wk	383
	wean: 9 wk	383
	wean: 3 mo	1189
Rhynchonycteris naso	wean: 3 mo	1189
Saccopteryx bilineata	solid food: 2 wk	1186
	solid food: 18-30 d	10669
	wean: 8-10 wk	1186
	wean: ~2 mo	10669
Saccopteryx leptura	wean: 2.5 mo	1186
Taphozous longimanus	wean: 2-3 wk	6030
Taphozous georgianus	independent: 18-33 d	5380
Taphozous melanopogon	wean: 5-6 wk	9491
	nurse: 2 mo	1342
	wean: 2 mo	7161
LITTERS/YEAR		
Balantiopteryx plicata	1	1187
	1	6568
Coleura afra	1	5759
	2; r1-2	7122
Peropteryx kappleri	3	10785
Rhynchonycteris naso	max 2	1187
	>1	2230
Saccopteryx bilineata	1	1187
	1	10669
Saccopteryx leptura	max 2	1187
Taphozous longimanus	1	1340
	2	6029
	2	3912, 3937
Taphozous melanopogon	1	5714, 5716
	1	1342
Taphozous nudiventris	1	9175
	min 2	10006
Taphozous perforatus	2	9297

SEASONALITY

Balantiopteryx io	preg f: Apr; n2; Veracruz, Mexico	2230a, 4228
	lact f: July; n3; Oaxaca, Mexico	523
Balantiopteryx plicata	sperm: yr rd, low May, Aug; Central America	6568
	mate: end Jan-mid Feb; Central America	6568
	ovulate: Mar; Central America	6568
	preg f: May; n3; El Salvador	3204
	preg f: May-July; n10; Mexico	1937
	preg f: June; n21; Sinaloa, Mexico	5409
	preg f: June-July; n14; Sinaloa, Mexico	5413a
	birth: June-mid July; Central America	6568
	birth: synchronous end June; Costa Rica	1187
	birth: end June-early July; Mexico	2341
	lact f: July; Oaxaca, Mexico	523
	lact f: Sept; n1; Jalisco, Mexico	11471a
Centronycteris maximiliani	preg f: May; n1; Nicaragua	4048_1
	preg f: May; n2; Costa Rica	6260
Coleura afra	mate: end June, end Dec; Kenya	7122
	ovulate: July, Dec; Kenya	7122
	preg f: end Dec; Tanzania	5759
	birth: end Jan; central, e Africa	252
	birth: Mar-June, Oct-Dec; Kenya	7122
	birth: Oct, end of rains; Sudan	5759
Coleura seychellensis	birth: Nov-Dec; Seychelles	7893
Cormura brevirostris	preg f: Aug-Sept; French Guiana	1350a
Diclidurus albus	preg f: Jan-June; Jalisco, Mexico	4671
	preg f: Mar, May; n2; Michoacan, Mexico	4671
Emballonura monticola	preg f: Feb-Mar, Oct-Nov; Selangor, Malaysia	7161
	preg f: Oct, Dec; n2; Malaysia	6446
	birth: Mar; Kuala Lumpur, Malaysia	3974
Emballonura nigrescens	preg f: Feb, May-July; n9; Papua, New Guinea	7073
Peropteryx kappleri	conceive: May-June, Dec-Jan; Colombia	8948
	preg f: Apr; Costa Rica	6260
	birth: peak Mar-May, Oct-Dec; Colombia	8948
	birth: mid May-early June; Costa Rica	1187
Peropteryx macrotis	preg f: Feb-Mar; n3; Guatemala	5405e
	preg f: Apr; n16; Yucatan, Mexico	5424f
	preg f: Apr, Aug-Sept; n20; Peru	3999a, 10974
	preg f: Oct; French Guiana	1350a
	lact f: Aug; n1; Quintana Roo, Mexico	5424f
Rhynchonycteris naso	preg f: Jan-Feb, Apr; Veracruz, Mexico	2230, 4228
	preg f: Feb; n3; Campeche, Mexico	5424f
	preg f: Mar; n2; Trinidad	3900a
	preg f: Mar; n3; Guatemala	5405e
	preg f: May; n3; El Salvador	1475
	preg f: May, July-Sept; n9; Peru	3999a, 10974
	preg f: mid July-mid Aug; n4; Trinidad	1635
	preg f: Aug; n1; Guatemala	2552
	preg f: Oct-Dec; French Guiana	1350a
	birth: Apr; none Nov-Mar; Costa Rica	1187
	lact f: July; Chiapas, Mexico	1641
	lact f: Oct; Veracruz, Mexico	4228
Saccolaimus peli	epididymal sperm: Aug; Gabon	252
	preg f: June, Dec; n3; ne Zaire	191, 252, 1207
	birth: Jan; Gabon	252
	birth: Jan, July; ne Zaire	252
Saccolaimus saccolaimus	preg f: Sept; n1; Malacca, se Asia	7161
	preg f: Sept-Nov; Sri Lanka	8501
	lact f: Sept-Nov; Sri Lanka	8493, 8494
Saccopteryx bilineata	mate: Dec; Panama	10669
	preg f: Feb; n2; Veracruz, Mexico	4228
	preg f: Feb, Apr; n2; Guatemala	5405e
	preg f: Mar; n2; Jalisco, Mexico	11471a
	preg f: Mar; n1; El Salvador	3204

	preg f: Mar-May, July; Trinidad	3900a
	preg f: Apr; n1; Panama	4228a
	preg f: June; n4; Veracruz, Mexico	6148a
	preg f: June-Aug, Oct; n38; Peru	3999a, 10974
	birth: May-June; Panama, Costa Rica	3322
	birth: synchronous early May; Panama	10669
	birth: end May-mid June; Trinidad	1186
	lact f: June-July; Trinidad	3900a
	lact f: July; n1; Guatemala	2552
Saccopteryx leptura	preg f: Feb; n2; Nicaragua	5405c
	preg f: Feb; n1; Panama	2345
	preg f; Mar; n1; Nicaragua	5425
	preg f: Apr-May; n2; Nicaragua	2345
	preg f: May; Trinidad	3900a
	birth: May; Costa Rica	1187
	birth: May, Nov; Costa Rica	1187
	birth: June-July; Trinidad	1186
Taphozous georgianus	accessory glands: enlarge spring; central QLD, Australia	5381
	epididymal sperm: yr rd; central QLD, Australia	5381
	sperm: yr rd, low mid fall-mid winter; WA, Australia	5829, 5831
	scrotal testes: summer; WA, Australia	5829
	scrotal testes: Nov-Feb; QLD, Australia	5380, 5382
	birth: Oct-Feb; WA, Australia	5829, 5831
	birth: end Nov-early Dec; QLD, Australia	5380
Taphozous longimanus	quiescent: Sept-Dec; India	6030
	mate: mid Jan-early Feb, May-early June; Varanasi, India	6029
	mate: mid Jan, end Apr-mid May; India	6030
	mate: yr rd; India	3912
	preg f: Sept; n1; Sri Lanka	8501
	preg/lact f: yr rd; India	3911
	birth: end Apr-early May; Varanasi, India	6029
	birth: yr rd; India	1340, 1345, 3912
Taphozous mauritianus	testes: large Dec; ne Zaire, Tanzania	252
	mate: July; ne Zaire, Angola, Tanzania, s Sudan	252
	mate: Oct-Nov; Katanga, Zambia	252
	mate: Dec-Jan; ne Zaire; Tanzania	252
	birth: Feb; Katanga, Zambia	252
	birth: est Apr-May; ne Zaire., Tanzania	252
	birth: Oct-Nov; ne Zaire, Angola, Tanzania, s Sudan	252
	young found: yr rd; Zaire	191
	repro: yr rd; equatorial Africa	253
Taphozous melanopogon	anestrus: Dec; India	5262
	regression genital system: Oct-Nov; India	5262
	preg f: Jan-May; India	9491
	preg f: Jan-June, peak Feb-Apr; n522; India	5716
	preg f: Jan-Mar, Oct-Nov; no data other mo; Malaysia	7161
	preg f: only Feb-May; India	5715
	preg f: May-June; India	5262
	preg/lact f: Sept; Sri Lanka	8501
	birth: mid Apr-mid May; India	1340, 1342
	birth: end May-early June; India	9491
	birth: end July-early Aug; India	5262
	lact f: Sept; India	5262
	wean: June-mid July; India	9491
	young of all ages found: Sept; Sri Lanka	8494, 8501
Taphozous nudiventris	hibernate: Dec?-Mar; Iraq	111
	epididymal sperm: Jan-Feb, Oct; Cairo, Egypt	6721
	spermatogenesis: Feb-May; Cairo, Egypt	6721
	mate: Sept; Pakistan	9175
	mate: fall; Iraq	110, 111
	ovulate: Mar; Iraq	111
	preg/lact f: July; few data; Burkina Faso, w Africa	5903
	birth: mid Apr; Pakistan	9175
	birth: end May; Iraq	111

	birth: mid June-mid July; India	1340
Taphozous perforatus	mate: Jan-Mar; Africa	255
	preg f: Feb-Mar; Niger	8611
	birth: Apr-May; India	1340
	birth: Apr-May; Kenya	252, 255
	birth: May-June; Nigeria	252, 255
	birth: end May-early June; few data; Nigeria	4399
	wean: mid June; Africa	252
	nearly full grown: mid Aug; Burkina Faso, w Africa	5903
	repro: est not seasonal; equatorial Africa	253

Order Chiroptera, Family Craseonycteridae, Genus *Craseonycteris*

Few reproductive data have been reported for *Craseonycteris thonglongyai*, a 1.7-2 g bat endemic to Thailand. Females have 2 pectoral mammae and 2 inguinal false teats (4746). Details of its natural history are available (3576, 4748, 7109).

LITTER SIZE		
Craseonycteris thonglongyai	1 yg	2747
INTERLITTER INTERVAL		
Craseonycteris thonglongyai	1 yr	2747
SEASONALITY		
Craseonycteris thonglongyai	repro: end Apr-?; Thailand	2747

Order Chiroptera, Family Nycteridae, Genus *Nycteris*

Nycterids (*Nycteris*) are 5-43 g insectivores primarily from Africa, but also from Malaysia and Indonesia. Implantation may preferentially occur in the right horn of the bicornuate or duplex uterus (824, 6966, 10154). A postpartum estrus is present in tropical species (255, 6962, 6966) after the birth of a furred neonate (11270). Temperate forms are monoestrous, but have two periods of follicular development (824). Females have 2-4 mammae (331, 502, 4079, 6720, 7557). Males have ampullary glands and a baculum (6065, 6966).

NEONATAL MASS		
Nycteris arge or *N major*	4 g; n1 emb	11273
LITTER SIZE		
Nycteris arge or *N major*	1 emb; n1 f	11273
Nycteris grandis	1 emb; n1 f	11273
	1.5 emb; r1-2; n4 f	1347
Nycteris hispida	1	11270
	1 emb; n1 f	191
Nycteris javanica	1 emb, yg; n2 lt	7161
Nycteris macrotis	1 emb; n1 f	252
	1 emb; n1 f	8611
	1 yg; n1 f	8611
	1 emb ea; n2 f	4924
Nycteris nana	1	11270
Nycteris thebaica	1	10158
	1	8965
	1	10154
	1 emb; n1 f	331
	1 emb; n1 f	9945
	1 emb ea; n3 f	7719
	1.5 emb; r1-2; n2 f	185, 186
GESTATION		
Nycteris hispida	~3 mo	11270
Nycteris macrotis	2.5 mo	255
Nycteris nana	~3.5 mo; field data	11270
Nycteris thebaica	2.5 mo	255
	5 mo	824

LACTATION		
Nycteris arge	nurse: ~2 mo	5759
Nycteris macrotis	wean: 2 mo	255
Nycteris nana	wean: 45-60 d	252
	wean: 45-60 d	11270
Nycteris thebaica	wean: 2 mo	824
	wean: 2.5 mo	255
LITTERS/YEAR		
Nycteris hispida	1 or 2	11270
Nycteris macrotis	2-3	252, 255
Nycteris nana	1	11270
Nycteris thebaica	1-2?	255
SEASONALITY		
Nycteris arge	mate: Jan-Feb; Zaire	252
	mate: Oct; Zaire	252
	birth: Jan; Zaire	252
	birth: Apr-May; Zaire	252
	preg f: Mar, Dec; n3; Albert Park, Zaire	11272
	lact f: Apr; n1; Albert Park, Zaire	11272
	wean: Mar-June/July; Garamba Nat Park, Zaire	252
Nycteris grandis	birth: Jan; Cameroun	252
	birth: mid Apr, early Aug, mid Dec; Gabon	252
	birth: May; Garamba Nat Park, Zaire	252
	lact f: July-Aug; n2; Mt Nimba, Liberia	11811
Nycteris hispida	mate: early Jan; Zaire	11270
	mate: Jan; Tanzania, Zaire	252
	mate: June-July; Gabon	252
	mate: Dec; Cameroun	252
	preg f: Aug; n1; Rio Muni, W Arfica	5387
	preg f: Aug; n1; Zaire	191
	birth: Jan, June; Zaire	11270
	birth: Feb-Mar; Cameroun	252
	birth: Mar-Apr; Tanzania, Zaire	252
	birth: end Mar-early Apr; Zaire	11270
	birth: Sept-Oct; Gabon	252
	birth: Dec; n3; w Great Rift Valley, e Africa	6966
	lact f: Oct, Dec; Rio Muni, W Arfica	5387
	wean: May; Cameroun	252
	wean: June; Garamba Nat Park, Zaire	11270
	wean: June-July; Tanzania, Zaire	252
	wean: Nov; Gabon	252
	wean: Dec; Rio Muni, Guinea	252
Nycteris javanica	preg f: Mar; n1; Perlis, Malaysia	7161
	birth: Apr; n1; Kuala Lumpur, Malaysia	3974
	lact f: Nov; n1; Selangor, Malaysia	7161
Nycteris macrotis	mate: Aug, Oct; central Africa	252, 255
	mate: Dec; Garamba Nat Park, Zaire	252
	preg f: May; n1; Niger	8611
	preg f: end Dec; n1; Tanzania	5759
	birth: Jan; Kenya	252
	birth: Jan-Mar; Rwanda	252, 255
	birth: Mar; Garamba Nat Park, Zaire	252
	birth: Oct-Nov, early Jan; Rwanda	252
	birth: Oct-Dec; Liberia	255
	lact f: Dec; few data; Tanzania	186
Nycteris nana	mate: early Dec; Zaire	11270
	mate: end Dec-early Jan; central Africa	252
	birth: simultaneously mid Mar; central Africa	252
	birth: mid Mar; Zaire	11270
	wean: May-mid June; Zaire	11270
	wean: mid May; central Africa	252
Nycteris thebaica	estrus: July; Natal, S Africa	824
	mate: Apr; Rwanda	252
	mate: June; Natal, S Africa	824

mate: July; se Zaire, Tanzania	252
mate: Aug; Zambia	252
mate: Oct-Nov; Kenya, Tanzania	252
conceive: early Nov, early June; Natal, S Africa	824
preg f: end Jan-early Feb; n2; e Africa .	185
preg f: May; n3; Saudi Arabia	7719
preg f: Aug; n1; Zambia	331
preg f: Aug-Oct only; Zimbabwe	10158
preg f: Oct; n1; Namibia	9945
preg f: Dec; n1; Tanzania	186
birth: Jan-Feb; Kenya, Tanzania	252
birth: July; Rwanda	252
birth: ~Sept-Feb; few data; Botswana	10154
birth: mid Oct; Katanga, Zaire	252
birth: early Nov; Zambia	252
birth: early Nov; Natal, S Africa	824
birth: early summer; Transvaal, S Africa	8965
birth: yr rd; se Zaire, Rwanda	255
wean: Jan-Feb; Zambia	252
wean: Apr; Kenya, Tanzania	252
wean: end Sept; Rwanda	252
wean: end Dec; se Zaire, Tanzania	252

Order Chiroptera, Family Megadermatidae

False vampire and yellow-winged bats are 20-170 g predators on insects and small vertebrates in equatorial Africa (*Cardioderma*, *Lavia*), tropical Australia (*Macroderma*), India, and southern and southeastern Asia (*Megaderma*). Females may be larger than (*Cardioderma* 7121; *Lavia* 4924) or equal to (*Megaderma* 8494) males in size. Ovulation in *Megaderma* is predominantly from the left ovary (8872, 8876, 8890), while *Cardioderma* uses both ovaries (6966). Multiovular follicles are common (3937). Connective tissue links the ovaries with the uterus (3937). Embryos enter the bipartite or bicornuate uterus in the morula stage where they develop endotheliochorial, discoid placentae with superficial, circumferential, implantation that can be either mesometrial, antimesometrial, or lateral (3910, 3917, 3928, 3929, 6966, 11799). The left uterine horn is used preferentially (3928, 8872, 8876, 8887, 9490; cf 3910).

Placentophagia occurs occasionally (3930, 11270) after the birth of either naked and blind (*Cardioderma* 4924, 5759) or furred (*Lavia* 11270) and open-eyed (*Megaderma* 3930) neonates. Females have 2 functional, pectoral teats and both sexes have 2 nonfunctional pubic teats (2698, 4669, 6966, 7161, 9297). Males have urethral glands, ampullary glands or seminal vesicles, and a baculum (5704, 6065, 6966, 7663, 9178). Details of birth (3930) are available.

NEONATAL MASS

Megaderma lyra	7-8 g	3915
	10.25 g; r8.6-11.38; n11	3930

LITTER SIZE

Lavia frons	est 1	11663
	1	9297
	1	11270
	1 yg	11233, 11234
	1 emb; n1 f	191
Macroderma gigas	1 yg	7075, 7076
Megaderma lyra	1	8890
	1	8876
	1 yg	1340, 1343
	mode 1 yg; stillborn twins; n1 lt	3930, 3931
	1 emb; 0.3% twins; r1-2; n330 f	3927
	2; n1 lt	8887
Megaderma spasma	1	8494, 8501
	1 yg; 2 rare	1340, 1343, 1346
	mode 1; r1-2	7161

SEXUAL MATURITY		
Megaderma lyra	f: 1st mate: 1.5 yr	8876
	m: 15 mo	8890
	f: 18 mo	8890
GESTATION		
Cardioderma cor	3 mo	252
	~3 mo	5759
Lavia frons	~3 mo	5759
	~3.5 mo; field data	11270
Megaderma lyra	5 mo; field data	3915, 3928
	~150 d: field data	8876
	150-160 d	7161
	150-160 d	8890
Megaderma spasma	70-90 d	1340
	~5 mo	7161
LACTATION		
Lavia frons	wean: >50 d	11234
	wean: 2.5 mo	252
Megaderma spasma	wean: 4-5 wk; few data	8876
Megaderma lyra	wean: 2 mo	8890
LITTERS/YEAR		
Cardioderma cor	min 2	252
Lavia frons	1	11270
SEASONALITY		
Cardioderma cor	mate: Apr-May; Kenya	252
	mate: Sept-Oct; Tanzania	252
	preg f: mid-May-mid-Aug, Nov-Jan; Kenya	252
	preg f: mid Oct-mid Jan; Tanzania	252
	preg/lact f: yr rd; few data; Kenya	7121
	birth: early Jan; Tanzania	252
	birth: Jan, Aug; Kenya	252
	birth: Aug+; Eritrea to n Zambia	5759
	lact f: Aug; n1; e Africa	4924
	wean: mid Apr, end Oct; Kenya	252
	wean: end Mar; Tanzania	252
Lavia frons	mate: Jan, n1; Seronera, Tanzania	11663
	mate: end May; se Zaire	252
	mate: end Dec; Rwanda	252
	mate: end Dec; Zaire	11270
	preg f: Mar; n1; Zaire	191
	birth: end Mar; Rwanda	252
	birth: end Aug; se Zaire	252
	birth: early Apr; Zaire	11270
	birth: early Apr; n2; Kenya	11233
	birth: mid-Apr, mid-Aug, mid-Nov, mid-Jan; Tanzania	252
	repro: yr rd; equatorial Africa	253
Macroderma gigas	mate: end Oct-early Nov; n Australia	10809
	preg f: June; n1; nw Australia	2697a, 2698
	preg f: Aug; n1; QLD, Australia	7076
	birth: Oct-Nov; Pilbara, WA, Australia	2698
	birth: Sept; n1; nw Australia	2698
	young seen: Sept-Nov; Australia	10698
Megaderma lyra	spermatogenesis: July-Dec; Maharashtra, India	8890
	sperm: July-Jan; India, Sri Lanka	8876
	testes: large Oct, small Dec; India, Sri Lanka	8876
	mate: early Oct; India	3937
	mate: early-mid Nov; India	3928
	mate: end Nov; India, Sri Lanka	8876
	preg f: Feb-Apr; central India	5705
	preg f: end Nov-mid Apr; Maharashtra, India	8890
	preg f: end Nov-Apr; India, Sri Lanka	8876
	preg f: Nov-May; India	3915
	birth: Mar; n1; Kuala Lumpur, Malaysia	3974
	birth: end Mar; India	1340

	birth: Apr; n1; India	8887
	birth: mid Apr; India	3928
	birth: end Apr; India, Sri Lanka	8876
	lact f: May-June; Maharashtra, India	8890
Megaderma spasma	mate: est Nov; few data; India, Sri Lanka	8876
	mate: Dec-Jan; India	1343
	preg f: Feb-Apr, Aug; no data other mo; Selangor, Malaysia	7161
	birth: Feb; n16/21; Kuala Lumpur, Malaysia	3974
	birth: Mar-May; India	1340, 1343
	birth: est end Apr; few data; India, Sri Lanka	8876
	birth: May; Sri Lanka	8493
	lact f: May; few data; Sri Lanka	8501

Order Chiroptera, Family Rhinolophidae

Horseshoe, leaf-nosed, and trident bats (*Anthops* ~50 mm; *Asellia* 6-10 g; *Aselliscus* 3-8 g; *Cloeotis* 3-8 g; *Coelops* 6-9 g; *Hipposideros* 5-180 g; *Paracoelops* 7 g; *Rhinolophus* 2-32 g; *Rhinonycteris* 8-10 g; *Triaenops* 8-25 g) are old-world insectivores. Females are often larger than males (4968, 8494, 9297, 9945, 11270; cf 191).

Ovulation may be induced (1640). Asymmetry of female reproductive function is common and either side may predominate (202, 826, 828, 832, 906, 908, 3573, 3918, 3919, 3925, 3934, 3936, 3941, 3942, 5527, 5528, 5529, 5538, 6627, 6718, 6956, 7161, 7208, 7681, 8057, 8875, 8882, 9241, 9489, 10068; cf 6966). Vaginal plugs (3573, 6956, 8055, 8830, 8918, 9238, 9239, 9241, 9437, 11318), sperm storage (826, 828, 830, 831, 3573, 3574, 6956, 7208, 7519, 8918, 9437, 11018, 11019), delayed development (832, 3942, 8882, 9489), delayed implantation (3942, 7208, 8882), monoestry (832, 1348, 8875), and polyovular follicles (11318) are reported.

The endotheliochorial or vasochorial, labyrinthine, bidiscoidal placenta has a mesometrial or circumferential, superficial implantation into a bicornuate or bipartite uterus (202, 828, 832, 905, 906, 907, 908, 1228, 1524, 3574, 3913, 3917, 3918, 3919, 3925, 3934, 3936, 3942, 5527, 5528, 5529, 5538, 6718, 6966, 7563, 7681, 9178, 10068, 11318, 11799) and is often eaten after birth (8873, 8877, 11270). A bicornuate vagina is reported (5529, 5538).

Neonates, up to 25% of adult mass (7681), are blind and either naked or covered with downy hair (897, 1343, 1349, 3571, 6718, 7681, 8798, 8873, 8875, 8877, 9175, 10285, 11270, 11623, 12021). Eyes open at 3-15 d (*Rhinolophus* 3571, 12021). Most females have 2 non-functional pubic teats as well as 2 pectoral mammary glands (331, 3571, 4669, 6720, 6956, 7161, 9297, 10158). Seminal vesicles, ampullary or urethral glands and a baculum are present in males (828, 3574, 5704, 6065, 6957, 6966, 7440, 9178).

Details of female anatomy (832, 3937, 6718, 6956, 8878, 9178, 11093, 11318), male anatomy (831a, 2830, 6957, 8054, 11293), endocrinology (830, 3936, 4661), spermatogenesis (8056), spermiophagy/abnormal sperm (827, 829, 11019), and conception (8057) are available.

NEONATAL MASS

Hipposideros ater	1-1.5 g	3915, 3934
Hipposideros bicolor	1.2 g	3915
Hipposideros caffer	2.8 g; ±0.9; n8 term emb	7681
Hipposideros commersoni	28 g; n1	1349
Hipposideros fulvus	2.2 g	3915, 6718
Hipposideros lankadiva	8 g; n1; smallest neonate	9489
	8-11 g; term emb	897
Hipposideros larvatus	r3.5-4.0 g; n3	10872
Hipposideros ruber	3.0 g; term emb	11273
Hipposideros speoris	2.1 g; n1	8873
Rhinolophus cornutus	2.4 g	6115
	2.55 g; r2.4-2.7; n2	12021
Rhinolophus euryale	3.9 g; r3-4.5; n11	2402
Rhinolophus ferrumequinum	5.8 g	6115
Rhinolophus rouxi	1.8-2.1 g	8877
	4.2 g; r3.7-4.7; n4	10285

NEONATAL SIZE

Asellia tridens	27.7-29.8 mm	110
Hipposideros caffer	FA: 26 mm; ±5.0; n17	7681

WEANING MASS

Rhinolophus cornutus	7.1 g; r6.5-7.9; 55 d	12021
Rhinolophus rouxi	11.0 g; 3 wk	10285

LITTER SIZE

Aselliscus tricuspidatus	1 yg	10114
	1 emb ea; n12 f	7073
Cloeotis percivali	1 emb	10158
Coelops frithi	1 emb ea; n2 f	7983
Hipposideros abae	1 or 2; large number of young suggests 2	252
	1-2	11270
Hipposideros ater	1 yg	3915, 3927, 3931, 3934
	1 emb or yg; n2 lt	5340
	1 emb *ea*; n208 f	3927
Hipposideros bicolor	1 yg; n many	1340, 1343
	1 emb; n1 f	10872
	1 emb ea; n24 f	3915, 3936
Hipposideros caffer	1	11270
	1 emb	10158
	1 yg	1348, 1351
	1 emb *ea*; n3 f	8611
	1 emb ea; n>60 f	7681
Hipposideros cineraceus	1 emb; n1 f	10872
	1 emb; n1 f	5705
Hipposideros commersoni	1 emb; n1 f	331
	1 yg	1351
Hipposideros cyclops	1	11270
Hipposideros diadema	1 yg	7161
	1 yg ea; n6 lt	7075
Hipposideros fulvus	1 emb ea; n3 f	8494, 8501
	1 emb ea; n12 f	5527
	1 emb *ea*; n123 f	3927
	1.03 emb; r1-2; n64 f	3915, 3927, 3931, 5528, 6718
	1.25 emb; r1-2; n4 f	10872
Hipposideros galeritus	1 yg	8494, 8501
	1 emb ea; n3 f	7073
	1 emb ea; n4 f	6447
	1 yg ea; n5 lt	7075
Hipposideros lankadiva	1	3927
	1 emb *ea*; n66 f	3927
	1 emb *ea*; n169 f	9489
Hipposideros larvatus	1 emb ea; n3 f	10872
Hipposideros ridleyi	1 emb; n1 f	3974
Hipposideros ruber	1 emb, yg; n4 lt	11273
Hipposideros speoris	1	3918, 3927, 3931
	1; n "several"	8494, 8501
	1 yg *ea*; n200 lt	1340, 1343
	1 emb ea; n280 f	3927
Rhinolophus affinis	1 emb	7161
Rhinolophus alcyone	1 emb; n1 f	11273
Rhinolophus capensis	1	10158
Rhinolophus clivosus	1 emb, yg	10158
	1 yg	9945
	1 emb	8965
	1; seldom 2	252
Rhinolophus cornutus	1	6115
Rhinolophus darlingi	1	8965
	2 emb; n1 f	10158
Rhinolophus euryale	1	436
Rhinolophus ferrumequinum	1	6115
	1	11318
	1; seldom 2	3571
	1 ovum	11019
	1 emb	8049

Rhinolophus fumigatus	1	11270
	1 emb	10158
	est 1 emb ea; n2 f	8611
Rhinolophus hildebrandti	mode 1; 2 emb; n1 f	252
Rhinolophus hipposideros	mode 1; r1-2; 35% twins based on population dynamics	10068
	1-3; 3 eggs exceptional	436
	1; r1-2	3571
	1 emb ea; n17 f	1524
	1.02 emb; r1-2; n46 f	3573, 3574
Rhinolophus landeri	1	252, 255
	1 emb; n1 f	10158
Rhinolophus lepidus	1	1340
	mode 1 emb	7161
	2 emb; n1 f	5705
Rhinolophus luctus	1	1340
	1 emb; n1 f	8494, 8501
Rhinolophus macrotis	mode 1 emb	7161
Rhinolophus maclaudi	1 emb; n1 f	73
Rhinolophus megaphyllus	1 yg; n many	7074, 7075
Rhinolophus rouxi	1	8882
	1 yg	8494, 8501
	1 yg; n150 lt	1340, 1343
	1 emb; 0.4 % twins; r1-2; n285 f	3927
	1 yg; 2 rare	3927, 3941, 3942
Rhinolophus simulator	1 emb; n1 f	331
Rhinolophus stheno	mode 1 emb	7161
Rhinolophus swinnyi	1 emb	252
SEXUAL MATURITY		
Hipposideros bicolor	f/m: 18-20 mo	1341
Hipposideros caffer	f/m: 2nd yr	5759
	f/m: 2 yr	1349
Hipposideros commersoni	f/m: 2 yr	1349
	f/m: ~2 yr	5759
Hipposideros lankadiva	f/m: 16-17 mo min	9489
Hipposideros speoris	f: 7.5-8 mo	3918
	f/m: 18-20 mo	1341
Rhinolophus capensis	f/m: > 1 yr	828
Rhinolophus clivosus	f: 1st preg: 2 yr 20%; 3 yr 67%	11818
	f: 1st estrus: 6 mo	826
	m: 1st mate: 2nd yr	826
Rhinolophus euryale	f: 1st birth: 2-3 yr	2586
	m: spermatogenesis: 2-3 yr	2586
Rhinolophus ferrumequinum	f: ovary development: 18 mo	9239
	f: 1st mate: 15 mo	6956
	f: 1st mate: 2 yr	9241
	f: 1st birth: 3-4 yr	2586
	f: 1st birth: 5 yr; r3-7	8835
	m: testes development: 18 mo	9239
	m: spermatogenesis: 3-4 yr	2586
	m: urethral gland size: 3 yr	9241
Rhinolophus hipposideros	f: 1st estrus: some 4 mo; most end 2nd yr	3573
	f: large follicle, sperm in vagina: 1.5 yr	10068
	f: 1st birth: 1-2 yr	2586
	f: 1st birth: 15 mo	6956
	f: 10-15 mo	3574
	f: 2 yr	9241
	m: sperm: 12-14 mo	3574, 3580
	m: spermatogenesis: 1-2 yr	2586
	f/m: 1st mate: 1-1.5 yr	3571
	f/m: 1st mate: 2nd yr	1524
Rhinolophus rouxi	f: min 19 mo	3942
	m: min 16 mo	3942
	f/m: 1 yr	1343

GESTATION

Asellia tridens	9-10 wk	110
Cloeotis percivali	3 mo	255
Hipposideros abae	3 mo	11270
Hipposideros ater	155-160 d	3915
	190-200 d	3934
Hipposideros bicolor	40-45 d	3915
Hipposideros caffer	preimplant: ~21 d; r14-28	832
	postimplant: 199 d	832
	3 mo; field data	7681
	80-100 d; field data	1349
	220 d	832
Hipposideros commersoni	5 mo with 2 mo delay	252, 254, 255
	5 mo; field data	1349
Hipposideros cyclops	3.5 mo	11270
Hipposideros fulvus	150-160 d	3915, 6718
	150-160 d	5527
Hipposideros lankadiva	~260 d with delay	9489
Hipposideros ruber	3 mo	252, 255
Hipposideros speoris	~90 d	1340
	135-140 d	3918
Rhinolophus blasii	3 mo	255
Rhinolophus capensis	3-4 mo; field data	828
Rhinolophus clivosus	3 mo	255
	3.5-4 mo	826
Rhinolophus euryale	est 3 mo; field data	2402
Rhinolophus ferrumequinum	10-11 wk	9238
	est ~3 mo	9437
	3-4 mo; est ovulation to birth	8918
Rhinolophus fumigatus	3 mo	255
	3 mo; field data	11270
Rhinolophus hildebrandti	~3 mo	252, 255
Rhinolophus hipposideros	~7 wk	9175
	~75 d	3571
	~2.5 mo	3573, 3574
Rhinolophus landeri	delay: 2 mo	252, 255
	postdelay gestation: 3 mo	252, 255
	5 mo	252, 255
Rhinolophus rouxi	implant: 40-45 d	8882
	3 mo; visible emb to birth	10285
	150 d; ±8; with delay; field data	3942, 8875, 8882
Rhinolophus simulator	3 mo	255
Rhinolophus swinnyi	~3 mo	252, 255

LACTATION

Hipposideros ater	wean: 2.5-3 mo	3934
Hipposideros caffer	nurse: 2+ mo	832
	wean: 3+ mo	1349
Hipposideros commersoni	wean: 5+ mo	1349
Hipposideros cyclops	solid food: ~2 mo	252, 11270
Hipposideros fulvus	wean: 22 d	5527
	wean: ~2 mo	6718
Hipposideros lankadiva	wean: ~3 mo	9489
Hipposideros speoris	wean: 50-55 d	3918
Rhinolophus capensis	wean: 2 mo	828
Rhinolophus clivosus	nurse: 2 mo	826
Rhinolophus cornutus	independent: ~55 d	12021
Rhinolophus euryale	wean: est 3.5 mo	9241
Rhinolophus ferrumequinum	wean: est ~2 mo	9437
Rhinolophus fumigatus	wean: 2 mo	11270
Rhinolophus hipposideros	wean: 4-6 wk	3573, 3574
Rhinolophus landeri	wean: 2 mo	255
Rhinolophus megaphyllus	wean: 2 mo	2830
Rhinolophus rouxi	nurse: 1-3 mo	1640
	nurse: ~2 mo; field data	3942

LITTERS/YEAR

Hipposideros abae	1	11270
Hipposideros caffer	1	1348, 1351
	1	11270
Hipposideros commersoni	1	1351
Hipposideros cyclops	1	11270
Hipposideros galeritus	1	8494
Hipposideros speoris	1	1343
Rhinolophus euryale	1	9241
Rhinolophus ferrumequinum	1	3571
	1	9241
Rhinolophus fumigatus	1	11270
Rhinolophus hildebrandti	1	252
Rhinolophus hipposideros	1	9241
	1	3571
Rhinolophus rouxi	1	1343

SEASONALITY

Asellia tridens	mate: fall; Iraq	110
	birth: May-June; few data; Burkina Faso, w Africa	5903
	birth: June-July; Iraq	110
	young found: July, Nov; Burkina Faso, w Africa	5903
Aselliscus tricuspidatus	preg f: July; n12; New Guinea	7073
	preg f: Oct; n1; New Hebrides	518
	f with yg; June, July, or Aug; New Guinea	10114
Cloeotis percivali	birth: Oct; se Zaire, Zambia	252, 255
	birth: est Oct-Dec; S Africa	10158
Coelops frithi	preg f: Jan; n2; Java, Indonesia	7983
Hipposideros abae	mate: Dec; Cameroun, Zaire, Uganda	252, 11270
	birth: Mar; Cameroun, Zaire, Uganda	252, 11270
Hipposideros armiger	preg f: Feb-May, Sept-Oct; Malaysia	7161
	preg f: Feb, June; n2; Malaysia	6446
	birth: Mar; few data; Kuala Lumpur, Malaysia	3974
Hipposideros ater	preg f: mid Nov-June; India	3915, 3934
	lact f: Jan; QLD, Australia	7076
Hipposideros beatus	birth: mid Mar; boreal; Nov; austral, Gabon	252
	birth: mid Mar; central Africa	11270
	wean: end May; central Africa	11270
Hipposideros bicolor	mate: Mar; India	3937
	mate: Oct; Selangor, Malaysia	7161
	preg f: Mar-May; India	3915
	birth: Apr; few data; Kuala Lumpur, Malaysia	3974
	birth: Apr; Selangor, Malaysia	7161
	birth: end Apr-early May; India	1340, 1343
Hipposideros caffer	sperm: only Nov; Nigeria	7208
	testes: peak Nov; Nigeria	7208
	mate: end Apr-early May; Natal, S Africa	832
	mate: Oct-Jan; central Africa	11270
	mate: Nov; e Africa	5759
	mate: Nov-Dec; boreal, central Africa	252
	mate: July-Aug; austral, central Africa	252
	preg f: Feb, May; n3; Niger	8611
	preg f: Mar; n15; Nigeria	7208
	preg f: Mar-Apr, none Nov; Zaire	11272
	preg f: Aug; n1; Rio Muni, W Africa	5387
	preg f: Aug; Zimbabwe	331
	preg f: Dec-Mar; Uganda	7678, 7681
	birth: Feb-Mar; Uganda	5759
	birth: Mar; Uganda	7681
	birth: Mar, Oct; ne Gabon	1348, 1349, 1351
	birth: Mar-Apr; Nigeria	255
	birth: Apr; Uganda	7678
	birth: Oct; Gabon	255
	birth: Oct/Nov-early Dec; Natal, S Africa	832

	birth: Oct-Dec, Feb-Apr; central Africa	11270
	birth: Oct-Dec; S Africa	10158
	lact f: Nov; Rio Muni, W Africa	5387
Hipposideros camerunensis	preg f: Oct; n12; Cameroun	2955
	lact f: Oct; n2; Cameroun	2955
Hipposideros cineraceus	mate: Oct; Malaysia	7161
	birth: end Apr; Malaysia	7161
	preg f: Mar; n1; central India	5705
Hipposideros commersoni	mate: May; Gabon	254
	mate: May; ne Gabon	1349
	emb development: June-?; central Africa	252
	birth: Oct; ne Gabon	1349, 1351
	birth: Oct; Gabon	254
	birth: Oct; se Zaire, Rwanda	255
Hipposideros coxi	lact f: Jan; Sarawak, Malaysia	10314
Hipposideros cyclops	mate: early Dec; central Africa	11270
	preg f: Feb-Mar; n3; Zaire	11272
	birth: mid Mar; central Africa	11270
	wean: mid May; central Africa	11270
Hipposideros diadema	mate: Oct; Kuala Lumpur, Malaysia	3974
	preg f: Apr, Oct; n3; Malaysia	6446
	preg f: May; n1; Philippines	7586
	birth: Mar; Kuala Lumpur, Malaysia	3974
	lact f: Mar-Apr; Selangor, Malaysia	7161
Hipposideros fuliginosus	birth: Mar; central Africa	252
	young found: Apr; few data; Cameroun	71
Hipposideros fulvus	mate: early Nov; India	6718
	mate: mid Nov; Maharashtra, India	5527
	ovulate: mid Nov; Maharashtra, India	5527
	conceive: mid Nov; India	5528
	preg f: Mar; n3; Sri Lanka	8501
	preg f: Mar-Apr; n7; India	5705
	preg f: mid Nov-Apr; India	6718
	preg f: Nov-May; India	3915
	birth: end Apr-early May; India	5528
	birth: May; n1; central India	5705
Hipposideros galeritus	preg f: Apr; n1; Sri Lanka	8494, 8501
	preg f: June; n4; Malaysia	6447
	preg f: July, Sept; n3; Papua, New Guinea	7073
	birth: Feb; few data; Kuala Lumpur, Malaysia	3974
Hipposideros lankadiva	mate: Aug; Chandrapur, Maharashtra, India	908
	conceive: Aug; Chandrapur, Maharashtra, India	908
	preg f: Mar-May, Feb-Apr; 2 areas; Chandrapur, Maharashtra, India	897
	implant: end Sept; Chandrapur, Maharashtra, India	908
	birth: mid-end May; Chandrapur, India	9489
	birth: early June; Chandrapur, Maharashtra, India	908
Hipposideros ridleyi	birth: end Apr; Kuala Lumpur, Malaysia	3974
Hipposideros ruber	mate: Dec; Uganda; early July; central Africa	252
	birth: Mar; Liberia	11881
	birth: Mar; Zaire	252
	birth: Oct; se Zaire	252, 255
Hipposideros speoris	conceive: end Dec-mid Mar; Chandrapur, Maharashtra, India	8878
	preg f: Jan-July; Chandrapur, Maharashtra, India	3918
	birth: Jan; n1; s India	11507b
	birth: May; India	1340, 1343
	birth: early Aug; Sri Lanka	8494, 8501
Rhinolophus affinis	preg f: Feb-Mar; Pahang, Malaysia	7161
	preg f: peak Apr-May, Oct; Selangor, Malaysia	7161
Rhinolophus alcyone	mate: Aug; central Africa	252
	preg f: June; n3; w Uganda	5759
	birth: Sept; n1; Uganda	5759
	birth: end Nov-early Dec; central Africa	252
	lact f: June; n1; w Uganda	5759

Rhinolophus blasii	birth: June; n Africa	72
	birth: Dec; se Zaire	255
	lact f: May; n1; Iran	2404a
Rhinolophus capensis	spermatogenesis: Oct-May; Cape Province, S Africa	828
	spermatogenesis: Jan-Apr; inactive: June-Sept; Cape Province, S Africa	830
	epididymal sperm: Apr-May; Cape Province, S Africa	828
	mate: Aug-Sept; Cape Province, S Africa	828, 830
	ovulate: Aug-Sept; Cape Province, S Africa	828
	birth: Nov-Dec; Cape Province, S Africa	828
	birth: Dec; S Africa	831
	birth: mid Dec-?; Cape Province, S Africa	10158
	lact f: Dec-Jan; Cape Province, S Africa	828
Rhinolophus clivosus	mate: Apr-May, Sept-Oct; Natal, S Africa	826
	mate: July-Aug; se Zaire, Rwanda	255
	preg f: Nov; Transvaal, S Africa	8965
	birth: Oct-Nov; se Zaire, Rwanda	252, 255
	birth: end Nov-early Dec; Natal, S Africa	11818
	birth: Dec-Feb; Natal, S Africa	826, 831
Rhinolophus cornutus	birth: early July; Japan	12021
Rhinolophus darlingi	preg f: Oct; n1; Transvaal, S Africa	8965
	birth: est early summer; S Africa	10158
Rhinolophus euryale	mate: Oct?; France	9241
	birth: end June-July; France	9241
	wean: mid Sept; France	9241
Rhinolophus ferrumequinum	sperm in uterus: Nov-Mar; Gloustershire, England	6956
	epididymal sperm: yr rd; France	9437
	spermatogenesis: May-Oct; France	9437
	spermatogenesis: May-Oct; Bologna, Italy	11293
	m: sex organs: developed Sept-May, regressed May-Aug; France	9241
	estrus: ?-Oct; Gloustershire, England	6956
	mate: Aug, early winter; Gloustershire, England	8918
	mate: mid-end Oct; Japan	7519, 8057
	mate: Oct; France	9241
	mate: Oct-Nov; France	9437
	copulatory plug: ?-Mar; France	9437
	ovulate: early Apr; Japan	7519, 8057
	ovulate: early-mid Apr; Gloustershire, England	6956
	ovulate: Apr; France	9437
	ovulate: ~Apr; Gloustershire, England	8918
	implant: before Mar 15; Sardinia, Italy	3517
	preg f: Mar-Apr; Sardinia, Italy	3517
	birth: May-July; Germany	3571
	birth: June; Algeria	5988
	birth: June; France	9238, 9241
	birth: June-July; Gloustershire, England	6956
	birth: end June-early July; France	9437
	birth: July-Aug; Gloustershire, England	8918, 8919
Rhinolophus fumigatus	mate: Dec: Zaire, Ethiopia	255
	preg f: Feb; n2; Niger	8611
	birth: Mar-Apr; Ethiopia, Zaire	252, 255
	birth: Oct-Nov; Kenya	252, 255
	birth: est Oct-Dec; S Africa	10158
Rhinolophus hildebrandti	mate: Aug-Sept; central Africa	252
	preg f: est Oct-early Mar; Uganda	7680
	birth: Dec-early Jan; central Africa	252
	birth: Dec-Jan; se Zaire, Rwanda	255
	birth: est summer Oct-Dec; S Africa	10158
Rhinolophus hipposideros	sperm in uterus: Nov-Mar; Gloustershire, England	6956
	spermatogenesis: May-Sept; Czechoslovakia	3574
	spermatogenesis: July-Oct; Europe	2075
	testes: developed Sept; France	9241
	estrus: Oct; Gloustershire, England	6956

	mate: Jan; Netherlands	10068
	mate: Oct-Apr; Moravia, Czechoslovakia	3573, 3580
	mate: Oct; France	9241
	ovulate: Apr; Moravia, Czechoslovakia	3573, 3574
	ovulate: Apr; Gloustershire, England	6956
	birth: (May) June-July; Europe	3571
	birth: June-July; Gloustershire, England	6956
	birth: June-July; France	9241
	birth: June-Aug; Netherlands	10068
	birth: July; Czechoslovakia	3574
Rhinolophus landeri	sperm: July-Dec; Nigeria	7208
	mate: June; central Africa	252
	mate: Nov; Nigeria	255
	preg f: Mar; Nigeria	7208
	preg f: Oct; n1; Zimbabwe	10158
	unimplanted trophoblasts: Nov; Nigeria	7208
	implant: Feb; Nigeria	255
	implant: June; Tanzania	255
	birth: end Apr; Nigeria	255
	birth: Sept-Oct; Tanzania	252, 255
	birth: Oct-Jan; Cameroun	252
	lact f: May; Nigeria	7208
Rhinolophus lepidus	preg f: Feb-Apr; Pahang, Malaysia	7161
	birth: mid Apr-mid May; India	1340
Rhinolophus luctus	preg f: Jan; n1; Sri Lanka	8494, 8501
	birth: Apr; India	1340
Rhinolophus maclaudi	preg f: Aug; n1; central Africa	73
	birth: Sept-Oct; se Zaire, Rwanda	255
Rhinolophus macrotis	preg f: Feb-Mar; Malaysia	7161
Rhinolophus megaphyllus	testes: peak Apr-May; ne NSW, Australia	2830
	birth: mid Nov-Dec; ne NSW, Australia	2830
	wean: Jan; ne NSW, Australia	2830
Rhinolophus mehelyi	mate: Mar; Sardinia, Italy	3517
	preg f: end Mar-ear June; Algeria	5988
Rhinolophus rouxi	testes: peak Sept-Nov, regress Dec; Bangalore, India	8875
	mate: Nov-early Dec; Bangalore, India	8875
	mate: Nov-Dec; Bangalore, India	8882
	mate: Dec; w Ghats, India	8882
	preg f: Jan-Apr; India	10285
	preg f: Jan-May; India	3942
	birth: Apr; India	1340, 1343
	birth: Apr-May; Sri Lanka	8494, 8501
	birth: end Apr-early May; Bangalore, India	8875, 8882
	lact f: Apr-June; India	10285
Rhinolophus silvestris	birth: Oct; few data; central Africa	252
Rhinolophus simulator	mate: Aug; central Africa	252
	birth: est Oct-Nov; S Africa	10158
	birth: Nov; se Zaire	252, 255
	birth: est Nov; Cameroun	2955
Rhinolophus stheno	preg f: Feb-Apr; Pahang, Malaysia	7161
Rhinolophus swinnyi	mate: July; se Zaire	252
	birth: Oct; se Zaire	252, 255
Rhinolophus trifoliatus	preg f: May-June, Sept-Nov; Selangor, Malaysia	7161
	preg f: Mar; Perlis, Malaysia	7161
	preg f: Apr, Oct; n2; Malaysia	6446
Triaenops persicus	sperm: high June-Oct, low Dec-Jan; Tanzania	6735
	spermatogenesis: end Dec; central Africa	252
	preg f: Dec; Mhulumuzi, e Africa	5759
	birth: Jan; Tanzania	252

Order Chiroptera, Family Noctilionidae, Genus *Noctilio*

Bulldog bats (*Noctilio albiventris, N. leporinus*) are 30-80 g, piscivores and insectivores from Central and South America. Males are larger than females and may be differently colored (178a; cf 2808a). Ovulation may alternate between left and right ovaries and conception is reported to occur in the ipsilateral oviduct (8944, 8945). Blastocysts begin differentiation in the oviduct before the superficial, lateral to antimesometrial implantation of an endotheliochorial, discoid placenta near the cranial end of the endometrial ridge of the bipartite or bicornuate, symmetrical uterus (275, 4957, 6627, 9178, 11799). Neonates are naked, but have open eyes (1371). The clitoris is large (9178). Seminal vesicles are present (9178), but a baculum is not. Pockets of skin associated with the scrotum may be glandular (2808a, 3900a). Sperm have very large heads (3414). Details of anatomy (4957) and reproductive failure (8949) are available.

LITTER SIZE		
Noctilio albiventris	1 cl; 1 emb; no twins	275
	1 emb; n1 f	1006
	1 emb; n1 f	11471a
	1 emb ea; n2 f	10974
	1 emb ea; n9 f	3999a
	1 emb ea; n13 f	4966
	1 emb ea; n17 f	2345
	1 emb ea; n26 f	5425
	1.01 emb; mode 1; r1-2; n72 f	8944, 8949
Noctilio leporinus	1 emb; n1 f	5425
	1 emb; n1 f	10974
	1 emb ea; n2 f	3900
	1 emb ea; n2 f	2345
	1 emb ea; n3 f	4671
	1 emb ea; n3 f	4966
	1 emb ea; n3 f	3999a
	1 emb ea; n7 f	1641
	1 emb ea; n30 f	178a
LACTATION		
Noctilio albiventris	solid food: 44.7 d; sd4.3; r36-49; n9	1371
	nurse: 80.5 d; sd5.2; r75-90; n8	1371
SEASONALITY		
Noctilio albiventris	testes: small Apr; n1; Guatemala	2552
	mate: Mar; Cauca Valley, Colombia	8944
	mate: end Nov-Dec; Central America	275
	mate: end Nov-early Jan; Panama, Costa Rica	3322
	ovulate: Mar-Apr; Cauca Valley, Colombia	8944
	conceive: Mar; Cauca Valley, Colombia	8945
	preg f: Feb-mid Apr; Cauca Valley, Colombia	8944, 8949
	preg f: Apr; n1; Jalisco, Mexico	11471a
	preg f: Apr; Nicaragua	5425
	preg f: Apr; n1; Costa Rica	2345
	preg f: June-Aug; n9; Peru	3999a, 10974
	preg f: Dec; central America	275
	birth: Apr; central America	275
	birth: end Apr-May; Panama, Costa Rica	3322
	lact f: Jan; n1; Jalisco, Mexico	11471a
	lact f: Feb-Mar; neotropics	4966
	lact f: Mar; n1; Costa Rica	2345
	lact f: end May; Central America	275
Noctilio leporinus	preg f: Feb; Trinidad	3900a
	preg f: Feb; n3; Michoacan, Mexico	4671
	preg f: Feb; n4; Panama, Guatemala	1641
	preg f: Feb; n30; Panama	178a
	preg f: Mar; n1; Nicaragua	5425
	preg f: Apr; n1; Guatemala	2552
	preg f: July-Aug; n3; Peru	3999a, 10974
	preg f: Sept-Jan; Brazil	11736
	preg f: Oct; n3; Chiapas, Mexico	1641

	birth: Dec-Feb; Mexico, Cuba	1346
	lact f: Feb-Mar; Trinidad	3900a
	lact f: June; n3; Sinaloa, Mexico	5413a
	lact f: July; n7; Campeche, Mexico	5424f
	lact f: July; n2; Lesser Antilles	5411a
	lact f: Sept; n2; Guatemala	2552

Order Chiroptera, Family Mormoopidae

The insectivorous, ghost-faced bats (*Mormoops* 13-18 g) and moustached bats (*Pteronotus* 7.5-25 g) range from southwest United States to Brazil, Bolivia, and Peru. Ovulation occurs only from the right ovary and implantation is predominantly in the cranial portion of the right uterine horn (*Pteronotus* 3613, 4957, 11804). The endotheliochorial placenta implants in a bicornuate uterus (4957, 11799). Males have inguinal testes (*Pteronotus* 3613). Embryos can reach 30% of maternal mass. Details of anatomy (4957) are available.

NEONATAL MASS		
Pteronotus fuliginosus	1.8 g; large emb	9213
NEONATAL SIZE		
Pteronotus davyi	25 mm; term emb	62
LITTER SIZE		
Mormoops megalophylla	1 emb	577
	1 emb; n1 f	690a
	1 emb; n1 f	1475
	1 emb; n1 f	4671
	1 emb ea; n3 f	1937
	1 emb ea; n5 f	522
	1 emb ea; n15 f	950
Pteronotus davyi	1 emb; n1 f	1937
	1 emb; n1 f	5405e
	1 emb ea; n3 f	3205
	1 emb ea; n4 f	5425
	1 emb ea; n16 f	11471a
Pteronotus fuliginosus	1 emb; n1 f	5873
Pteronotus gymnonotus	1 emb; n1 f	3205
	1 emb ea; n3 f	3999a
	1 emb ea; n3 f	5425
	1 emb ea; n4 f	1475
Pteronotus parnellii	1; n1 lt	452
	1 emb; n1 f	10112a
	1 emb ea; n2 f	3705c
	1 emb ea; n2 f	5405b
	1 emb ea; n2 f	5425
	1 emb ea; n3 f	950
	1 emb ea; n5 f	3999a
	1 emb ea; n6 f	1937
	1 emb ea; n9 f	1475
	1 emb ea; n11 f	11471a
	1 emb ea; n21 f	5405e
	1 emb ea; n40 f	3205
Pteronotus personatus	1 emb; n1 f	1937
	1 emb; n1 f	5405e
	1 emb; n1 f	5413a
GESTATION		
Pteronotus parnellii	~3.5 mo	3613
LITTERS/YEAR		
Mormoops megalophylla	1	1474
SEASONALITY		
Mormoops megalophylla	preg f: Feb; n2; Campeche, Mexico	5424f
	preg f: Mar; n1; El Salvador	1475
	preg f: Mar, May; n5; Coahuila, Mexico	522
	preg f: May; n1; Michoacan, Mexico	4671
	preg f: June; n1; AZ	690a

	birth: June-July; sw US	1474
	lact f: Mar; Trinidad	3900a
Pteronotus davyi	preg f: Mar; n1; Guatemala	5405e
	preg f: Mar, May; n3; none Jan; El Salvador	3205
	preg f: Apr-May; n10; Yucatan, Mexico	950
	preg f: May; n1; N America	1937
	preg f: May; n4; Nicaragua	5425
	preg f: May-June; n16; Jalisco, Mexico	11471a
	preg f: June; n1; Sinaloa, Mexico	5413a
Pteronotus fuliginosus	preg f: May; n1; Haiti	5873
	preg f: Feb-June, peak May; Cuba	9213
	preg f: peak May; Puerto Rico	9213
Pteronotus gymnonotus	mate: Dec-Jan; few data; El Salvador	3205
	preg f: May; n1; none Aug-Sept, Dec; El Salvador	3205
	preg f: May; n4; El Salvador	1475
	preg f: May; n3; Nicaragua	5425
	preg f: Aug; n3; Peru	3999a
	lact f: Aug; n1; Costa Rica	2670a
Pteronotus parnellii	preg f: Jan, none Sept, Dec; El Salvador	3205
	preg f: Feb; n45; s Compeche, Mexico	5424f
	preg f: Feb-Mar; n2; Nicaragua	5425
	preg f: end Feb-early Mar; n1-3; Jamaica	7103a
	preg f: end Feb-May; Chinameca, Mexico	3613
	preg f: Mar; n1; El Salvador	1475
	preg f: Mar; n2; Michoacan, Mexico	4671
	preg f: Mar; n8; Panama	1475
	preg f: Mar; n21; Guatemala	5405e
	preg f: Apr; n1; Sonora, Mexico	1939
	preg f: Apr-May; n3; Yucatan, Mexico	950
	preg f: Apr-May; Costa Rica	6261
	preg f: May; n6; N America	1937
	preg f: May-June; n2; Sinaloa, Mexico	5413a
	preg f: May-June; n11; Jalisco, Mexico	11471a
	preg f: June; n1; Sinaloa, Mexico	523a
	preg f: June; n2; Durango, Mexico	5405b
	preg f: June, Aug; n5; Peru	3999a
	preg f: July; n1; Margarita I, Venezuela	10112a
	preg f: July; n2; Suriname	3705c
	preg/lact f: peak Mar-June, low July-Jan; Panama	1070
	birth: Mar-Apr; Panama, Costa Rica	3322
	lact f: Apr; Nicaragua	5425
	lact f: June; n1; Sinaloa, Mexico	523a
	lact f: July; n1; Guatemala	9113
Pteronotus personatus	mate: Jan; n1; El Salvador	3205
	preg f: Mar; n1; Guatemala	5405e
	preg f: May; n1; N America	1937
	preg f: June; n1; Sinaloa, Mexico	5413a

Order Chiroptera, Family Phyllostomidae

New-world, leaf-nosed bats (*Ametrida* 8-10 g; *Anoura* 4-20 g; *Ardops* 15-19 g; *Ariteus* 9-13 g; *Artibeus* 10-87 g; *Brachyphylla* 45-67 g; *Carollia* 10-25 g; *Centurio* 17-28 g; *Chiroderma* 13-29 g; *Choeroniscus* 5-9 g; *Choeronycteris* 10-22 g; *Chrotopterus* 73-90 g; *Desmodus* 15-50 g; *Diphylla* 24-34 g; *Ectophylla* 6-13 g; *Erophylla* 65-75 mm; *Glossophaga* 9-16 g; *Hylonycteris* 6-9 g; *Leptonycteris* 17-30 g; *Lichonycteris* 5-8 g; *Lionycteris* ~50 mm; *Lonchophylla* 7-16 g; *Lonchorhina* 11-17 g; *Macrophyllum* 6-10 g; *Macrotus* 12-20 g; *Micronycteris* 4-16 g; *Mimon* 9-23 g; *Monophyllus* 8-17 g; *Musonycteris* 70-79 mm; *Phylloderma* 57-71 g; *Phyllonycteris* 14-21 g; *Phyllops* ~48 mm; *Phyllostomus* 20-100 g; *Platalina* 72 mm; *Pygoderma* to 27 g; *Rhinophylla* 43-48 mm; *Scleronycteris* 57 mm; *Sphaeronycteris* ~53 mm; *Stenoderma* 25 g; *Sturnira* 15-67 g; *Tonatia* 7-36 g; *Trachops* 24-39 g; *Uroderma* 14-23 g; *Vampyressa* 7-10 g; *Vampyrodes* 33 g; *Vampyrops* 12-72 g; *Vampyrum* 120-190 g) have diverse diets and range throughout the tropics and subtropics of North, Central, and South America. Males of *Anoura* and *Phyllostomus* may be larger than females (1070, 7745), while female

Artibeus, *Choeroniscus*, *Desmodus*, *Phyllops*, *Pygoderma*, *Stenoderma*, and *Uroderma* are recorded as larger than males (3608, 3900a, 5423a, 5873, 7696, 10657). Sexual dichromatism may occur (6639; cf 279a).

Artibeus, *Carollia*, *Choeroniscus*, and *Pygoderma* males have an extra chromosome (3492, 7696). Ovulation and implantation in *Macrotus* occur in the right ovary and uterine horn, respectively (999, 1001, 1199, 1647, 11804). Unfertilized ova degenerate within the oviduct (*Glossophaga* 8946). The hemochorial or endotheliochorial, labyrinthine, discoid placenta has an interstitial implantation, usually into the intramural cornua of the simplex or bicornuate uterus (967, 1000, 1038, 1039, 3318, 3917, 4289, 4290, 4293, 4957, 6627, 7563, 8816, 8941, 8942, 8943, 8946, 9178, 10661, 11798, 11799, 11813, 11859) and occasionally is eaten after birth (5433, 9615, 11859).

Delayed development occurs in *Artibeus* (3318) and *Macrotus* (1198, 1199, 1460, 1461, 1462). *Macrotus*, *Leptonycteris*, and *Choeronycteris* are monoestrus (11754), *Glossophaga*, *Carollia*, *Sturnira*, *Uroderma*, *Vampyrops*, *Vampyrodes*, *Vampyresssa*, *Artibeus*, and *Desmodus* are continuously or more often bimodally polyestrous (2109, 3318, 3322, 9113, 10654, 10658, 10785, 11754, 11813; cf 8816). Concurrent gestation and lactation occur (*Artibeus* 4671). Neonates are 14-50% of adult mass (5857), have fur, and have open eyes (577, 2408, 3973, 4523a, 5432, 5433, 7353, 9615, 10661, 10663, 10664). Twinning is rare (595). *Diphylla* females have 2 mammae (4494). Males have a multipartite sex-gland complex whose components can be resolved histologically (6065, 6072, 9178).

Details of female anatomy (2374, 4957, 8941, 8942), endocrinology (6072), sperm (3413), mating (4048a), birth (902, 903, 7353, 10664), milk composition (5087, 5296, 10524), and female reproductive cycles (11754) are available.

NEONATAL MASS

Anoura cultrata	3.35 g; r3.0-3.7; n2 aborted term emb	6372a
Anoura geoffroyi	5.0 g; r4.8-5.2; n2 largest emb	3900a
	5.2 g; r4.5-5.7; n3	3900a
Artibeus jamaicensis	5.9 g; n1; with placenta	903
Artibeus lituratus	7.8 g; n1 smallest young	10658
	8.9 g; n1	10658
	10.7 g; n1	10663
	12.2 g; n1; est r7.8-15.5 g; term emb, 1st observed yg	10661
Brachyphylla nana	9.9 g; largest emb	10594
	15.1 g; smallest yg	10594
Carollia perspicillata	5.0 g; r4.1-5.9; n13	5857
Choeronycteris mexicana	4.4 g; r4.3-4.5; n2 emb	7612
Diphylla ecaudata	4.4 g; with some membranes	902
Desmodus rotundus	5-7 g	4049
	6.9 g; r6.7-7.1; n2 smallest yg	3900a
	11.5 g; n1	11813
Leptonycteris sanborni	7.1 g; r6.2-7.9; n4	4523a
	8.3 g; n1 largest emb	4523a
Micronycteris minuta	1.8 g; n1 smallest yg	3900a
Phyllostomus discolor	7 g; n1; may not be neonate	3900a
Stenoderma rufum	7 g	10664
Sturnira aratathomasi	13.6 g; n1 term emb	10785a
Sturnira tildae	2.2 g; n1 large emb	3900a
Vampyressa pusilla	3.2 g; n1 large emb	8412
Vampyrops lineatus	8.3 g; n1	11737

NEONATAL SIZE

Anoura cultrata	CR: 20.6 mm; r20.0-21.3; n2 aborted term emb	6372a
Artibeus jamaicensis	HB: 31 mm; n1; FA: 15 mm; n1	903
Artibeus lituratus	HB: 58 mm; n1	10661
Choeronycteris mexicana	CR: 29 mm; n2 emb	7612
Stenoderma rufum	HB: 45 mm; n1; FA: 29.4 mm; n1	10664
Sturnira aratathomasi	FA: 23 mm; n1 term emb	10785a
Vampyrops lineatus	HB: 46.5 mm; n1	11737

WEANING MASS

Carollia perspicillata	~10 g; 4 wk from graph	5857
	~15 g; 8 wk from graph	5857
Leptonycteris sanborni	~15 g; 1st flight	4523a
Uroderma bilobatum	8.15 g; r7.7-8.6; n2 nursing juveniles	2552

LITTER SIZE

Ametrida centurio	1 emb ea; n17 f	1635
Anoura caudifer	1	10618
	1; n1	10663
	1; n1	452
	1 emb ea; n8 f	3999a
Anoura cultrata	1 emb; n1 f	3608
	1 emb ea; n3 f	3999a
	1 emb ea; n4 f	6372a
	2 yg; n1 lt	6372a, 7745
Anoura geoffroyi	1 emb ea; n2 f	10974
	1 emb ea; n22 f	3999a
Ardops nichollsi	1 emb ea; n5 f	5424d
Artibeus anderseni	1 emb ea; n4 f	3999a
Artibeus aztecus	1 emb; n1 f	523a
	1 emb ea; n>2 f	2342a
	1 emb ea; n17 f	5405d
Artibeus cinereus	1	1346
	1	10785
	1 emb; n1 f	5405e
	1 emb; n1 f	5425
	1 emb ea; n7 f	1475
	1 emb ea; n7 f	1635
	1 emb ea; n21 f	3999a
Artibeus fraterculus	1 emb ea; n46 f	3999a
Artibeus fuliginosus	1 emb ea; n23 f	3999a
Artibeus hartii	1	10785
	1 emb; n1 f	6258
	1 emb ea; n3 f	3608
	1 emb ea; n6 f	3999a
Artibeus hirsutus	1 emb; n1 f	280a
	1 emb; n1 f	3701
	1 emb ea; n4 f	1937
	1 emb ea; n8 f	11471a
	1 emb ea; n15 f	11938
Artibeus jamaicensis	1 emb	6307
	1; n1 lt	11859
	1 emb; n1 f	1162
	1 emb; n1 f	3701
	1 emb; n1 f	4671
	1 emb; n1 f	9113
	1 yg; n1 f	5411a
	1 emb ea; n3 f	1937
	1 emb ea; n5 lt	5405e
	1 yg; n5 lt	7978
	1 emb ea; n6 f	2390
	1 emb ea; n7 f	950
	1 emb ea; n8 f	1635
	1 emb ea; n9 f	11471a
	1 ea; n10	452
	1 emb ea; n13 f	5873
	1 emb ea; n16 f	1475
	1.02 emb; n195 f	595
Artibeus lituratus	1	10618
	1 emb; n1 f	1937
	1 emb; n1 f	4671
	1 emb ea; n2 f	1635
	1 emb ea; n2 f	2390
	1 yg ea; n2 f	8953
	1 emb ea; n3 f	3701
	1 emb ea; n5 f	5405e
	1 emb ea; n8 f	11471a
	1 emb ea; n10 f	217
	1 emb ea; n36 f	3999a

	1 emb ea; n40 f	10657, 10658, 10661, 10660, 10663
	1.4 cl or mature follicles; n49 f	10661
Artibeus phaeotis	1; n1 lt	452
	1 emb; n1 f	5405e
	1 emb; n1 f	7624
	1 emb ea; n3 f	9113
	1 emb ea; n18 f	11471a
Artibeus planirostris	1	10618
	1 emb ea; n4 f	10112a
	1 emb ea; n49 f	3999a
Artibeus toltecus	1 emb; n1 f	3701
	1 emb; n1 f	2390
	1 emb ea; n4 f	2345
	1 emb ea; n≥8 f	2342a
	1 emb ea; n33 f	11471a
Brachyphylla nana	1 emb; n1 f	10594
	1 emb ea; n12 f	1429c
Carollia brevicauda	1 emb; n1 f	11526
	1 emb ea; n28 f	3999a
Carollia castanea	1	10785
	1 emb ea; n2 f	5425
	1 emb ea; n3 f	11526
	1 emb ea; n24 f	3999a
Carollia perspicillata	1	3024
	1	6123
	1	10618
	1 emb; n1 f	1475
	1 yg ea; n2 lt	10660
	1 emb ea; n3 f	10974
	1 emb ea; n3 f	5405e
	1 ea; n13 lt	452
	1.02 emb; r1-2; n57 f	3999a
	1.05 emb; r1-2; n22 f	1635
	1.51 emb; r1-2; n11 f	181
Carollia subrufa	1 emb; n1 f	5405e
	1 emb; n1 f	6639
	1 emb ea; n49 f	3206
Centurio senex	1 yg; n1 f	3900a
	1 emb ea; n2 f	2345
	1 emb ea; n2 f	11471a
	1 emb ea; n3 f	5425
Chiroderma doriae	1	10618
Chiroderma salvini	1 emb; n1 f	1641
	1 emb ea; n2 f	6258
	1 emb ea; n7 f	11471a
Chiroderma trinitatum	1 emb; n1 f	3900a
	1 emb ea; n2 f	1635
	1 emb ea; n4 f	3999a
Chiroderma villosum	1; n1 lt	1635
	1 emb ea; n5 f	2345
	1 emb ea; n7 f	3999a
Choeroniscus godmani	1 emb; n1 f	5405d
	1 emb; n1 f	5425
Choeroniscus minor	1 emb ea; n2 f	3999a
Choeronycteris mexicana	1	1474
	1 yg	577
	1 emb; n1 f	522
	1 emb; n1 f	11471a
Chrotopterus auritus	1 emb; n1 f	4228
	1 yg; n1 f	10618
Desmodus rotundus	1	1346
	1 yg	2229
	1; n1 lt	9614

	1 emb; n1 f	2064
	1 yg; n1 lt	7353
	1 emb ea; n2 f	950
	1 emb ea; n4 f	280a
	1 emb ea; n4 f	5405e
	1 ea; n5 f	452
	1 emb ea; n10 f	3207
	1 emb ea; n16 f	1937
	1 emb ea; n31 lt	3999a
	mode 1 cl; sometimes 2	11813
	twins prob bom premature	1463
Diphylla ecaudata	1	1346
	1 yg	2229
	1 emb; n1 f	3999a
	1 emb ea; n3 f	2228a
Ectophylla alba	1 emb; n1 f	3608
	1 emb ea; n2 f	1652
Ectophylla macconnelli	1 emb; n1 f	10974
	1 emb ea; n6 f	3999a
	1 emb ea; n7 f	1635
Erophylla sezekorni	1 emb; n1 f	11055a
	1 yg; n1 f	983a, 11754
	1.1 emb; r1-2; n10 f	595
Glossophaga commissarisi	1 emb; n1 f	5405e
	1 emb ea; n4 f	11471a
Glossophaga leachii	1 emb; n1 f	8285a
	1 emb ea; n5 f	4228
	1 emb ea; n≥5 f	11471a
	1 emb ea; n16 f	1937
Glossophaga longirostris	1 emb ea; n4 f	3900a
	1 emb ea; n8 f	10112a
Glossophaga mexicana	1 yg	11527
Glossophaga soricina	1	10618
	1	10785
	1; n1 f	452
	1 emb; n1 f	10974
	1 yg; n1 lt	1475
	1 emb ea; n3 f	5405e
	1 emb ea; n16 f	1937
	1 yg ea; n21 lt	10660
	1 emb ea; n38 f	3999a
	1.002; r1-2; n614 lt	595
Hylonycteris underwoodi	1 yg	5424
	1 emb ea; n3 f	8483a
	1 emb ea; n3 f	11471a
Leptonycteris curasoae	1 emb ea; n7 f	10112a
Leptonycteris nivalis	1-2	1474
	1-2 yg	577
	1 emb ea; n12 f	217
Leptonycteris sanborni	1 yg	577
	1 emb or yg; n97+ lt	4523a
Lichonycteris obscura	1 emb; n1 f	3608
	1 emb ea; n2 f	1641
Lionycteris spurrelli	1 emb; n1 f	10974
	1 emb ea; n2 f	3999a
Lonchophylla mordax	1 emb; n1 f	3608
Lonchophylla robusta	1 emb; n1 f	3999a
	1 emb; n1 f	10657
Lonchophylla thomasi	1 emb ea; n2 f	3999a
Lonchorhina aurita	1 emb ea; n2 f	3999a
Macrophyllum macrophyllum	1 yg	4401
Macrotus californicus	1	1474
	1	1199
	1 yg	1346

	1 yg	577
	1 emb ea; n4 f	1471a
	1 emb ea; n61 f	4079
	1.3 emb; r1-2; n3 f	1937
Macrotus waterhousii	1 emb ea; n5 f	1937
	1 emb ea; n5 f	5424e
	1 emb ea; n5 f	5873
	1 emb ea; n5 f	11471a
Micronycteris brachyotis	1 emb; n1 f	7154a
	1 emb; n1 f	9113
Micronycteris daviesi	1 emb; n1 f	10974
	1 emb; n1 f	3999a
Micronycteris hirsuta	1 emb; n1 f	10974
	1 emb; n1 f	3999a
	1 yg; n1 f	3900a
Micronycteris megalotis	1 emb; n1 f	4671
	1 emb; n1 f	1475
	1 emb ea; n2 f	10974
	1 emb ea; n2 f	5425
	1 emb ea; n8 f	3999a
Micronycteris minuta	1 emb; n1 f	3608
	1 ea; n5 lt	452
Micronycteris nicefori	1; n1 lt	452
Mimon cozumelae	1 emb; n1 f	9113
	1 emb; n1 f	1641
	1 emb ea; n2 f	4228
Mimon crenulatum	1; n1 lt	452
	1 emb; n1 f	3705c
	1 emb; n1 f	5405d
	1 emb ea; n2 f	3608
	1 emb ea; n2 f	10974
	1 emb ea; n7 f	3999a
Monophyllus redmani	est 1	3902
	1 emb ea; n2 f	5873
	1 emb ea; n4 f	4941
Monophyllus plethodon	1 emb	9685a
Phylloderma stenops	1 emb; n1 f	6260
Phyllonycteris aphylla	1 emb; n1 f	3902
Phyllonycteris poeyi	1 emb ea; n3 f	5873
	1 emb ea; n many	7328a
Phyllops haitiensis	1 emb ea; n19 f	5873
Phyllostomus discolor	1	10618
	1 emb; n1 f	10112a
	1 emb ea; n2 f	3999a
	1 emb ea; n2 f	5405c
	1 yg ea; n5 lt	10660
	1 yg ea; n6 f	8953
Phyllostomus elongatus	1 ea; n2 lt	452
	1 emb ea; n5 f	10974
	1 emb ea; n15 f	3999a
Phyllostomus hastatus	1	10785
	1 emb ea; n2 f	5425
	1 emb ea; n12 f	10974
	1 emb ea; n31 lt	3999a
Platalina genovensium	1 emb; n1 f	3999a
Pygoderma bilabiatum	1 emb ea; n>60 f	7696
Rhinophylla fischerae	1 emb ea; n3 f	3999a
Rhinophylla pumilio	1 emb ea; n9 f	3999a
Stenoderma rufum	1; n1 lt	10664
	1 emb ea; n6 f	5423a
Sturnira aratathomasi	1 emb ea; n2 f	10785a
Sturnira bidens	1 emb ea; n3 f	3608a
	1 emb ea; n15 f	3999a
Sturnira bogotensis	1 emb ea; n22 f	3999a

Stumira erythromos	1 emb; n1 f	6372
	1 emb ea; n47 lt	3999a
Stumira lilium	1	10618
	1 emb; n1 f	11938
	1 emb; n1 f	9113
	1 emb ea; n2 f	134
	1 emb ea; n2 f	950
	1 emb ea; n2 f	10974
	1 emb ea; n3 f	3701
	1 emb ea; n3 f	1635
	1 ea; n5 lt	452
	1 emb ea; n15 f	5405e
	1 emb ea; n32 f	3999a
Stumira ludovici	1	10785
	1 emb ea; n5 f	5424c
	1 emb ea; n8 f	11471a
Stumira magna	1 emb ea; n6 f	3999a
Stumira mordax	1 emb ea; n2 f	3608
Stumira nana	1 emb; n1 f	3999a
Stumira tildae	1 emb; n1 f	3900a
	1 emb ea; n4 f	3999a
	1 emb ea; n5 f	1635
Tonatia bidens	1 emb; n1 f	3608
	1 emb ea; n2 f	3999a
Tonatia brasiliense	1 emb; n1 f	6258
	1 emb; n1 f	2345, 11055
Tonatia silvicola	1 emb ea; n4 f	10974
	1 emb ea; n10 f	3999a
Trachops cirrhosus	1 emb; n1 f	10974
	1 emb; n1 f	3999a
	1 emb ea; n3 f	1475
	1 emb ea; n5 f	1641
	1 emb ea; n10 f	5405e
Uroderma bilobatum	1 emb; n1 f	5405e
	1 emb; n1 f	10974
	1 emb ea; n4 f	1475
	1 emb ea; n4 f	3207
	1 emb ea; n4 f	5405c
	1 yg ea; n6 lt	10660
	1 emb ea; n8 f	2342
	1 emb ea; n28+ f	3999a
Uroderma magnirostrum	1 emb; n1 f	5425
Vampyressa bidens	1 emb; n1 f	3705a
	1 emb ea; n2 f	2343a
	1 emb ea; n27 f	3999a
Vampyressa melissa	1 emb ea; n4 f	3999a
Vampyressa nymphaea	1	10785
	1 emb ea; n2 f	3608
Vampyressa pusilla	1 emb; n1 f	8412
	1 emb; n1 f	2345
	1 emb; n1 f	9113
	1 emb; n1 f	11055
	1 emb; n1 f	5425
	1 emb ea; n15 f	3999a
Vampyrodes caraccioli	1 emb; n1 f	2345
	1 emb; n1 f	5405d
	1 emb; n1 f	10974
	1 emb ea; n3 f	5425
	1 emb ea; n4 f	3999a
Vampyrops brachycephalus	1 emb ea; n3 f	9318a
	1 emb ea; n12 f	3999a
Vampyrops dorsalis	1	10785
	1 emb ea; n33 f	3999a

Vampyrops helleri	1	10785
	1 emb; n1 f	2345
	1; n3 lt	452
	1 emb ea; n3 f	6258
	1 emb ea; n3 f	10974
	1 emb ea; n14 f	1635
	1 emb ea; n15 f	3999a
Vampyrops infuscus	1 emb ea; n6 f	3999a
Vampyrops lineatus	1	10618
	1 emb ea; n25 f	3999a
Vampyrops vittatus	1 emb; n1 f	3608
	1 emb ea; n2 f	2345
	1 emb ea; n4 f	3999a
Vampyrum spectrum	1; n1 lt	2109
SEXUAL MATURITY		
Artibeus lituratus	f: 46.3 g	10661
Carollia perspicillata	f: 1st birth: 13.5 mo; r11-16; n2	8683
	f/m: 1st mate: 8-9 mo	8683
Macrotus californicus	f: 1st mate: 3-4 mo	1199
	f: 1st conceive: 4 mo, no emb dev until 9 mo	1199
	m: 1st mate: 15-16 mo	1199
CYCLE LENGTH		
Desmodus rotundus	estrus: 2-3 d	9615
Glossophaga soricina	24 d; mode 24; r22-26; n43 cycles, 39 f	8941, 8942
GESTATION		
Anoura caudifer	est > 3.5 mo	10618
Artibeus jamaicensis	2.5 mo delay; ≤4 mo no delay, 7 mo with delay	3318
Artibeus lituratus	est > 3.5 mo	10618
Artibeus planirostris	est > 3.5 mo	10618
Carollia perspicillata	1.5-4 mo	6123
	est > 3.5 mo	10618
Chiroderma doriae	est > 3.5 mo	10618
Chiroderma villosum	~4 mo	10618
Chrotopterus auritus	99 d; n1; capture to birth	10618
Desmodus rotundus	>5 mo; n1; capture to birth	11813
	205 d; n1	9614, 9615
Glossophaga soricina	implant: 12-14 d	8941, 8943
	est > 3.5 mo	10618
Macrotus californicus	diapause: ~4.5 mo	999
	9 mo; with delayed development	1461, 1462
	~8 mo; with delayed development; field data	1199
	8.5 mo	999
	9 mo; with delayed development	1038, 1039
Micronycteris megalotis	~4 mo	10618
Phyllostomus discolor	est > 3.5 mo	10618
Phyllostomus hastatus	~4 mo	10618
Sturnira lilium	est > 3.5 mo	10618
Uroderma bilobatum	est 4-5 mo	527
Vampyrops lineatus	est > 3.5 mo	10618
LACTATION		
Brachyphylla cavernarum	1st fly: ~2 mo	7828a
Carollia perspicillata	1st fly: 2.5-4 wk	8683, 8684
	thick milk: 33 d; r21-49	5857
	thinner milk: until ~56 d; r42-72	5857
Desmodus rotundus	solid food: 2 mo	4049
	solid food: 6 mo	9615
	wean: 9 mo	9615
	nurse: 300 d	4049
Glossophaga soricina	1st fly: 25-28 d; n1	5857
	wean: ~2 mo; n1	5857
Leptonycteris sanborni?	wean: 4-8 wk	5296
Macrotus californicus	nurse: 1 mo	1199
Uroderma bilobatum	independent: 1 mo	527

INTERLITTER INTERVAL		
Artibeus lituratus	220 d; r200-240; n2 f	2552
Carollia perspicillata	159 d; r119-417	8683
Desmodus rotundus	9-10 mo	11703
SEASONALITY		
Ametrida centurio	preg f: July-Aug; Trinidad	1635
Anoura caudifer	epididymal sperm: yr rd, peak Apr-June; n57; São Paulo, Brazil	10618
	preg f: Jan; n2; Brazil	8368
	preg f: Jan-Feb; French Guiana	1350a
	preg f: May, July-Sept, Nov; n8; Peru	3999a
	preg f: May, Nov; n2; Colombia	10785
	preg f: June; n1; Colombia	10663
	preg f: est yr rd; São Paulo, Brazil	10618
	lact f: Feb; n1; Brazil	6084a
Anoura cultrata	preg f: Apr-June; n3; Peru	3999a
	preg f: July-Aug; n5; Colombia	6372a, 7745
	preg f: Aug; n1; Costa Rica	3608
	lact f: July; n1; Colombia	6372a, 7745
	lact f: Aug; n1; Peru	1639b
	lact f: Aug; few data; Peru	7745
Anoura geoffroyi	preg f: Mar; n2; Costa Rica	6809
	preg f: May-Aug; n8; Peru	3999a, 10974
	preg f: July; n1; Nicaragua	5425
	preg f: Nov; n56; Trinidad	3900a
	lact f: July; n3; Hidalgo, Mexico	1640a
	lact f: Nov; n5; Colima, Mexico	11754
Ardops nichollsi	preg f: Mar-Apr; n5; Dominica, Lesser Antilles	5424d
	preg f: July; n2; Guadeloupe, Lesser Antilles	527a
	lact f: Apr; n1; Dominica, Lesser Antilles	5424d
	lact f: July; n2; Guadeloupe, Lesser Antilles	527a
Artibeus anderseni	preg f: June, Aug, Nov; n4; Peru	3999a
Artibeus aztecus	preg f: Mar-Apr; middle America	2342a
	preg f: July; n1; Durango, Mexico	523a
	preg f: July; n18; Sinaloa, Mexico	5405d, 5413a
	preg f: July-Aug; n2; Tamaulipas, Mexico	217
	lact f: Sept; n1; Mexico	11754
Artibeus cinereus	preg f: Jan; n7; El Salvador	1475
	preg f: Feb-Apr, July-Aug, Nov; middle America	2342b
	preg f: Feb, Aug; n2; Nicaragua	5425
	preg f: Mar; n1; Guatemala	5405e
	preg f: Apr-May, July-Aug, Oct-Jan; n20; Colombia	10785
	preg f: June-July; n2; Brazil	11754
	preg f: June-Sept, Nov; n21; Peru	3999a
	preg f: June, Dec-Feb; n4; Panama	3322
	preg f: July; n1; Venezuela	11754
	preg f: Aug; n1; Colombia	352a
	lact f: Jan, Mar, May, Aug-Sept; n5; Colombia	10785
	lact f: Apr, Aug; n4; Panama	3322
	lact f: July-Aug; n4; Venezuela	11754
	lact f: Aug; n5; Colombia	352a
Artibeus concolor	preg f: Feb; n1; Colombia	10785
Artibeus fraterculus	preg f: June, Aug-Dec; n46; Peru	3999a
Artibeus fuliginosus	preg f: June-Dec; n24; Peru	3999a
Artibeus hartii	preg f: Jan, May-June; n3; Costa Rica	3608
	preg f: Apr-May; n19; Colombia	10785
	preg f: Aug-Sept, Nov-Dec; n6; Peru	3999a
	lact f: Apr; n1; Michoacan, Mexico	4671
	lact f: May-July; Costa Rica	3608
	lact f: Aug; Costa Rica	366b
	lact f: July, Sept; n2; Colombia	10785
Artibeus hirsutus	preg f: Feb, June, Aug; n8; Jalisco, Mexico	11741a
	preg f: Apr-May; n3; Sonora, Mexico	1137, 1139
	preg f: May; n8; Sonora, Mexico	279a

	preg f: May; n2; Guerrero, Mexico	279a
	preg f: July; n1; Chihuahua, Mexico	280a
	preg f: Aug; n5; Sinaloa, Mexico	5413a
	preg f: Sept; n15; Sonora, Mexico	3254b
	lact f: June; n1; Sinaloa, Mexico	5413a
	lact f: June, Aug; Jalisco, Mexico	11741a
	lact f: Sept; n6; Sonora, Mexico	3254b
Artibeus inopinatus	preg f: May; Honduras	11524
	lact f: July; n1; Honduras	2670a
Artibeus jamaicensis	preg f: Jan; n4; Providencia, Caribbean Sea, Colombia	10654
	preg f: Jan-Feb, Apr-July; Sinaloa, Mexico	5413a
	preg f: Jan-June; Jalisco, Mexico	11471a
	preg f: Feb; n1; Guerrero, Mexico	11754
	preg f: Feb; Veracruz, Mexico	4228
	preg f: Feb; n1; Tobago, West Indies	5433a
	preg f: Feb; n6; Tobago, West Indies	3900a
	preg f: Feb-Mar, Sept, Nov-Dec; n8; Michoacan, Mexico	4671
	preg f: Feb-Mar; n5; Guatemala	5405e
	preg f: Feb-Mar, June; St Croix, Virgin I	16052a
	preg f: Feb, Apr, July, peak July; Yucatan, Mexico	5424f
	preg f: end Feb-early Mar; n5; Jamaica	7103a
	preg f: Feb-May, July; Trinidad	3900a
	preg f: Mar; n6; Tamaulipas, Mexico	2390
	preg f: Mar-Apr; n7; Yucatan, Mexico	950
	preg f: Mar, May-Sept, Nov-Jan; n34; Colombia	10785
	preg f: Apr; n1; Yucatan, Mexico	1162
	preg f: Apr, July; Virgin I	595
	preg f: May; n1; Tamaulipas, Mexico	217
	preg f: May; n1; Guatemala	9113
	preg f: June; n3; San Luis Potosi, Mexico	1937
	preg f: June; n3; Sinaloa, Mexico	11754
	preg f: June; Puerto Rico	344a
	preg f: June; Trinidad	5433
	preg f: June; n1; Colombia	10657
	preg f: June, Aug, Dec-Jan; n13; Haiti	5873
	preg f: June or July; n3; Venezuela	8562a
	preg f: July; Hidalgo, Mexico	1640a
	preg f: July; n4; Morelos, Mexico	7978
	preg f: July; n1; Lesser Antilles	5411a
	preg f: July; n9; Guadeloupe, Lesser Antilles	527a
	preg f: July; n1; Colombia	352a
	preg f: Aug; n1; Veracruz, Mexico	595
	preg f: Nov; n1; Paraguay	7699a
	preg f: Nov-July, Sept; n42; Costa Rica	3322
	preg f: Dec; n16; El Salvador	1475
	preg f: Dec; n6; Jamaica	3902
	preg f: Dec-July; n138; Panama	3322
	preg f: end wet season; Brazil	11736
	preg f: yr rd, peak Jan-Mar, low Sept-Nov; Panama	1070
	preg/lact f: Apr-May 10% n289; July-Aug 9% n243; Sept 0% n152; Guatemala	2552
	birth: Mar, July; Guana, Virgin Islands	6307
	birth: peaks Mar-Apr, July-Aug; Panama	3318
	birth: July; n1; Lesser Antilles	5411a
	lact f: Jan-Apr, June-Aug, Oct; n29; Colombia	10785
	lact f: Feb-May, July, Dec; n48; Costa Rica	3322
	lact f: May, July-Aug; Yucatan, Mexico	5424f
	lact f: Mar-Apr; St Croix, Virgin I	16052a
	lact f: Mar-Oct; n134; Panama	3322
	lact f: Mar, Nov; Paraguay	7699a
	lact f: Apr-May, July-Oct; Jalisco, Mexico	11471a
	lact f: Apr, June-July, Sept; Trinidad	3900a
	lact f: June or July; n2; Venezuela	8562a
	lact f: July; n1; Guadeloupe, Lesser Antilles	527a

	lact f: July; n7; Lesser Antilles	5411a
	lact f: July-Aug; n4; Colombia	352a
	lact f: Sept, Nov; Sinaloa, Mexico	5413a
Artibeus lituratus	spermatogenesis: yr rd; Melgar, Colombia	10658, 10662
	preg f: Jan; n1; Querétaro, Mexico	10261a
	preg f: Jan-Feb, Apr, July; n6; Yucatan, Mexico	5424f
	preg f: Jan-Feb, May, July; n5; Costa Rica	3322
	preg f: Jan, Mar-May, Aug; n14; Panama	3322
	preg f: Jan, Mar-June, Aug, Oct-Nov; n31; Colombia	10661
	preg f: Feb; n1; Tobago	3900a
	preg f: Feb-Mar; n5; Guatemala	5405e
	preg f: Feb, Apr, June-July; Sinaloa, Mexico	5413a
	preg f: Feb-May, Nov; n9; Michoacan, Mexico	4671
	preg f: Mar; n2; Tamaulipas, Mexico	2390
	preg f: Mar; n3; Panama	1006
	preg f: Mar-Apr, June-July; n8; Jalisco, Mexico	11471a
	preg f: Mar-July; Trinidad	3900a
	preg f: Apr; n5; Oaxaca, Mexico	11754
	preg f: Apr, July-Aug, Oct-Dec; n16; Peru	3999a
	preg f: May; n10; Tamaulipas, Mexico	217
	preg f: May; n1; Morelos, Mexico	1937
	preg f: May; n2; Panama	4228a
	preg f: July; n5; Colombia	352a
	preg f: July; n1; Margarita I, Venezuela	10112a
	preg f: July; n2; Brazil	11754
	preg f: July-Aug; Brazil	8368
	preg f: Aug; n1; Venezuela	11754
	preg f: Aug-Sept; French Guiana	1350a
	preg f: few end wet season; Brazil	11736
	preg f: yr rd, low Aug-Sept; n183; Colombia	10785
	preg f: yr rd; Melgar, Colombia	10658
	preg f: est yr rd; São Paulo, Brazil	10618
	preg/lact f: Jan-June, Sept-Nov; Panama	1070
	preg/lact f: July-Aug; Hidalgo, Mexico	1640a
	preg/lact f: July-Aug 8.4% n107; Sept 8.5% n47; Apr-May 17.5% n143; Guatemala	2552
	lact f: Mar-Apr, Sept-Oct; n11; Panama	3322
	lact f: Mar, May-July, Oct-Nov; n11; Colombia	10661
	lact f: Apr-May, July; n5; Costa Rica	3322
	lact f: Apr-May, Oct; Trinidad	3900a
	lact f: May; n2; Guatemala	9113
	lact f: June; n2; Durango, Mexico	5405b
	lact f: July; n10; Colombia	352a
	lact f: July-Oct; Jalisco, Mexico	11471a
	lact f: July, Oct; Sinaloa, Mexico	5413a
	lact f: Sept; n1; Tobago	3900a
	lact f: yr rd; n100; Colombia	10785
Artibeus phaeotis	preg f: Jan-Mar; n10; Campeche, Mexico	5424e
	preg f: Jan-Apr, June, Aug; n15; Panama	3322
	preg f: Jan-Apr, June-Aug; middle America	2342b
	preg f: Jan-June, none Sept-Dec; Panama	1070
	preg f: Jan, Apr, June; n18; Jalisco, Mexico	11471a
	preg f: Mar; n1; Guatemala	5405e
	preg f: Mar-May, Dec; n13; Michoacan, Mexico	4671
	preg f: Apr; n1; Guatemala	7624
	preg f: Apr, Aug-Sept; s Mexico	11754
	preg f: Apr, Aug-Sept; La Selva, Costa Rica	6261
	preg f: May; n2; Guatemala	9113
	preg f: July; n4; Sinaloa, Mexico	5413a
	preg f: Aug; n2; Cozumel I, Mexico	5424f

	lact f: Mar; n1; Campeche, Mexico	5424f
	lact f: Apr; n1; Quintana Roo, Mexico	950
	lact f: peak May-Sept; Barro Colorado I, Panama	1070
	lact f: June, Oct; s Mexico	11754
	lact f: Aug; n1; Jalisco, Mexico	11471a
Artibeus planirostris	epididymal sperm: yr rd; n35; São Paulo, Brazil	10618
	preg f: Feb; n6; Tobago, W Indies	5433a
	preg f: Apr, June-Sept, Nov-Jan; n49; Peru	3999a
	preg f: July; n4; Margarita I, Venezuela	10112a
	preg f: est yr rd; São Paulo, Brazil	10618
	lact f: July; n2; Margarita I, Venezuela	10112a
Artibeus toltecus	preg f: Jan; n2; Puebla, Mexico	6258a
	preg f: Jan; n7; El Salvador	1475
	preg f: Jan-Apr, June-July; n33; Jalisco, Mexico	11471a
	preg f: Jan, May, Oct; Sinaloa, Mexico	5413a
	preg f: Jan-Aug; middle America	2342a
	preg f: Mar, Sept; n3; Michoacan, Mexico	4671
	preg f: Apr, June; n9; Nicaragua	5425
	preg f: Apr, Aug; Monte Verde, Costa Rica	6261
	preg f: end June-early July; n4; Chiapas, Mexico	2345
	preg f: July; n1; Tamaulipas, Mexico	2390
	preg f: July; n4; Hidalgo, Mexico	1640a
	lact f: Apr-Nov, peak Aug-Oct; Costa Rica	2587
	lact f: May; Sinaloa, Mexico	5413a
	lact f: May, Aug; n2; Chiapas, Mexico	2345
	lact f: June-Sept; Jalisco, Mexico	11471a
Brachyphylla cavernarum	preg f: Feb; Puerto Rico	7983, 11754
	preg f: Mar; St Croix, Virgin I	7983, 11754
	birth: end May-early June; Virgin I	7828a
	lact f: Apr; n1; Puerto Rico	7983, 11754
	lact f: Apr; n1; St Croix, Virgin I	7827a
	lact f: July; Puerto Rico	344a
	lact f: July; n3; Guadeloupe, Lesser Antilles	527a
	lact f: July; n4; Lesser Antilles	5411a
Brachyphylla nana	preg f: Mar; n12; Caicos I, Greater Antilles	1429c
	preg f: Dec-May; Cuba	10594
	lact f: Aug; n1; Haiti	5873
Carollia brevicauda	preg f: Jan-Mar; n20; Panama	8551
	preg f: Feb; n1; Guatemala	9113
	preg f: Feb-Mar; n5; Guatemala	8551
	preg f: Mar; Veracruz, Mexico	4228
	preg f: Mar; n1; Ecuador	8551
	preg f: Mar; n2; Oaxaca, Mexico	8551
	preg f: Mar; n18; Veracruz, Mexico	8551
	preg f: Mar-Apr; n4; Costa Rica	8551
	preg f: Mar-Apr, Aug; La Selva, Costa Rica	6261
	preg f: Apr; n1; Quintana Roo, Mexico	8551
	preg f: Apr; n1; Yucatan peninsula, Mexico	5424f
	preg f: Apr; n2; San Luis Potasi, Mexico	8551
	preg f: May-June; n6; Honduras	8551
	preg f: June-July; n7; Chiapas, Mexico	8551
	preg f: July; n1; Hidalgo, Mexcio	1640a
	preg f: July; n1; Nicaragua	8551
	preg f: Aug; n1; Guatemala	5405e
	preg f: Aug-Nov; n28; Peru	3999a
	preg f: Oct; n1; Peru	8551
	lact f: Feb; n1; Panama	8551
	lact f: Mar-Apr; n2; Costa Rica	8551
	lact f: Apr-May; n3; Tabasco, Mexico	8551
	lact f: Apr-June; n9; Honduras	8551
	lact f: July; n1; Nicaragua	8551
Carollia castanea	preg f: Jan-Mar; n15; Panama	5425
	preg f: Jan-Mar; French Guiana	1350a
	preg f: Jan, Mar-Apr, July-Aug; n10; Panama	3322

	preg f: Jan, Mar-Apr, Sept-Nov; n14; Colombia	10785
	preg f: end Jan-mid Mar; n15; Panama	8551
	preg f: Feb-Mar; n4; Costa Rica	8551
	preg f: Feb-Mar, Aug; n7; Costa Rica	5425
	preg f: Mar, July; Nicaragua	5425
	preg f: Apr, Aug; La Selva, Costa Rica	6261
	preg f: May, July; n4; Honduras	8551
	preg f: June-Sept, Nov-Dec; n24; Peru	3999a
	lact f: Jan, Apr-May; n5; Colombia	10785
	lact f: June; n1; Panama	3322
	lact f: Aug; n1: Costa Rica	8551
Carollia perspicillata	epididymal sperm: yr rd; n28; São Paulo, Brazil	10618
	m repro: May, Oct-Nov; Rio de Janeiro, Brazil	8368
	mate: Nov, Jan; El Salvador	3206
	preg f: Jan-Apr, June-Aug; n32; Costa Rica	3322
	preg f: Jan, Mar; n2; Colombia	10660
	preg f: Jan, Aug; Amazon Valley	181
	preg f: Jan, Aug; n7; Brazil	8551
	preg f: Jan, Sept-Oct; Rio de Janeiro, Brazil	8368
	preg f: Feb-Apr, June; n13; Panama	8551
	preg f: Feb-Apr, July-Aug; n14; Costa Rica	8551
	preg f: Feb-Aug, Oct; Trinidad	3900a
	preg f: Feb, Apr-Aug; n16; Nicaragua	8551
	preg f: few Feb-June; Brazil	11736
	preg f: Feb, Aug; La Selva, Costa Rica	6261
	preg f: Mar; n1; Ecuador	8551
	preg f: Mar; n1; El Salvador	1475
	preg f: Mar; n3; Guatemala	5405e
	preg f: Mar; n2; Guatemala, El Salvador	8551
	preg f: Mar; n1; Panama	3024
	preg f: Apr; Tuxtepec, Mexico	4201a
	preg f: May; n2; Campeche, Mexico	5424f
	preg f: May; n3; Veracruz, Mexico	11754
	preg f: May, July; n4; Honduras	8551
	preg f: June; Trinidad	4201a
	preg f: June-July; n2; Veracruz, Mexico	6148a
	preg f: June-July; n5; Colombia	8551
	preg f: June-July; n7; Venezuela	8562a
	preg f: July-Aug; n5; Quintana Roo, Mexico	8551
	preg f: July-Aug; n6; Quintana Roo, Mexico	5424f
	preg f: July-Sept; n55; Colombia	352a
	preg f: July-Nov; French Guiana	1350a
	preg f: July-Dec; n57; Peru	3999a, 10974
	preg f: Aug; n1; Peru	8551
	preg f: Aug-Sept; n2; Bolivia	8551
	preg f: Oct-Jan, Mar-Aug; n29; Colombia	10785
	preg f: Dec-Sept; n98; Panama	3322
	preg f: est yr rd; São Paulo, Brazil	10618
	preg/lact f: Jan-June, few Sept-Nov; Barro Colorado I, Panama	1070
	birth: Mar-Apr, July-Aug; captive	8684
	birth: Mar-Apr, Aug-Sept; Costa Rica	11713
	birth: mid Mar-mid May, mid Aug-mid Nov; zoo	6123
	birth: yr rd, peaks Mar-Apr, Aug-Sept; captive	8683
	lact f: Jan-June, Sept, Nov; n15; Colombia	10785
	lact f: Feb-Apr, Aug; n8; Costa Rica	8551
	lact f: Mar-Apr; n2; Panama	8551
	lact f: Mar-Oct; n47; Panama	3322
	lact f: Apr; n1; El Salvador	3206
	lact f: Apr-May, Aug; n26; Costa Rica	3322
	lact f: Apr, Oct; n2; Colombia	10660
	lact f: May; n1; Nicaragua	8551
	lact f: May, July; n7; Honduras	8551
	lact f: May-Oct; Trinidad	3900a
	lact f: June or July; n1; Venezuela	8562a

	lact f: Aug; n1; Chiapas, Mexico	8551
	lact f: Aug; n16; Colombia	
	repro: Jan-Aug, Oct-Nov; no data Dec; sw Colombia	10785
Carollia subrufa	preg f: Jan-Mar, Oct; n51; El Salvador	3206
	preg f: Feb; n6; Chiapas, Mexico	11754
	preg f: Feb; n6; Guatemala	8551
	preg f: May; n2; Oaxaca, Mexico	11754
	preg f: July-Aug; n7; Chiapas, Mexico	8551
	preg f: July-Aug; n2; Nicaragua	8551
	preg f: Aug; n1; Guatemala	5405e
	preg f: Dec; n1; Guerrero, Mexico	6639
	preg f: Dec-Mar; Panama	7893, 11754
	lact f: May, Oct; n2; Chiapas, Mexico	8551
	lact f: Nov; n2; Guatemala	8551
Centerio senex	preg f: Feb-Mar; n3; Nicaragua	5425
	preg f: Feb, July; n2; Yucatan, Mexico	5424f
	preg f: Mar; n2; Jalisco, Mexico	11471a
	preg f: Mar; n2; Oaxaca	11754
	preg f: Apr; n1; Michoacan, Mexico	4671
	preg f: Apr, July; n2; Chiapas, Mexico	2345
	preg f: June; n1; Tamaulipas, Mexico	217
	preg f: Aug; n2; Honduras	6258
	birth: Jan; n1; Trinidad	3900a
	lact f: Mar; n1; Oaxaca, Mexico	11754
	lact f: Mar, Aug; n2; Jalisco, Mexico	11471a
	lact f: Apr; n1; Michoacan, Mexico	4671
	lact f: Apr; n5; Veracruz, Mexico	11754
	lact f: July; n2; Hidalgo, Mexico	1640a
Chiroderma doriae	epididymal sperm: Aug; n9; São Paulo, Brazil	10618
Chiroderma villosum	preg f: Jan; n1; Colombia	10785
	preg f: Mar, July; n8; Nicaragua	5425
	preg f: Apr; n1; Panama	3322
	preg f: Apr-May, Aug; Guatemala	2552
	preg f: May; n1; Nicaragua	4048_1
	preg f: July, Dec; n5; Chiapas, Mexico	2345
	preg f: Aug-Sept; n2; Trinidad	3900a
	lact f: Mar-Apr; n4; Panama	3322
	lact f: May; n1; Chiapas, Mexico	2345
Chiroderma salvini	preg f: Jan; n1; Sinaloa, Mexico	5413a
	preg f: Feb, June; n7; Jalisco, Mexico	11471a
	preg f: Mar-June, Oct, Dec-Jan; n11; Colombia	10785
	preg f: July; n1; Honduras	1641
	preg f: July-Aug; n2; Honduras	6258
	lact f: Mar-Apr, July; n6; Colombia	10785
	lact f: July-Aug; n2; Honduras	6258
Chiroderma trinitatum	preg f: Feb, May; n3; Panama	3322
	preg f: Mar; n1; Trinidad	3900a
	preg f: June-July; n2; Brazil	11754
	preg f: July; n1; Colombia	10785
	preg f: Oct-Nov; Peru	2345a
	preg f: Nov; n4; Peru	3999a
	lact f: May, Sept; n2; Panama	3322
Chiroderma villosum	preg f: June-Dec; n7; Peru	3999a
Choeroniscus godmani	preg f: Mar; n1; Nicaragua	5425
	preg f: Mar; n1; Costa Rica	6809
	preg f: July; n1; Sinaloa, Mexico	5405d
	lact f: Nov/Dec; n1; Oaxaca, Mexico	9542a
Choeroniscus intermedius	preg f: Aug; n1; Trinidad	3900a
Choeroniscus minor	preg f: Feb; n1; British Guiana	3900a
	preg f: June; n2; Peru	3999a
	lact f: Dec; n1; Colombia	10664a
Choeronycteris mexicana	preg f: Feb; n1; Sinaloa, Mexico	5413a
	preg f: Feb; n2; Guerrero, Mexico	11754
	preg f: Mar; n1; Coahuila, Mexico	522

	preg f: June; n2; NM	7612
	preg f: Sept; n1; Jalisco, Mexico	11471a
	lact f: June; n4; NM	7611, 7612
	lact f: June; n4; Coahuila, Mexico	522
	lact f: July-Aug; AZ	1568a
	lact f: Aug; Tamaulipas, Mexico	217
	repro: June-July; sw US	577, 1474
Chrotopterus auritus	preg f: Apr; n1; Veracruz, Mexico	4228
	preg f: Apr; n1; Yucatan, Mexico	5424f
	perg f: July; n1; Argentina	11754
	lact f: July; n1; Qunitana Roo, Mexico	5424f
Desmodus rotundus	sexually active: yr rd; Panama	11813
	mate: Mar, Aug-Dec; El Salvador	3207
	mate: mid Sept, mid Apr; lab	9614, 9615
	preg f: Jan; n1; Nayarit, Mexico	1937
	preg f: Jan; n1; Mexico, Mexico	1463
	preg f: Jan; n1; Morelos, Mexico	1463
	preg f: Jan; n2; Puebla, Mexico	6258a
	preg f: Jan; Brazil	8368
	preg f: Jan-Apr, June-July; n10; Yucatan, Mexico	5424f
	preg f: Jan-May, July, Nov; n17; Costa Rica	3322
	preg f: Jan-July, Sept, peak Jan-May; Jalisco, Mexico	11471a
	preg f: Feb; Veracruz, Mexico	4228
	preg f: Feb, Apr-May, July, Nov; n15; Panama	11813
	preg f: Mar, Aug; n4; Guatemala	5405e
	preg f: Mar, May, June; n4; Tamaulipas, Mexico	217
	preg f: Mar, May, July; n~90; Colima, Mexico	1465a
	preg f: Mar, May, July-Aug, Oct; n10; El Salvador	3207
	preg f: Mar, May, Dec-Jan; n4; Sinaloa, Mexico	5413a
	preg f: Apr; n2; Yucatan, Mexico	950
	preg f: Apr-May; n3; Panama	3322
	preg f: May; n4; Chihuahua, Mexico	280a
	preg f: May, Oct; n2; Colombia	10785
	preg f: June, Aug-Sept, Nov; n41; Guerrero, Mexico	3414a
	preg f: June-July; n5; Veracruz, Mexico	6148a
	preg f: June-Nov; n31; Peru	3999a
	preg f: July; n1; Hidalgo, Mexico	1640a
	preg f: July; n1; Michoacan, Mexico	4229a
	preg f: July; n3; Colombia	352a
	preg f: Oct; n1; Zacatecas, Mexico	1937
	preg f: Nov; Colombia	10657
	preg f: Dec; Querétaro, Mexico	9603a
	preg f: yr rd; Panama	11813
	preg f: yr rd; Trinidad	3900a
	preg f: yr rd; Brazil	11736
	preg/lact f: Feb-Sept; Veracruz, Mexico	4228
	birth: Dec; n1; captive	2109
	lact f: Jan; Brazil	8368
	lact f: Feb-Mar; n3; Costa Rica	3322
	lact f: Apr-May; n2; Panama	3322
	lact f: Apr, July-Aug; Yucatan, Mexico	5424f
	lact f: Apr, Oct; n2; Colombia	10785
	lact f: July; n1; Hidalgo, Mexico	1640a
	lact f: July; n3; Colombia	352a
	lact f: Aug; Jalisco, Mexico	11471a
	lact f: Nov; Panama	11813
	repro: yr rd: n Argentina	11304
Desmodus youngi	preg f: June; n1; Peru	3999a
	preg f: Oct; n4; Trinidad	3900a
	lact f: Aug, Oct; n3; Trinidad	3900a
Diphylla ecaudata	preg f: Mar; San Luis Potosí, Mexico	2228a
	preg f: Aug; n2; Honduras	11055
	preg f: Aug, Oct-Nov; n10; Mexico, Mexico	11754
	preg f: Oct; n1; Peru	3999a

	preg f: Nov; n2; Tamaulipas, Mexico	217
	preg f: Nov; n1; Yucatan, Mexico	4494
	lact f: May; n1; Yucatan, Mexico	950
	lact f: Aug; n1; Honduras	11055
	lact f: Nov; n1; Tamaulipas, Mexico	217
	lact f: Nov; n2; Yucatan, Mexico	4494
	lact f: Nov; n2; Mexico, Mexico	11754
Ectophylla alba	preg f: Feb, Aug; Costa Rica	6260
	preg f: Mar; n1; Costa Rica	3608
	lact f: Mar; n1; Costa Rica	6260
	lact f: Apr; n1; Costa Rica	3608
Ectophylla macconnelli	preg f: Jan; n1; Colombia	10785
	preg f: June-Aug, Oct; n6; Peru	3999a, 10974
	preg f: July; n1; Peru	11754
	lact f: May; n1; Peru	11754
Erophylla sezekorni	preg f: Feb-May; West Indies	1429b
	preg f: Mar-Apr; Puerto Rico	7983, 11754
	preg f: Apr; Puerto Rico	395
	preg f: June; n1; Puerto Rico	11055a
	birth: June; n1; Bahamas	983a, 11754
	lact f: May, July; Puerto Rico	7983, 11754
	lact f: June; n8; Bahamas	983a, 11754
	lact f: June; West Indies	1429b
Glossophaga commissarisi	preg f: Jan, July; n5; Sinaloa, Mexico	5413a
	preg f: Feb; n1; Guatemala	5405e
	preg f: Feb; n1; Nicaragua	5405c
	preg f: Feb, Apr, Sept; n4; Jalisco, Mexico	11471a
	preg f: Mar-Apr, Sept; La Selva, Costa Rica	6261
Glossophaga leachii	preg f: Jan-Mar, July-Sept, Nov-Dec; El Salvador	3206
	preg f: Feb-Apr, Sept-Oct; Jalisco, Mexico	11471a
	preg f: Aug; n1; Yucatan, Mexico	8285a
	preg f: Sept; n1; Veracruz, Mexico	4228
Glossophaga longirostris	preg f: Feb-Apr, Aug; n10; Trinidad	3900a
	preg f: Mar; n4; Patos I, Venezuela	3900a
	preg f: June or July; n3; Venezuela	8562a
	preg f: July; n8; Margarita I, Venezuela	10112a
	lact f: July; n2; Venezuela	11754
	lact f: Aug; n103; Bonaire, Dutch West Indies	3705b
Glossophaga mexicana	preg f: Mar; n1; Central America	11527
	lact f: May; n1; Central America	11527
Glossophaga soricina	epididymal sperm: yr rd; São Paulo, Brazil	10618
	preg f: Jan; n1; Puebla, Mexico	6258a
	preg f: Jan; n1; Jamaica	3902
	preg f: Jan-Feb, Aug; n7; Nayarit, Mexico	1937
	preg f: Jan-July; n40; Panama	3322
	preg f: Jan, Apr-June, Dec; Trinidad	3900a
	preg f: Feb; Chiapas, Mexico	11754
	preg f: Feb; n1; Panama	1006
	preg f: Feb-Mar; French Guiana	1350a
	preg f: Feb, Apr, July; Yucatan peninsula, Mexico	5424f
	preg f: Feb-Apr, Sept, Nov; n14; Michoacan, Mexico	4671
	preg f: Mar-Apr; n7; Oaxaca, Mexico	11754
	preg f: Mar-Apr, Aug, Oct-Jan; n23; Colombia	10660
	preg f: Mar-Apr, Sept; n5+; Veracruz, Mexico	4228
	preg f: Mar, May, Aug-Jan; Sinaloa, Mexico	5413a
	preg f: Mar, Aug; n3; Guatemala	5405e
	preg f: May; n3; Sonora, Mexico	1937
	preg f: May; Tres Marias I, Mexico	7235a
	preg f: May-July; Tabasco, Mexico	11754
	preg f: June-Nov; n38; Peru	10974
	preg f: June-July, Oct, Dec-Feb; n18; Colombia	10785
	preg f: June-Mar; São Paulo, Brazil	10618
	preg f: July; n1; Hidalgo, Mexico	1640a
	preg f: July; n7; El Salvador	10313a

	preg f: July-Aug; n2; Costa Rica	10313a
	preg f: July-Aug; n40; Colombia	352a
	preg f: July-Sept, Nov-Mar; n61; El Salvador	3205
	preg f: Aug; Chiapas, Mexico	595
	preg f: Aug; n1; Honduras	10313a
	preg f: Aug; n1; Venezuela	11754
	preg f: Sept; n1; Oaxaca, Mexico	1937
	preg f: Sept; n1; El Salvador	1475
	preg f: Nov-Dec; Colima, Mexico	11754
	preg f: Nov-Dec; Mato Grosso, Brazil	4291, 4293
	preg f: yr rd; Brazil	11736
	preg/lact f: Jan; Brazil	8368
	birth: Sept; n1; El Salvador	1475
	birth: yr rd; Mexico	1937
	lact f: Jan, Mar-Apr, Aug, Oct-Nov; n13; El Salvador	3205
	lact f: Jan-Mar, June; Trinidad	3900a
	lact f: Mar; n5; Michoacan, Mexico	4671
	lact f: Mar-Apr, Sept-Oct, Dec; n18; Colombia	10785
	lact f: Mar, Aug; n5; Panama	3322
	lact f: Apr; n1; Yucatan, Mexico	950
	lact f: June-July; Tabasco, Mexico	11754
	lact f: June-July, Sept, Dec; n5; Colombia	10660
	lact f: July; n1; Chihuahua, Mexico	280a
	lact f: Oct; French Guiana	1350a
	lact f: Oct; n1; Paraguay	11754
	lact f: Nov; Veracruz, Mexico	4228
	lact f: Nov; n1; El Salvador	1475
	lact f: Dec; n1; Sonora, Mexico	1939
Hylonycteris underwoodi	preg f: Jan-Apr, Oct-Nov; n7; Costa Rica	6260
	preg f: Sept; n3; Jalisco, Mexico	8483a
	lact f: Mar; n1; Guatemala	1641
	lact f: May; n1; Tabasco, Mexico	11754
	lact f: Nov; n1; Oaxaca, Mexico	11754
Leptonycteris curasoae	preg f: Nov; n7; Margarita I, Venezuela	10112a
Leptonycteris nivalis	preg f: Aug; n12; Tamaulipas, Mexico	217
	preg f: Sept; n6; Veracruz, Mexico	4228
	lact f: June; n3; TX	2865a
	repro: Apr-June; TX	2343
Leptonycteris sanbomi	preg f: Feb, July, Nov; n3; Sinaloa, Mexico	5413a
	preg f: Mar-Apr; n12; Sonora, Mexico	1939
	preg f: Sept, Nov; n2; Mexico	11754
	birth: May; AZ	577
	birth: May; s AZ	4523a
	birth: Nov; Mexico	4523a
	lact f: Aug; AZ	4900a
	lact f: Dec; n1; Michoacan, Mexico	4671
Lichonycteris obscura	preg f: Feb; n2; Guatemala	1641
	preg f: Mar; n1; Costa Rica	3608
	lact f: Jan; n1; Costa Rica	3608
Lionycteris spurrelli	preg f: Aug; n2; Peru	3999a, 10974
Lonchophylla mordax	preg f: Aug; n1; Costa Rica	3608
	lact f: Mar; n1; Costa Rica	2345
Lonchophylla robusta	preg f: Nov; n1; Peru	3999a
Lonchophylla thomasi	preg f: July; n2; Peru	3999a
Lonchorhina aurita	preg f: Feb; Oaxaca, Mexico	7983, 11754
	preg f: Feb-Mar; n4; Panama	1006
	preg f: Feb-Mar; n4; Panama	3322
	preg f: Apr; n1; Trinidad	3900a
	preg f: Aug; n2; Peru	3999a
	lact f: July; n1; Peru	10974
	repro: Feb-Apr; Venezuela, Trinidad, Antilles	1346
Macrophyllum macrophyllum	mate: Dec; n3; El Salvador	3206
	preg f: Mar, May; Costa Rica	6260
	preg f: Oct; n2; El Salvador	4401

	preg f: Oct-Nov; French Guiana	1350a
Macrotus californicus	testes: enlarge July-Aug, regress end Nov; AZ	1199
	testes: enlarge June-Aug; AZ	6072
	mate: Aug-Nov; AZ	6072
	mate: Sept; s AZ	1462
	mate: Sept-Nov; AZ	1199
	mate: early Oct; s AZ	999
	preg f: Mar-Apr; n3; CA	1937
	preg f: Mar-Apr; n7; Sonora, Mexico	1939
	preg f: Apr; n9; CA	11754
	preg f: Apr; n60; CA	4079
	preg f: May; CA	5067a
	preg f: May; n4; Sonora, Mexico	1471a
	conceive: Sept-Nov; AZ	1199
	conceive: Sept; s AZ	1462
	conceive: Sept-Oct; AZ	1038, 1039
	implant: Sept-Oct; AZ	1000
	implant: Oct; placentation starts Feb; AZ	1038
	implant: end Oct; AZ	999
	birth: May-early July; s AZ	1198, 1199
	birth: end May-early June; s AZ	1462
	birth: June; s AZ	999
	birth: mid June; AZ	1038
	lact f: July; Sonora, Mexico	1471a
Macrotus waterhousii	preg f: Feb-Mar; n5; Jalisco, Mexico	11471a
	preg f: Feb, Apr; n4; West Indies	1429a
	preg f: Mar; n4; Cuba	280
	preg f: May; n5; Haiti	5873
	preg f: May; Tres Marias I, Mexico	7235a
	preg f: July; n5; Baja CA, Mexico	5424e
	lact f: May; Jalisco, Mexico	11471a
	lact f: July; Sinaloa, Mexico	5413a
	lact f: Dec; n4; Jamaica	3902
Micronycteris brachyotis	preg f: May, Aug; n2; Veracruz, Mexico	7154a
	preg f: July; n1; Guatemala	2520
	lact f: July; n6; Guatemala	2520
	lact f: Aug; n12; Veracruz, Mexico	7154a
Micronycteris daviesi	preg f: Aug; n1; Peru	10974
Micronycteris hirsuta	preg f: Mar, May; n3; Trinidad	3900a
	preg f: July; n1; Peru	10974
Micronycteris megalotis	preg f: Feb-Mar; n2; Trinidad	3900a
	preg f: Mar; n1; El Salvador	1475
	preg f: Mar, June; n3; Nicaragua	5425
	preg f: Apr; n1; Michoacan, Mexico	4671
	preg f: Apr; n1; Yucatan, Mexico	5424f
	preg f: May; n2; Oaxaca, Michoacan, Mexico	4671, 11754
	preg f: June, Aug; n3; Peru	3999a, 10974
	preg f: July; n1; Venezuela	11754
	lact f: May; n1; Panama	3024
	lact f: June; n2; Nicaragua	5425
	lact f: June; n1; Trinidad	3900a
	lact f: June; n1; Colombia	10785
	lact f: Aug; n3; Venezuela	11754
Micronycteris minuta	preg f: Mar; n1; Costa Rica	3608
	preg/lact f: May; n5; Trinidad	3900a
Micronycteris nicefori	lact f: July; n1; Nicaragua	527b
Micronycteris sylvestris	preg f: Feb-Mar; French Guiana	1350a
Mimon cozumelae	preg f: Mar; n1; Chiapas, Mexico	1641
	preg f: Mar; n1; Guatemala	9113
	preg f: Apr; n2; Veracruz, Mexico	4228
	preg f: Apr; n19; Yucatan, Mexico	5424f
	preg f: Apr; Costa Rica	6260
	lact f: May; Campeche, Mexico	5424f
	lact f: July; n1; Yucatan, Mexico	5424f

	lact f: July; n1; Honduras	11055
	lact f: Aug; n1; Guatemala	9113
	lact f: Aug; Costa Rica	6260
Mimon crenulatum	preg f: Feb; n1; Campeche, Mexico	5405d, 5424f
	preg f: Mar; n2; Trinidad	3900a
	preg f: Apr; n1; Costa Rica	6260
	preg f: Apr; n2; Costa Rica	3608
	preg f: June-July; n7; Peru	3999a, 10974
	preg f: July; n1; Suriname	3705c
Monophyllus redmani	preg f: Feb; n11; Jamaica	8134a
	preg f: end Feb-early Mar; n6; Jamaica	7103a
	preg f: May; n2; Haiti	5873
	preg f: Dec-Jan; n17; Jamaica	3902
	preg f: Dec-Feb; n5; Caribbean	4941
Monophyllus plethodon	preg f: Mar-Apr; Dominica, Lesser Antilles	9685a
	lact f: July; n1; Guadeloupe, Lesser Antilles	527a
Musonycteris harrisoni	preg f: Sept; n2; Colima, Mexico	11754
Phylloderma stenops	preg f: Feb; n1; Costa Rica	6260
Phyllonycteris aphylla	preg f: Jan; n1; Jamaica	3902
Phyllonycteris poeyi	preg f: June; Cuba	7328a
	preg f: Dec; n3; Haiti	5873
Phyllops haitiensis	preg f: Jan, May, Aug; n19; Haiti	5873
Phyllostomus discolor	preg f: Feb-Mar, June, Aug; Trinidad	3900a
	preg f: Feb, June, Aug-Sept, Nov; El Salvador	3206
	preg f: Feb, Oct; n5; Colombia	10660
	preg f: Mar; n2; Nicaragua	5405c
	preg f: Mar; n1; Costa Rica	3322
	preg f: July; n1; Margarita I, Venezuela	10112a
	preg f: Oct; n2; Colombia	11055b
	preg f: Nov; n2; Peru	3999a
	preg f: Dec; n1; El Salvador	1475
	preg/lact: Aug-Sept; Guatemala	2552
	lact f: Feb-Mar, May, Sept; n7; Colombia	10660
	lact f: July; n2; Margarita I, Venezuela	10112a
	lact f: July; n1; Brazil	7983, 11754
	lact f: Aug-Oct; Trinidad	3900a
	lact f: Dec; n11; Costa Rica	3322
Phyllostomus elongatus	preg f: June; n1; Colombia	10785
	preg f: July-Aug; n15; Peru	3999a, 10974
Phyllostomus hastatus	preg f: Mar; n2; Nicaragua	5425
	preg f: Mar, Sept-Oct; n4; Colombia	10785
	preg f: end Mar-Apr; Trinidad	3900a
	preg f: June-Oct; n31; Peru	3999a, 10974
	preg f: July; n1; Colombia	352a
	preg f: Aug; n1; Brazil	8368
	birth: Apr-Sept; Trinidad	1346
	birth: end Apr-early May; Trinidad	7036
	lact f: Apr-May; n2; Panama	3322
	lact f: Apr, June; Trinidad	3900a
	lact f: June-Aug; Nicaragua	5425
	lact f: Aug; Venezuela	11754
	preg f: Sept, Dec; n3; Colombia	10785
Platalina genovensium	preg f: Sept; n1; Peru	3999a
Pygoderma bilabiatum	preg f: Mar, July-Aug; Paraguay	7696
	preg f: Aug; n1; Brazil	8368
Rhinophylla fischerae	preg f: June-July; n3; Peru	3999a
	preg? f: Aug; n1; Peru	1639a
Rhinophylla pumilio	preg f: May, July; n2; Colombia	10785
	preg f: July-Sept, Nov; n9; Peru	3999a
	preg f: Dec; n1; Venezuela	7983, 11754
	lact f: Apr, June, Dec; n3; Colombia	19785
Stenoderma rufum	preg f: Feb-Mar, May, July-Aug; Puerto Rico	16052a
	preg f: July; n5; Puerto Rico	5423a
	preg f: July-Aug; n2; Puerto Rico	3699

	preg f: Aug; n1; Puerto Rico	10664
	birth: July; n1; Puerto Rico	5423a
	lact f: Feb, Nov; Puerto Rico	16052a
	lact f: July; n1; Puerto Rico	5423a
Sturnira aratathomasi	preg f: Feb, Aug; n2; Colombia	10785a
Sturnira bidens	preg f: June-Aug, Oct; n15; Peru	3999a
	preg f: Aug; n3; Peru	3608a
Sturnira bogotensis	preg f: June, Aug-Nov; n22; Peru	3999a
Sturnira erythromos	preg f: June-Oct, Dec; n47; Peru	3999a
	preg f: Aug; n10; Peru	3608a
	preg f: Dec; n2; Colombia	10785
Sturnira lilium	m repro: May, Aug, Nov; Rio de Janeiro, Brazil	8368
	preg f: Jan; n1; Querétaro, Mexico	10261a
	preg f: Jan, Mar-Apr, June-July, Sept; Jalisco, Mexico	11471a
	preg f: Jan, July; n6; Campeche, Mexico	5424f
	preg f: peak Feb-Mar, July-Sept, Nov-Dec; Mexico	4670, 4671
	preg f: Feb-Mar, June-Aug; Guatemala	5405e
	preg f: Mar-Apr, Aug; n24; Lesser Antilles	5424b
	preg f: Apr; n1; Sinaloa, Mexico	1939
	preg f: Apr; n2; Quintana Roo, Mexico	950
	preg f: Apr; n1; Oaxaca, Mexico	11754
	preg f: May; n1; Guatemala	9113
	preg f: May-June, Aug; n4; Sinaloa, Mexico	5413a
	preg f: June; n1; Nicaragua	5405c
	preg f: June or July; n3; Venezuela	8562a
	preg f: June-Aug; French Guiana	1350a
	preg f: June-Dec; n32; Peru	3999a
	preg f: July; n1; Durango, Mexico	523a
	preg f: July; n4; Veracruz, Mexico	6148a
	preg f: July; n1; Nicaragua	10313a
	preg f: July; n1; Brazil	11754
	preg f: July-Aug; n10; Colombia	352a
	preg f: July-Aug; n2; Peru	10974
	preg f: Aug; n1; Quintana Roo, Mexico	5424f
	preg f: Sept; n1; Sonora, Mexico	3254b
	preg f: Oct-Jan, Mar-Apr; n22; Colombia	10875
	preg f: Dec-Feb, May-June; n12; Costa Rica	3322
	preg/lact f: July; Hidalgo, Mexico	1640a
	lact f: Jan-May, July-Aug; n24; Costa Rica	3322
	lact f: Feb, Apr-Oct, Dec; n19; Colombia	10875
	lact f: Apr, June-Oct; Jalisco, Mexico	11471a
	lact f: May; n2; Guatemala	9113
	lact f: May-June; n5; Chiapas, Mexico	11754
	lact f: June; n3; Durango, Mexico	5405b
	lact f: July; n1; Durango, Mexico	523a
	lact f: July; n1; El Salvador	10313a
	lact f: July; n1; Nicaragua	10313a
	lact f: Aug; Rio de Janeiro, Brazil	8368
	lact f: Dec; n1; Oaxaca, Mexico	9542a
Sturnira ludovici	preg f: Feb-May, Aug, Oct-Dec; n38; Colombia	10785
	preg f: Apr, July; n8; Jalisco, Mexico	11471a
	preg f: Apr, Aug; Monte Verde, Costa Rica	6261
	preg f: Aug; n1; Chipas, Mexico	11754
	preg f: Nov; n5; Jalisco, Mexico	5424c
	lact f: Apr-Nov, peaks May, Sept-Oct; Monteverdi, Costa Rica	2587
	lact f: May-Sept; n13; Colombia	19785
	lact f: May, Aug; n3; Jalisco, Mexico	11471a
Sturnira magna	preg f: Nov; n3; Colombia	10653
	preg f: Aug, Oct, Dec; n6; Peru	3999a
	preg f: Dec; n1; Peru	10653
	lact f: May; n1; Peru	3607a
Sturnira mordax	preg f: Feb; n2; Costa Rica	3608
	preg f: Aug; n1; Costa Rica	366b
	lact f: May; n1; Costa Rica	3608

	lact f: May; Costa Rica	6260
Sturnira nana	preg f: Aug; n1; Peru	3999a
Sturnira thomasi	lact f: July; n2; Guadeloupe, Lesser Antilles	527a, 3705a
Sturnira tildae	preg f: Mar; n1; Trinidad	3900a
	preg f: June, Aug, Nov; Peru	3999a
	preg f: July; n1; Brazil	11754
	preg f: Aug; n5; Trinidad	1635
Tonatia bidens	preg f: Jan; n1; Costa Rica	3608
	preg f: Feb; n1; Guatemala	1641
	preg f: May; n2; Trinidad	3900a
	preg f: July; n2; Peru	3607a
	preg f: Aug; n2; Honduras	11055
Tonatia brasiliense	preg f: Feb; n1; Panama	2345
	preg f: Feb, Apr; n2; Costa Rica	6260
	preg f: July; n1; Nicaragua	11055
	lact f: Aug; n1; Honduras	11055
Tonatia evotis	preg f: Jan; Chiapas, Mexico	7154
	preg f: Mar; Belize	7154
	preg f: Apr; Quintana Roo, Mexico	7154
	lact f: Apr; n2; Quintana Roo, Mexico	9457a
Tonatia silvicola	preg f: Jan; n1; Colombia	10785
	preg f: Mar; n2; Panama	3322
	preg f: July; n1; Paraguay	7699a
	preg f: July-Oct; n10; Peru	3999a, 10974
	lact f: Mar; n1; Paraguay	7699a
Trachops cirrhosus	preg f: Feb; n3; El Salvador	1475
	preg f: Mar; n1; Oaxaca, Mexico	11754
	preg f: Mar; n1; Chiapas, Mexico	1641
	preg f: Mar; n2; Trinidad	3900a
	preg f: Mar-Apr; n10; Guatemala	5405e
	preg f: May; n4; Nicaragua	1641
	preg f: July; n1; Peru	10974
	preg f: Aug; n1; Panama	3322
	preg f: Dec; n1; Chiapas, Mexico	11754
	lact f: Mar; n1; Costa Rica	1641
	lact f: Aug; n1; Costa Rica	366b
	lact f: Aug; n1; Honduras	11055
Uroderma bilobatum	preg f: Jan; n4; El Salvador	3207
	preg f: Jan; n8; Panama	2342
	preg f: Jan-July; n54; Panama	3322
	preg f: Jan, May; n4; El Salvador	1475
	preg f: Jan, July, Sept, Nov; n6; Colombia	10660
	preg f: Feb; n1; Guatemala	5405e
	preg f: Feb; n4; Nicaraqua	5405c
	preg f: Feb-Apr; Panama	7983, 11754
	preg f: May; n3; Trinidad	3900a
	preg f: June-Jan; n28+; Peru	3999a
	preg f: July; n1; Veracruz, Mexico	6148a
	preg f: July; n72; El Salvador, Honduras, Nicaragua	526a
	preg f: July; n3; Brazil	11754
	preg f: Aug; n1; Peru	10974
	preg/lact f: Apr-May, 3.3% preg, 4.9% lact; n123; July-Aug, 1.8% preg, 14.0% lact; n171; Sept, 20% preg, 0% lact; n80; Guatemala	2552
	birth: yr rd; Costa Rica to Peru, s Brazil	1346
	lact f: Feb, May; n4; Trinidad	3900a
	lact f: Mar; Panama	1006
	lact f: Mar; n1; Colombia	10660
	lact f: Mar-May, July-Aug; n39; Panama	3322
	lact f: Apr; n1; Quintana Roo, Mexico	9457a
	lact f: May; Chiapas, Mexico	11754
	lact f: June or July; n1; Venezuela	8562a
	lact f: July; n1; El Salvador, Honduras, or Nicaragua	526a
	lact f: Aug; n2; Nicaragua	2345

Uroderma magnirostrum	preg f: Mar; n1; Nicaragua	5425
	preg f: June; n1; El Salvador	2342
	preg f: June; n1; Brazil	11754
	preg f: July; n1; Nicaragua	2342
	preg f: Sept; n10; Bolivia	2342
Vampyressa bidens	preg f: Aug; n1; Suriname	3705c
	preg f: Aug, Oct-Nov; n27; Peru	3999a
	preg f: Dec; n2; Peru	2343a
Vampyressa melissa	preg f: Aug, Nov; n4; Peru	3999a
Vampyressa nymphaea	preg f: Feb; n1; Nicaragua	5425
	preg f: Apr; n2; Costa Rica	3608
	preg f: Oct-Aug; n111; Colombia	10785
	lact f: Jan-Aug; n96; Colombia	10785
Vampyressa pusilla	preg f: Jan, Apr; n2; Panama	3322
	preg f: Feb; n1; Campeche, Mexico	5424f
	preg f: Feb; n2; Costa Rica	6809
	preg f: Mar; n1; Belize	8412
	preg f: Mar; n4; Nicaragua	5425
	preg f: Mar-May, Aug-Nov; n6; Colombia	10785
	preg f: June; n1; Paraguay	7700a
	preg f: July; n1; Chiapas, Mexico	2345
	preg f: July; n1; Guatemala	9113
	preg f: July; n1; Nicaragua	10313a
	preg f: Aug; n1; Honduras	11055
	preg f: Aug; n1; Colombia	352a
	preg f: Oct-Nov; Peru	2345a
	preg f: Nov-Dec; n15; Peru	3999a
	lact f: Mar-Apr; n3; Panama	3322
	lact f: Apr; n1; Panama	4228a
	lact f: July; n1; Colombia	10785
	lact f: July-Aug; n2; Costa Rica	366a
	lact f: Aug; n3; Colombia	352a
Vampyrodes caraccioli	preg f: Jan; n2; Panama	3322
	preg f: Jan-Apr, July, Oct-Nov; n14; Colombia	10785
	preg f: June; n1; Chiapas, Mexico	5405d
	preg f: July; n1; Chiapas, Mexico	2345
	preg f: July, Nov; n4; Peru	3999a, 10974
	preg f: July-Aug; n3; Nicaragua	5425
	preg f: end July-early Aug; n14; Honduras	11055
	preg f: Oct-Nov; Peru	2345a
	lact f: Jan-June, Aug; n14; Colombia	10785
	lact f: May; n1; Honduras	2345
Vampyrops brachycephalus	preg f: Feb; n1; Venezuela	9318a
	preg f: July-Nov; n12; Peru	3999a
	preg f: Aug; n2; Peru	9318a
	lact f: Oct; n1; Venezuela	9318a
Vampyrops dorsalis	preg f: June-Nov; n33; Peru	3999a
	preg f: Nov-July; n45; Colombia	10785
	lact f: Feb-Mar, May-July, Oct; n18; Colombia	10785
	lact f: Aug; n3; Columbia	352a
Vampyrops helleri	preg f: Mar; n1; Costa Rica	6809
	preg f: Mar-Apr, June-Aug; Nicaragua	5425
	preg f: Apr, July, Dec-Jan; n4; Panama	3322
	preg f: June; n2; El Salvador	6258
	preg f: June; n1; Nicaragua	5405c
	preg f: July; n1; Chiapas, Mexico	2345
	preg f: July-Dec; n15; Peru	3999a, 10974
	preg f: Aug; n1; Honduras	6258
	preg f: Aug; n14; Trinidad	1635
	preg f: Aug; n2; Colombia	352a
	preg f: Aug-Sept; French Guiana	1350a
	preg f: Oct-June; n35; Colombia	10785
	lact f: Apr, Sept; n2; Panama	3322
	lact f: May; n1; Tabasco, Mexico	11754

	lact f: May; n1; Chiapas, Mexico	11754
	lact f: May; n1; Guatemala	9113
	lact f: Aug; n2; Colombia	352a
	lact f: Dec-Jan, Apr-May; n29; Colombia	10785
Vampyrops infuscus	preg f: Mar; n1; Colombia	6814a
	preg f: July-Aug, Nov; n6; Peru	3999a
	lact f: Mar; n3; Colombia	6814a
Vampyrops lineatus	epididymal sperm: yr rd; n39; São Paulo, Brazil	10618
	preg f: Jan, Mar, Dec; Brazil	8368
	preg f: May-Mar; São Paulo, Brazil	11737
	preg f: July-Aug; n25; Peru	3999a
	preg f: July-Mar; Brazil	11736, 11737
	lact f: Nov-May; São Paulo, Brazil	11737
Vampyrops vittatus	preg f: Mar; Costa Rica	6260
	preg f: Apr; n2; Costa Rica	2345
	preg f: May, Oct; n2; Colombia	10785
	preg f: June; n1; Costa Rica	3608
	preg f: Aug; n4; Peru	3999a
	lact f: May-July; n6; Costa Rica	3608
Vampyrum spectrum	birth: June; n1; captive	4048b
	birth: July; n1; captive	2109
	lact f: May; n1; Trinidad	3900a

Order Chiroptera, Family Natalidae, Genus *Natalus*

Funnel-eared bats (*Natalus* 5-10 g) are insectivores from the neotropics of North and Central America. Females have a bicornuate uterus (3255) and are monoestrus (7375). Embryo development is slow and the left uterine cornu is used preferentially (7375). Males have seminal vesicles and a baculum (6065). Neonates are over 50% of adult mass (7375).

NEONATAL MASS		
Natalus stramineus	1.45-1.75 g; term emb	7375
	2.1 g; r1.7-2.6	7375
LITTER SIZE		
Natalus stramineus	1 emb	7375
	1 emb; n1 f	527a
	1 emb; n1 f	950
	1 emb; n1 f	1475
	1 emb ea; n8 f	11471a
	1 emb ea; n16 f	1937
	1 emb ea; n25 f	5424e
GESTATION		
Natalus stramineus	8-10 mo; slow growth	7375
SEASONALITY		
Natalus stramineus	sperm: Sept-Mar; n hemisphere	6065
	preg f: Jan; n2; none May, Aug, Dec; El Salvador	3208
	preg f: Feb-Mar; n10; Michoacan, Mexico	4671
	preg f: Apr; n1; El Salvador	1475
	preg f: May-June; n16; Mexico	1937
	preg f: May, July; n25; Baja CA, Mexico	5424e
	preg f: June; n1; Sinaloa, Mexico	5413a
	preg f: June; n8; Jalisco, Mexico	11471a
	preg f: July; n1; Guadeloupe, Lesser Antilles	527a
	implant: Dec-early Jan; Sonora, Mexico	7375
	birth: end July-Aug; Sonora, Mexico	7375

Order Chiroptera, Family Furipteridae

The Central and South American furipterid bats, *Amorphochilus schnablii* and *Furipterus horrens* (3.3 g), are insectivorous forest dwellers. Females are larger than males (5125, 11026). Two abdominal mammary glands are a familial distinction (5125, 11026). Bacula are present (5125).

LITTER SIZE		
Amorphochilus schnablii	1 emb ea; n8 f	5125
	1 emb ea; n16 f	3999a
Furipterus horrens	1 yg	11026
SEASONALITY		
Amorphochilus schnablii	preg f: June, Aug, Oct; n16; Peru	3999a

Order Chiroptera, Family Thyropteridae, Genus *Thyroptera*

Thyroptera tricolor and *T. discifera* are small (3-4 g) bats from tropical Central and South America (11753, 11757). The labyrinthine, hemodichorial placenta has an interstitial implantation into a bicornuate uterus (11805, 11806). A large corpus makes the uterus appear simplex (3255). Both sexes have a pair of axillary nipples (11753). The testes are partially inguinal and partially abdominal, and the penis lacks a baculum (11753).

NEONATAL MASS		
Thyroptera tricolor	0.98-1.2 g; near term	11806
NEONATAL SIZE		
Thyroptera tricolor	CR: 18 mm; near term	11806
LITTER SIZE		
Thyroptera tricolor	1 emb; n1 f	217a
	1 emb ea; n3 f	3255
	1 emb ea; n3 f	3999a
	1 cl, emb; n8 f	11805, 11806
GESTATION		
Thyroptera tricolor	~5 mo	11805, 11806
SEASONALITY		
Thyroptera tricolor	preg f: Jan-May; n8; Veracruz, Mexico	11805, 11806
	preg f: Feb-Mar; French Guiana	1350a
	preg f: May, June, Dec; Colombia	11806
	preg f: July; n3; Peru	3999a
	preg f: Aug; Costa Rica	3255
	lact f: May; Panama	4228a

Order Chiroptera, Family Myzopodidae, Genus *Myzopoda*

The rainforests of Madagascar have yielded only a few specimens of and no reproductive information on *Myzopoda aurita* (~57 mm). The penis is short and lacks a baculum (10784). A short review of its natural history is available (9599).

There are no tabular data for *Myzopoda*.

Order Chiroptera, Family Vespertilionidae

Vespertilionids (*Antrozous* 14-39 g; *Barbastella* 6-13 g; *Chalinolobus* 6-15 g; *Eptesicus* 4-40 g; *Euderma* 14-20 g; *Eudiscopus* 40-45 mm; *Glischropus* 3.5-4.5 g; *Harpiocephalus* 55-75 mm; *Hesperoptenus* 6.1-32 g; *Histiotus* 54-70 mm; *Ia* 89-104 mm; *Idionycteris* 8-20 g; *Kerivoula* 4-10 g; *Laephotis* 4-8 g; *Lasionycteris* 6-14 g; *Lasiurus* 6-35 g; *Mimetillus* 6-11 g; *Miniopterus* 4-20 g; *Murina* 3-12 g; *Myotis* 2-45 g; *Nyctalus* 11-55 g; *Nycticeius* 4-20 g; *Nyctophilus* 7-18 g; *Otonycteris* 18-20 g; *Pharotis* 50 mm; *Philetor* 9-13 g; *Phoniscus* 9-10 g; *Pipistrellus* 2-20 g; *Plecotus* 5-20 g; *Rhogeessa* 3-10 g; *Scotoecus* 4-5 g; *Scotomanes* 72-78 mm; *Scotophilus* 15-40 g; *Tomopeas* 2-4 g; *Tylonycteris* 3-11 g; *Vespertilio* 14-33 g) are insectivores with a worldwide distribution.

Females are often larger than males (217, 2045, 3216, 3944, 4231, 5405b$_1$, 5759, 6717, 8494, 8965, 9297, 9945, 9989, 11270, 11714; cf 191, 8127, 11759). *Lasiurus* exhibits sexual dichromatism (178a).

Male *Glischropus* have an extra chromosome (11339). *Miniopterus* exhibits extensive reproductive asymmetry: only one ovary, either left or right, is functional; one uterine horn is often much longer than the other; and implantation occurs only on the right side (820, 823, 2075, 3945, 6966, 7163, 7518, 9102, 11015, 11128, 11131, 11133, 11134, 11418). Other vespertilionids also exhibit uterine asymmetry (*Chalinolobus* 12042; *Eptesicus* 5834; *Myotis* 825, 1417, 1524, 2947, 3517, 4177a, 6070, 6106, 7695, 7950, 8843, 9645, 10070, 11318; *Nyctalis* 11318; *Pipistrellus* 1524, 6064; *Vespertilio* 2824, 9240).

The endotheliochorial or hemochorial, labyrinthine, discoid placenta has accessory placentae and an antimesometrial, superficial or interstitial implantation (1228, 1735, 2163, 3020, 3730, 3908, 3909, 3923, 3943, 4402, 5530, 5531, 5648, 5728, 6740, 7563, 8879, 9178, 11133, 11145a, 11318, 11793, 11796, 11799) into a bipartite or bicornuate uterus (820, 2824, 3020, 3730, 3907, 3909, 3917, 3939, 3943, 3944, 3945, 5530, 5531, 6064, 6966, 7759, 7950, 8695, 8839, 8843, 8879, 8883, 9002, 9102, 10070, 11094, 11131, 11133, 11134, 11418, 11793; cf 1735, 9178). Placentophagia occurs (1043, 2945, 2948, 5386, 5939, 7610, 8127, 9398, 9895, 11270, 12042; cf 9306). Breech births are common (1043, 4038, 12042).

Monoestry (6028, 6064, 9102, 11137), postpartum estrus (4038, 5830, 6026, 6716, 7695, 11759), vaginal plugs (5833, 6025; cf 2420), sperm storage (825, 1524, 2134, 2135, 2420, 3933, 3935, 4102, 4449, 4819, 4820, 5860, 6025, 6026, 6027, 6028, 6064, 6068, 6070, 7164, 7512, 7513, 7515, 7695, 8198, 8828, 8843, 8998, 9240, 9679, 11095, 11138, 11139, 11792, 11792a, 11811), delayed implantation (822, 5727, 8463, 8464, 9102, 11128, 11129, 11132, 11418), and delayed development (2947, 3564, 8826, 8840, 11017, 11145b) occur.

Most neonates are naked and blind (577, 1344, 1474, 2228, 2593, 2828, 2842, 2948, 3973, 5024, 5759, 6175, 6714, 6715, 6717, 7163, 7164, 8049, 8314, 8492, 8493, 8494, 10158, 11132, 12042; cf 3944, 6000, 6106, 6722, 8049, 8658, 11270). Neonates open their eyes at 2-10 d (*Antrozous* 7523, 2335, 8127; *Myotis* 1474, 2945, 3571, 6000, 7424, 7523, 8798, 9895; cf 3216, 3973; *Pipistrellus* 3932, 5845, 7523, 8798, 8865, 9175, 11622), 4-11 d (*Nyctalus* 2945, 3571, 5845, 6722, 7523, 8798; *Vespertilio* 3563, 9240), 1 wk (*Eptesicus* 5759, 5845, 8798; *Miniopterus* 3571), or 8-10 d (*Plecotus* 1474). *Lasiurus* neonates have fine hair at birth and open their eyes at 10-12 d (577, 1043, 5210, 7523, 7619, 8656), while *Nycticeius* neonates have a few hairs and open their eyes within hours of birth (577, 5386, 7523, 9398, 11470). Litter mass at birth may reach 40% of maternal mass (6107, 10295).

Females usually have 2 mammae (577, 1474, 3563, 3571, 3944, 6720, 8314, 9175), although *Lasiurus* has 4 (3900a, 8656) and *Otonycteris* may have multiple nipples per gland (6720). A baculum, ampullary and urethral glands are present in the males of most species (5704, 6064, 6068, 6957, 6966, 7439, 7440, 7441, 7664, 9178). *Miniopterus* and *Scotophilus* males have postanal scrota and *Miniopterus* has no baculum (191, 3906, 6966, 9102).

Details of female anatomy (347, 759, 820, 821, 823, 1416, 3609, 3907, 3945, 4176, 4177, 4177a, 4946, 7695, 8175, 9002, 10070, 11093, 11130, 11132, 11148, 11794, 11810, 11812), male anatomy (831a, 2069, 3906, 3907, 4161, 6069a, 6957, 8054, 8208, 8554, 11809), endocrinology (343, 344, 1417, 1418, 1419, 1555, 2074, 2185, 2237, 2238, 2239, 2240, 2241, 2242, 3372, 4163, 4164, 4167, 4177, 4662, 4666, 5729, 6138, 8176, 8465, 10051, 10107, 11143), ovulation (1414, 5860, 7517, 8831, 10050, 10069, 11014), sperm/spermatogenesis (1272, 2073, 3523, 4661, 4826a, 4881, 5729, 6062a, 8829), mating/conception (3940, 5210, 7514, 7516, 10523b, 11015), birth (3932, 6122a, 8127, 9895, 11470, 11795, 11801), milk composition (5296, 6114, 10296), reproductive energetics (340, 685, 2099, 6121, 6122, 6140, 6141, 8841, 10520, 10521, 10522, 10604), and light (683, 686, 688, 2237, 8833) are available.

NEONATAL MASS

Antrozous pallidus	3.1 g; r2.9-3.3; n2	8127
	m: 3.0 g; se0.13; r2.4-3.5; n9	2335
	f: 3.2 g; se0.19; r2.4-4.4; n11	2335
Chalinolobus morio	1.7 g; r1.5-2.0; twins: 1.5 g ea; n4	12042
Eptesicus capensis	2 g; n1 emb	11273
Eptesicus fuscus	2.5 g; r2.5-2.6; n5	1829
	2.85 g; r2.75-2.95; term emb	8485
	3.14 g; r2.7-3.6; n5	6107
	3.3 g; n16	1455
	4 g; r3.4-4.5; n10	2354

	f: 7.8 g	11627
Eptesicus serotinus	4.81 g; n1	2945
	5.8 g; r5.2-6.2; n4	5845
Eptesicus tenuipinnis	a little bigger than a bee	9297
Euderma maculatum	2.9 g; r2.6-3.25; n2	2866
	4 g; n1	2865
Lasionycteris noctivagans	m: 1.1 g; n1	7235
	f: 1.24 g; r1.10-1.38; n2	7235
	1.85 g; r1.8-1.9; n2	6105a, 6109
	2.5 g ea with placenta; n2	6122a
Lasiurus borealis	3.175 g; n4 prob not neonates	10298
	5.8 g; r5.3-6.8; n4 probably not neonates	10298
Lasiurus cinereus	nearly 5 g	5210
	6 g	1043
Lasiurus intermedius	3 g	577
Miniopterus schreibersi	2.416+ g; 237+ d emb	11132
	2.744 g; sd0.142; n2, 240+ d emb	11132
	est 2.5-3.0 g; smallest juv: 2.6-3.4 g	2828
	2.5 g; se0.2; r2.1-2.8; n18 neonates	11129
	2.7 g; se0.1; r2.4-3.1; n25 term emb	11129
	3 g; se0.2	3945
	3.4 g	6115
Myotis austroriparius	1.12 g; r1.10-1.15; n2	9895
Myotis daubentoni	f: 2.09 g; n1	2945
Myotis evotis	r1.08-1.36 g; n5	6780
Myotis grisescens	2.9 g; sd0.24; r2.4-3.4; n94	10975
Myotis lucifugus	1.36 g; n6 term emb	3254
	1.50 g; r1.45-1.55; n2	7424
	1.5-1.9 g	11795
	2.4 g; r2.0-2.8; n3	1526
	2.08 g; sd0.18; n19	10106
	2.17 g; n3 term emb	2842
	2.2 g; sd0.14; n36 with umbilicus	6111
	2.3 g; sd0.26; n121 neonates	1455, 6111
Myotis macrodactylus	0.6-2.3 g; unknown age emb	7522
	1.5 g	6115
Myotis myotis	5-6 g	9596
	6.1 g	6000
	6.2 g; r5.6-7.6; n28	2402, 2403
Myotis nigricans	1.0 g; n4	11752
Myotis thysanodes	22% of adult mass	8005
Myotis velifer	3.2 g; se0.03; n89, 1-2 d	6106
Nyctalus lasiopterus	r3.4-6.4 g; n8 twins	6722
	6.9 g; r5.5-8.3; n2 singletons	6722
Nyctalus noctula	4.0-5.5 g	2593
	m: 5.65 g; n1	2945
	5.7 g; r4.7-6.7; n10	5845, 5860
Nycticeius balstoni	f: 2.4 g; n1	9398
Nycticeius humeralis	1.4-1.6 g	3629
	2.0 g; n11	5386
Pipistrellus ceylonicus	1.2 g; term emb	3915, 3932, 6715
	1.27 g; r~1.2-1.4; neonate	3915, 3932, 6715
Pipistrellus javanicus	0.88 g; r0.9-1.1; n15 term emb, neonates	7521
	1.1 g	11016
Pipistrellus mimus	0.5 g	3944
Pipistrellus pipistrellus	1 g	2420
	~1 g	8827
	1.0 g; se0.16; r0.6-1.4; n18	8865
	1.027 g ea; twins	1179
	1.323 g; singletons	1179
	1.4 g; r1.3-1.5; n9	5845
Pipistrellus subflavus	0.94 g; r0.93-0.96; n2	6175
	52% of maternal postpartum mass	3549
Plecotus rafinesquii	2.3-2.6 g	5397

Plecotus townsendii	2.4 g; r2.1-2.7; n10	8314
Scotophilus heathi	5.0 g	6028
	5.5-6.0 g	6717
Tylonycteris pachypus	f: 0.6 g; n1	7164
	0.7 g; term emb	7164
Tylonycteris robustula	1.25 g; n2 term emb	7164
Vespertilio superans	2 g	3563
NEONATAL SIZE		
Antrozous pallidus	m: FA: 17.4 mm; se0.38; r16-20; n9	2335
	FA: 17.5 mm; r17-18; n2	8127
	45.5 mm; r45-46; n2	8127
Chalinolobus argentatus	15.1 mm; r11-20; n2	71
Chalinolobus gouldii	CR: 21.7 mm; term emb	5830
Chalinolobus morio	CR: ~24 mm	5833
Chalinolobus tuberculatus	m: CR: 13.2 mm; r11.9-14.6; n2 emb	2259
Eptesicus fuscus	FA: 16.1 mm; r15.1-17.0; n5	6107
	FA: 16.8 mm; n16	1455
Eptesicus regulus	f: CR: 18 mm; largest emb	5834
Euderma maculatum	HB: 39 mm; FA: 21 mm	2865
Lasionycteris noctivagans	TL: 45.8 mm; HB: 32.3; FA: 12.5; n1	6105a
	TL: 45.9 mm; r44.7-47.1; FA: 14.1 mm; r14.0-14.2; n2	6122a
Lasiurus borealis	TL: 43 mm; r41-44; n3 large emb	9961
Lasiurus cinereus	m: HB: 40.5 mm; n1; FA: 18.5 mm	7619
	46 mm ea; n2, 2 d	7847
	FA: 18-19 mm	577
Lasiurus intermedius	FA: ~16 mm	577
Myotis evotis	TL: r40-47 mm; n5	6780
Myotis keenii	CR: 50 mm	2864
Myotis lucifugus	HB: 30 mm	2842
	FA: 15.8 mm; sd0.78; n36 with umbilicus	6111
	FA: 16.1 mm; sd0.76; n121 neonates	1455, 6111
Myotis myotis	46.6 mm	6000
Nyctalus noctula	65 mm; n1	2593
Nyctophilus geoffroyi	35 mm; r34-36; n2	7074
Nycticeius balstoni	f: 31.3 mm; n1	9398
Pipistrellus javanicus	HB: 23.6 mm; r19.5-25.5; n15 neonates, term emb	7521
Pipistrellus pipistrellus	17.5 mm	8827
Pipistrellus stenopterus	37 mm; n1 3 d	4625
Plecotus townsendii	CR: ~24 mm	577
WEANING MASS		
Eptesicus fuscus	17.2 g; r11.4-22.6; n33	6298
	fledge: 75% adult size	1455
Kerivoula phalaena	f found carrying yg 3/4 her size	9297
Lasiurus borealis	m: 6.26 g; r5.6-7.0; n4 juveniles	4260
	f: 7.12 g; r6.4-8.1; n6 juveniles	4260
Lasiurus cinereus	9-10 g; 3 wk	5210
Miniopterus schreibersi	12.5 g; n28	2828
Myotis grisescens	lighter at weaning the farther they have to fly to water	10977
Myotis lucifugus	1st flight: 84% adult size; 22 d	1455
Myotis nigricans	adult mass before weaning	11752
Myotis thysanodes	adult mass at 3 wk	8005
Nyctalus lasiopterus	22-30 g; 30 d; fly 40-45 d	6722
Nycticeius balstoni	5.6 g; 29 d	9398
Nycticeius humeralis	2.6 g; 4 wk	5386
Pipistrellus ceylonicus	4.5-6.5 g; ~1 mo	6715
Pipistrellus mimus	~1.5 g; detach from mother; 2 g; end of lactation	3944
Pipistrellus subflavus	5.2 g; r4.1-7.5; n17	6298
WEANING SIZE		
Myotis nigricans	adult size at weaning	11752

LITTER SIZE

Antrozous dubiaquercus	1 emb; n1 f	8554
Antrozous pallidus	1-2 emb	502
	1-2 emb	4079
	1-2 yg	2335
	1.33 yg; r1-2; n6 f	10455
	1.5 emb; r1-2; n4 f	1471a
	1.5 emb; r1-2; n27 f	502
	1.75 emb; r1-2; n4 f	10455
	1.8 emb, yg; mode 2; r1-3; n28 lt	8127
	1.8; r1-3, 1/61 triplets; n61 lt	639
	1.82 yg; r1-2; n70 lt	640
	2 emb; n1 f	1196a
	2 emb ea; n2 f	522, 523a
	2 emb ea; n2 f	11228a
	3 emb; n1 f	4079
Barbastella barbastellus	1-2	3571
	2	4645
	2	8049
Chalinolobus dwyeri	1 ea; n30 lt	7075
	1.8 yg; n13 f	2829
Chalinolobus gouldii	1 ea; n29 lt	5340
	mode 2; 2-9 ova/ovary; 2-9 cl	5830
Chalinolobus morio	mode 1 yg; 1/32 f with twins	5833
	1.3 yg; r1-2; n3 lt	12042
	1.5 emb; n24 f	12042
Chalinolobus picatus	2; n1 lt	7075
Chalinolobus tuberculatus	1 emb ea; n2 f	2259
Chalinolobus variegatus	1.3 emb; r1-2; n3 f	191
Eptesicus brasiliensis	1 emb ea; n12 f	3999a
	2.2; r2-4; n9	452
Eptesicus capensis	1 emb; n1 f	11273
	1.6; mode 1-2; r1-3; 3 rare	10158
	2 emb; several f	9945
	2.25 emb; n4 f	8965
Eptesicus furinalis	1 implantation ea; n4 f	9113
	2 emb; n1 f	4671
	2 emb; n1 f	5425
	2 emb ea; n5 f	950
	2.0 emb; mode 2 1st lt, mode 1 2nd lt; r1-4; n24 f	7695
Eptesicus fuscus	1 yg	502
	1-2 yg	1279
	1-2 yg; 3-7 ova shed	11795
	mode 1; but twins observed	6059
	1 emb; n1 f	1649a
	1 emb; n1 f	11471a
	1 emb ea; n2 f	5424_2
	1 emb ea; n4 f	1119
	1 yg ea; n~36 f	5027
	1+; r1-2	10954
	1.13 emb; r1-2; n115 f	9643
	1.25 emb; mode 1; r1-2; n4 f	3703
	1.27 emb; mode 1; r1-2; n15 f	2197, $5405b_1$
	1.75 emb; n8 f	5405e
	1.75 yg; n51 f	2354
	1.89 emb; mode 2; r1-2; n54 f	11627
	1.9 emb; r1-2; n11 f	950a
	1.9 emb; r1-2; n11 late gestation	6107
	1.95 yg; r1-2; n21 f	1850
	mode 2 yg 83%	1829
	2-7 cl	11794
	2 emb; n1 f	5415
	2 emb ea; n2 f	10954a
	2 ps ea; n2 f	11956

	2.1 yg; n11 f	3628
	3.0 emb; r2-5; n10 early gestation	6107
	4 emb; n1 f	4387
Eptesicus melckorum	2 emb ea; n2 f	10158
Eptesicus nilssoni	1; r1-2	3571
	r1-2 yg; n35-40 f	5102
	2	8049
Eptesicus pumilus	1	4038
	1-2	7075
	1.17 yg; r1-2; n6 lt	6713
	1.21; r1-2; n28 f	6714
	2 yg	7074
Eptesicus regulus	1 yg	5834
Eptesicus serotinus	1 yg	8049
	1-2	3571
	2	4645
Eptesicus somalicus	2; n1 lt	9219
Euderma maculatum	1 yg	577
Hesperoptenus tickelli	mode 1	8493, 8494, 8501
Histiotus macrotus	1 emb ea; n3 f	3999a
Idionycteris phyllotis	1 yg	577
	1 emb; n1 f	969
	1 emb; n1 f	3607$_1$
	1 emb; n1 f	5396
Kerivoula cuprosa	1 emb	252
Kerivoula lanosa	1 emb; n1 f	1347
Kerivoula papillosa	1 emb ea; n2 f	6447
Kerivoula picta	1	8494, 8501
Kerivoula smithi	1 emb; n1 f	252
Lasionycteris noctivagans	1 ps; n1 f	11956
	1+; r1-2; n11	10954
	1.78 emb; r1-2; n9 f	2868
	1.84 emb; n36 f	2733
	1.86 yg; r1-2; n7 f	8249
	2 emb; n1 f	261a
	2 emb; n1 f	4525
	2 emb; n1 f	10954a
	2 yg; n1 f	6122a
	2 emb ea; n4 f	495
	mode 2; r1-2	502
Lasiurus borealis	r1-4; 5 cases of 3; 3 cases of 4	181
	2-4 yg	4222
	2 emb; n1 f	1196a
	2 emb; n1 f	4525
	2 emb; n1 f	5396
	2 yg; n1 f	4054
	2 yg; n1 f	6105a
	2 ea; n2 lt	452
	2.67; mode 3; r2-4; n12 lt	11450
	2.78 yg; r1-4; n9 f	950a
	2.8 emb; mode 3; r2-3; n5 f	4079
	2.9 emb; mode 3; r2-3; n12 f	1290
	2.96 yg; r2-4; n27 f	5414a
	3 emb; n1 f	1973c
	3 emb; n1 f	4386
	3 yg; n1 lt	5405b$_1$
	3.0 emb; r2-4; n2 f	589a
	3.0; r2-4; n4 lt	6677
	3.17 emb; mode 3; r2-5; n12 f	11627
	3.17 emb; r2-4; n18 f	1937
	3.24 emb; r2-4; n29 f	5414a
	3.35 emb; r2-5; n34 f	7608, 7610, 7611
	3.38 emb; r3-4; n13 f	6263
	3.45; r3-4; n11 f	6298

	3.5; r3-4; n4 f	7557
	3.8 yg; r3-4; n6 f	2197
	4 emb; n1 f	1871a
	4 emb; n1 f	2197, 5405b$_1$
	4 yg; n1 f	1812a
	4 yg; n1 f	8187
	4 yg ea; n2 f	10298
Lasiurus cinereus	1+; r1-2	10954
	1.83; r1-2; n6 lt	1043
	1.9 yg; r1-2; n9 f	5414a
	2	181
	2-4 yg	502
	2 emb; n1 f	1196a
	2 emb; n1 f	10954a
	2 emb; n1 f	11244
	2 yg; n1 f	1812a
	2 yg; n1 f	7619
	2 yg; n1 f	7847
	2 yg; n1 f	8656, 8658
	2 ea; n2 lt	2197
	2 emb ea; n2 f	11627
	2 yg ea; n2 f	4222
	2 emb ea; n3 f	5414a
	2 emb ea; n4 f	1043
	2 yg ea; n4 f	9711
	2 emb ea; n8 f	8764
	max 4	2088
Lasiurus ega	1.5 emb; r1-2; n2 f	950
	2; n1 lt	190
	2 emb; n3 f	577
	2 emb ea; n3 f	5424e
	2.3; r1-3; n15	452
	2.9 emb; mode 3; r2-4; n17 f	7695
	3.0 emb; mode 3; r2-4; n7 f	528
Lasiurus intermedius	2 emb; n1 f	5413a
	2.0 emb; r1-3; n2 f	950
	3 emb; n1 f	6257
	3 yg ea; n2 f	9896a
	3.4 emb; r3-4; n13 f; est 2.8 bom	577
Lasiurus seminolus	2-4	1474
	2 emb; n1 f	1973c
	2.5 emb; r2-3; n2 f	589a
	3.0 emb; r2-4; n2 f	7498a
	3.3 emb; mode 3-4; r1-4; n21 f	577
Mimetillus moloneyi	1 emb; n1 f	1346
Miniopterus australis	1	7163
	1	9102
	1; n many lt	7075
Miniopterus inflatus	1 emb	252
	1 emb ea; n3 f	10158
Miniopterus medius	1 yg	9457
Miniopterus mullatus?	1 yg	1351
Miniopterus schreibersi	1 emb	8965
	1 emb; n1 f	8049
	1 mode; 1/150 f with twins	10158
	1 emb ea; n11 f	4402
	1 emb ea; n40 f	8464
	1.0 emb; r1-2; n312 f; <1% twins	11128, 11131, 11134, 11135
	1 emb ea; n447 f	3927
Murina cyclotis	2 emb; n1 f	7161
Murina tubinaris	1.4 emb; r1-2; n5 f	4813b
Myotis adversus	1 ea; n22 lt	7075
Myotis albescens	1 emb; n10 f	10974
	1 emb ea; n11 f	3999a

Myotis auriculus	1 yg	577, 11465
Myotis austroriparius	1.18 wean; n1523 f; 1st flight	9089
	1.72; r1-2; n29 f	9895
	1.72 yg; n112 f	9089
	1.90 emb; r1-2; n1489 f	9089
	mode 2 yg	3428, 5096a
	3 emb; n1 f	3428
Myotis bechsteini	1	3571
Myotis blythii	1 emb ea; n5 f	5988
Myotis bocagei	1 emb; n1 f	8611
	1 ea; n2 lt	191
	1 emb ea; n3 f	1347
Myotis californicus	1	6060
	1 emb; n1 f	5413a
	1 emb; n1 f	11228a
	1 yg; n1 f	7611
	1 yg; n1 f	12104a
	1 emb ea; n3 f	3607_1
	1 emb ea; n3 f	7611
	1 emb ea; n6 f	1937
Myotis capaccinii	1 emb; 1 f	5988
Myotis chiloensis	1 emb ea; n3 f	3608
Myotis daubentoni	1	3571
	1 yg; n1 lt	11622
	1-2 emb	8049
Myotis elegans	1 emb; n1 f	5424f
Myotis emarginatus	1	3571
	1 emb; n1 f	5988
	1 emb; n1 f	8049
Myotis evotis	1 emb; n1 f	1940
	1 emb; n1 f	2415
	1 emb; n1 f	4079
	1 emb ea; n2 f	5415
	1 emb ea; n2 f	5424_2
	1 emb ea; n2 f	11228a
	1 emb ea; n2 f	11646a
	1 emb ea; n2 f	1937
	1 emb ea; n≥3 f	261a
Myotis fortidens	1 emb ea; n4 f	5413a
Myotis frater	1	6723
Myotis grisescens	1 emb; n1 f	6257
Myotis hasseltii	1 emb; n1 f	10872
	1; n1 lt	7161
Myotis keaysi	1 emb ea; n7 f	3999a
Myotis keenii	1 emb ea; n4 f	2197
	1 emb ea; n7 f	3607_1
	1 ea; n23 f	2043
Myotis leibii	1 emb	5415
	1 yg	5921
	1 emb; n1 f	502
	1 emb; n1 f	5424_2
	1 emb; n1 f	8807a
	1 emb ea; n2 f	3703
	1 emb ea; n4 f	10954, 10954a
Myotis lucifugus	1 ps ea; n2 f	10839
	1 emb; n1 f	7607
	1 emb; n1 f	11956
	1 emb; n>1 f	168
	1 emb ea; n5 f	2842
	1 emb ea; n5 f	5210
	1 emb ea; n14 f	261a
	1.01 emb; r1-2; n312 f	9645
	2	7651
	2 yg; n1 f	2352

Myotis macrodactylus	1	6115
Myotis muricola	1	179
Myotis myotis	1	8049
	1	9220
	1; always 1 emb	436
	1; no twins	3517
	1 emb ea; n74 f	11318
	1; r1-2	3571
	mode 1; r1-2; 2 rare	10045
	2	4645
Myotis mystacinus	1	3571
	1 emb ea; n8 f	10516
	1-2	4645
Myotis nattereri	1 yg	8049
	1	2045
	1 emb ea; n8 f	5988
Myotis nigricans	1	7695
	1 emb ea; n2 f	10974
	1 emb ea; n2 f	5425
	1 ea; n9 lt	452
	1 emb ea; n58 f	3999a
	1 emb; n~140 f	11752, 11755
Myotis oxyotus	1 emb; n1 f	3999a
Myotis riparius	1 emb; n1 f	3999a
Myotis siligorensis	1 emb; n1 f	10872
Myotis simus	1 emb; n1 f	7699a
	1 emb ea; n3 f	3999a
Myotis sodalis	1 emb; n1 f	4231
	1 emb; n1 f	7609
	1 emb; n7 f	2867
Myotis thysanodes	1 emb; n1 f	1641
	1 emb ea; n2 f	280a
	1 emb ea; n2 f	1937
	1 emb ea; n2 f	11471a
	1 ea; n3 lt	10954
Myotis tricolor	1 emb ea; n3 f	825
	1 emb ea; n30 f	4402
Myotis velifer	1 emb; n1 f	11471a
	1 emb; n1 f	1196a
	1 emb ea; n2 f	1937
	1 emb ea; n9 f	5405b, 5409
Myotis vivesi	1	1346
	1 emb ea; n2 f	1469a
	1 emb ea; n27 f	9002a
Myotis volans	1	1937
	1 emb; n1 f	2415
	1 emb; n1 f	5424e
	1 emb; n1 f	11228a
	1 emb ea; n2 f	261a
	1 emb ea; n2 f	1940
	1 emb ea; n2 f	5415
	1 emb ea; n2 f	5424_2
	1 emb ea; n2 f	11471a
	1 emb ea; n4 f	8807a
	1 emb ea; n20+	2231a
Myotis yumanensis	1 emb ea; n7 f	3803a
	1 emb ea; n8 f	7607
	1 emb ea; n62 f	2227
Nyctalus aviator	1.91; n11 lt	3033
Nyctalus lasiopterus	1.67 yg; r1-2; n9 lt	6722
Nyctalus leisleri	1-2	4645
	1-2	3571
	2; mode 2	8883

Nyctalus noctula	1-2	181
	1-2	11147
	1-2	4645
	1-2	8137
	1-3 yg; 3 rare	3577
	1 ea; n2 lt	11622
	1.15; r1-3; n181 lt	754
	1.2; 20% twins; r1-2	10071
	1.3 yg; r1-2; n7 f	5860, 8836
	1.4; r1-2; n7 lt	2593
	1.4 emb; r1-2; n12 f	11318
	1.5; r1-2	8049
	2.0; r1-3; n4 lt	9399
Nycticeius balstoni	1 yg; n1 lt	9398
	2 yg; n1 lt	7075
Nycticeius humeralis	1 emb; n1 f	5414a
	1.5 emb; r1-2; n2 f	6474
	1.60; r1-2; n5 f	1290
	1.75 emb; r1-2; n4 f	2868
	1.98 emb; mode 2; r1-4; n58 f	577
	2 emb; n1 f	950a
	2 emb; n1 f	5803
	2 emb ea; n2 f	2197, 5405b$_1$
	2.0 emb; n3 f	11627
	2 yg ea; n several f	9709
	2 emb ea; n>9 f	3629
	2.0 yg; n14 lt	5386
	2 emb; r1-3; both 1 and 3 rare; n hundreds	11470
	2.3 yg; n115 lt	11472
	2.7 emb; r2-4; n3 f	589a
	3 emb; n1 f	4386
Nycticeius rueppellii	1; n1 lt	7075
Nycticeius schlieffeni	2.8 emb, morulae; mode 3; r1-5; n34 lt	11135, 11137, 11138
	2.8 cl; mode 3; r1-4; n32 f	11137, 11138
Nyctophilus geoffroyi	1-2	7075
	1.33; r1-2; n3 lt	9397
	mode 2; r1-2	4038, 4039
	2 yg ea; n4 f	7074
Nyctophilus gouldi	1.55; r1-2; n33 lt	8492
Nyctophilus timoriensis	2; n5 lt	7075
Otonycteris hemprichi	2 emb; n1 f	8049
Philetor brachypterus	1 emb; n1 f	6447
Pipistrellus spp	2 emb ea; n3 f	9136a
Pipistrellus ceylonicus	mode 2; r1-2	8493, 8494, 8501
	2 yg	1340, 1344
	2.01 emb; r1-3; n355 f	3915, 3927, 3932, 3933, 3939, 6715
Pipistrellus coromandra	1 yg ea; n2 f	8501
	2 emb; n1 f	1340, 1344
Pipistrellus dormeri	1+; r1-2; 1/12 twins	3943
	1.34 emb; r1-2; n271 f	3927
Pipistrellus hesperus	1.75; r1-2; n4 f	1937
	1.8 emb; mode 2; r1-2; n6 f	4079
	mode 2; r1-2	6064
	2 emb; n1 f	502
	2 emb; n1 f	11228a
	2 emb; n1 f	11471a
	2 emb ea; n2 f	4079
	2 emb ea; n4 f	7611
	2 emb ea; n6 f	280a
Pipistrellus javanicus	1-2	436
	2; r1-4	11016
	2.10 yg; mode 2; r1-3; n67 lt	11013, 11014
	2.50; r1-3; n32 f	10872

	2.94 emb; se0.13; mode 3; n17 f	12033
	3 emb; n1 f	179
	3 emb, yg; r2-4; n5 f	7521
Pipistrellus kuhlii	1-2	9175
	1 emb ea; n2 f	10158
	1.5; r1-2; n2 f	436
	1.5 emb; r1-2; n4 f	7720
Pipistrellus mimus	mode 1	9844
	1.93 emb; r1-3; n89 f	3927
	2	6026
	mode 2	5530
	mode 2; r1-2	8493, 8494, 8501
	mode 2 emb; r1-3	3944
	2 emb; n1 f	10872
	2.15 cl; 2.0 emb; r1.9-2.1; n115 lt	70
	3 emb; n1 f	5705
Pipistrellus nanulus	2 emb; n1 f	5387
Pipistrellus nanus	1+; r1-2; 2 rare	191
	1-2	6262
	1-2; occasionally 3	252
	1.5; r1-2; n6 f	1347
	1.9 emb; r1-2; n26 f	10158
	2 emb; n1 f	9945
	2 emb; n1 f	71
	2 emb, yg ea; n7 lt	6086
	2 ea; n16 f	8016
Pipistrellus nathusii	1-2	4645
	2	8049
	mode 2; r1-2	3571
Pipistrellus pipistrellus	1	8049
	1 ea; n3 lt	11622
	1 emb ea; n4 f	10872
	1 ea; n14 lt	8826
	1.05 emb; r1-2; n117 f	2420
	1.1 wean; r1-2; n35 lt	1179
	1.4; mode 1; r1-2; n12 lt	11318
	1.5 emb; r1-2; n64 f	8865
	mode 2	4645
Pipistrellus rusticus	mode 2 emb implant; r1-2	11139
	max 5 emb before implantation	11139
Pipistrellus savii	2	4645
	mode 2; r1-2	3571
Pipistrellus stenopterus	1 emb ea; n9 f	4625
Pipistrellus subflavus	1 emb ea ; n2 f	1937
	1.88 emb; r1-2; n24 f	1290
	2 emb; n1 f	4386
	2 emb; n1 f	4267
	2 emb; n1 f	8658
	2 yg; n1 f	2356
	2 emb ea; n2 f	6415
	2 emb ea; n7 f	6298
	2 emb ea; n17 f	2043
	2 emb ea; n19 f	6175
	2 yg; 2-4 ova	11795
	mode 2 yg; occasionally 3	2353
	3 emb; n1 f	7557
	3 emb; n1 f	4202
	4 emb; n1 f	6474
Pipistrellus tenuis	1; n1 lt	7075
	1 emb ea; n15 f	7073
Plecotus auritus	1	2045
	1-2	3571
	1-2	4645
Plecotus mexicanus	1 emb ea; n3 f	11471a

Plecotus rafinesquii	1 yg	587
	1 yg; n1 f	9578a
Plecotus townsendii	1	1937
	1	8314
	1 yg	5024
	mode 1	502
	1 emb; n1 f	10954a
Rhogeessa parvula	mode 2	11804
	2 yg ea; n2 lt	1937
	2 emb ea; n6 f	5413a
Rhogeessa tumida	1 emb ea; n2 f	5405e
	1.4; r1-2; n5 f	452
	2 emb ea; n2 f	950
	2 emb ea; n2 f	3608
Scotoecus hirundo	2 emb ea; n2 f	5759
Scotophilus dinganii	2 emb; n1 f	8084
	2 ea; n7 f	10158
	2.25 emb; r2-3; n4 f	10158
Scotophilus heathi	1.33 emb; r1-2; n3 f	10872
	1.93 emb; r1-2; n265	3927
	2 ova; 1 per ovary	6028
	mode 2 yg	1340, 1344
	mode 2	6717
Scotophilus kuhli	1-2	9175
	1.3 emb; r1-2; n3 f	8501
	2 emb	3905, 3915
	2 emb ea; n76 f	3927
Scotophilus leucogaster	1 emb; n1 f	8611
	2 emb	191
	2 emb; n4 f	5759
Scotophilus nigrita	1 emb ea; n2 f	191
	1-2 emb	9945
	1-2	5759
	1.57; r1-2; n7 f	252
	2; n1 f	5434
Scotophilus viridis	2 ea; 9 f	10158
Tylonycteris pachypus	1.92 emb; mode 2; n13 f	7161, 7164
Tylonycteris robustula	2.0 emb/yg; mode 2; n10 f	7161, 7164
Vespertilio murinus	1; r1-2	3571
	1-2	8049
	1 ea; n several hundred lt	2824
Vespertilio superans	2; r1-3	3563
SEXUAL MATURITY		
Eptesicus furinalis	f: preg: <1 yr	7695
	m: sperm: <1 yr	7695
Eptesicus fuscus	f: preg: 50% 1 yr	9643
	f/m: 2nd yr	8798
Lasionycteris noctivagans	f/m: 5 mo	2733
Lasiurus ega	f: preg: <1 yr	7695
	m: sperm: <1 yr	7695
Miniopterus australis	f: preg: 1st yr	7163
Miniopterus inflatus	f/m: 2 yr	1349
Miniopterus schreibersi	f: 19 mo	3945
	f: not 1st yr	2828
	m: 2 yr	3945
	f/m: 1.25-1.5 yr	3571
Myotis albescens	f: 1st mate: 2 mo	7695
	m: sperm: <1 yr	7695
Myotis austroriparius	f: 1st birth: 1 yr	9089
	m: 1st breeding season after birth	9089
Myotis blythii	f: may breed yr of birth	5988
Myotis daubentoni	f/m: 2nd yr	8798
Myotis emarginatus	f/m: ~15 mo	10070
Myotis frater	f: 1st birth: 2 yr	6723

Myotis grisescens	f: 1st birth: 2 yr	10976, 10978
	m: 1st mate: 15 mo	7330
Myotis lucifugus	f: 1st birth: 2 yr	9645
	f: 1st summer	4174
	m: 1st mate: 15 mo	7330
	m: spermatogenesis: 1 yr	4163
	f/m: 1st summer; 8 wk	4163
	f/m: 8 mo	5210
Myotis myotis	f: 1st birth: 2 yr	4663
	f/m: 1st mate: 1.25-1.5 yr	3571
	f/m: 2nd yr	10045
Myotis mystacinus	f: ~1 yr	10516
	f/m: 15 mo	10067
Myotis nigricans	m: spermatogenesis: 3.5-4 mo	11756
	f/m: 2-3 mo	11752
	f/m: 4 mo	7695
Myotis tricolor	f: 1st conceive: 4-16 mo	11818
Myotis velifer	f: 1st yr	6106
	f: 1 yr	6106
Myotis volans	f: 5 mo	2733
	m: yearlings	2733
Nyctalus noctula	f: 1st birth: 1 yr	5860, 8836
	f: 1st repro: 1 yr	3577
	f: 1st mate: 3 mo	5860
	m: 1st mate: 15 mo	5860
Nyctophilus gouldi	f: 7-9 mo	8492
	m: fertile: 12-15 mo	8492
Pipistrellus ceylonicus	f/m: 9-10 mo	6715
Pipistrellus mimus	f/m: 2 g	3944, 5530
Pipistrellus pipistrellus	m: spermatogenesis: 2 yr	2075
	f/m: 6 mo	6104
	f/m: ~2 yr	2420
Pipistrellus subflavus	f: 3-11 mo; n1	3549
Plecotus auritus	f: 1st birth: 75% 2nd yr, rest 3rd yr	2045
	m: 12 mo	2045
Scotophilus kuhli	m: epididymal sperm: 9 mo	3905, 3906
Tylonycteris pachypus	f/m: breed season following birth	7161
Tylonycteris robustula	m: 5 mo	7164
Vespertilio murinus	f/m: 2nd yr	9240
GESTATION		
Antrozous pallidus	~9 wk; r53-71 d	8127
Chalinolobus gouldii	~3 mo	5830
Eptesicus furinalis	3 mo; 1st lt; 2nd may be shorter	7695
Eptesicus fuscus	35 d	8798
	est 2 mo	1829
Eptesicus regulus	3 mo	5834
Eptesicus serotinus	2-2.5 mo	436
Lasionycteris noctivagans	implant: 10 d	2733
	50-60 d	2733
Lasiurus borealis	est 80-90 d growth	5210
	8 mo with delay	5210
Lasiurus cinereus	8 wk	2733
Lasiurus ega	3-3.5 mo	7695
Miniopterus australis	implant: <1 mo	9102
	4 mo without delay	2827, 2831, 2832
	~110 d	518
	4.5-5 mo; 134-149 d	7163
	6-6.5 mo with delay	2827, 2831, 2832
Miniopterus fraterculus	6.5 mo: ~4 mo growth + 2.5 mo delay	822
Miniopterus inflatus	3 mo	255
Miniopterus mullatus?	no delay	1351
Miniopterus schreibersi	implant: 4 mo	11128, 11129, 11131
	2 mo	5727
	~2.5 mo	11418

	120-125 d	3945
	~140 d	1340, 1344
	6 mo: 3 mo growth + 3 mo delay	2827, 2832
	7 mo	2828
	7.5 mo: 5 mo growth + 2.5 mo delay	9102
	8 mo: ~4 mo growth + 4 mo delay	821, 822
	8 mo: 4 mo growth + 4 mo delay	11128, 11129, 11131, 11132, 11135
	8 mo: 3 mo growth + 5 mo delay	8464
	260 d; with 2 mo delay	5727
	~9 mo: 2.5 mo growth + 6.5 mo delay	8463
	10-11 mo: 4 mo growth + 6-7 mo delay	8574
Myotis albescens	≤3 mo; later lt may be shorter	7695
Myotis daubentoni	~54 d	3571, 8798
Myotis lucifugus	implant: 10 d post ovulation	1414
	implant: > 10 d post ovulation	11793
	50-60 d	11795
	2.5 mo	343
	~80 d	1474
Myotis myotis	49-59 d	10045
	~50 d; can be longer if weather unfavorable	4663
	60-70 d; delayed implantation	3571
	2.5-3 mo	436
Myotis nigricans	50-60 d; field data	11752, 11755, 11759
	≤3 mo	7695
Myotis tricolor	63 d; n20; fetal developemnt	825
	3 mo	255
Mytois thysanodes	50-60 d	8005
Myotis velifer	est 60-70 d	6106
Nyctalus noctula	70-73 d; delayed	3571
	~70-73 d	2945
	73 d	8798
Nycticeius balstoni	~7 mo	9398
Nycticeius schlieffeni	11 wk; field data	11135, 11138
Pipistrellus ceylonicus	50-55 d; does not include sperm storage	3932, 3933, 6715
Pipistrellus javanicus	~70 d; field data	11013
Pipistrellus mimus	~60 d	6026
Pipistrellus pipistrellus	35 d	8798
	41-49 d	8049
	~44 d; delayed	2420
	45 d; r41-51; arousal to birth; field data	8840
Pipistrellus subflavus	postimplantation: 44 d	3549
Pipistrellus rusticus	11-12 wk; field data	11139
Plecotus auritus	60-70 d; arousal to birth; field data	10248
Plecotus townsendii	56-100 d	8314
Scotophilus heathi	3.5 mo	9175
	116 d	6028
Scotophilus kuhli	est 105-115 d; field data	3905
	~4.5 mo	7415
Scotophilus leucogaster	delay: 6-8 wk as morula, 2 mo as blastocyst	11145b
	7.5 mo	11145b
Tylonycteris pachypus	~89 d	7164
Tylonycteris robustula	89 d	7164
Vespertilio murinus	~50 d; arousal +5-10 d until birth	9240
LACTATION		
Antrozous pallidus	solid food: 6 wk	8127
	den emergence: 7 wk	8127
Eptesicus fuscus	1st fly: 3 wk	6107
	forage: 4 wk	1455, 6107
	wean: est 32-40 d	6105a, 6107
Eptesicus pumilus	1st fly: 3-4 wk	6714
Eptesicus regulus	wean: ~2 mo	5834

Lasionycteris noctivagans	1st fly: 3-4 wk	1871a
	wean: 36 d	1871a
Lasiurus borealis	1st fly: 4-6 wk	1871a
	wean: est 38 d	6105a
Lasiurus cinereus	solid food: 22 d	1043
	1st fly: 3-4 wk	1871a
	1st fly: 4 wk	1474
	nurse: ~30 d	2733
	wean: 34 d	1043
Miniopterus schreibersi	wean: 6-7 wk	3562
	wean: 2 mo	8464
	wean: 3 mo	2828
Myotis adversus	nurse: 2 mo	2834
Myotis albescens	wean: 1 mo	7695
Myotis austroriparius	1st fly: 5-6 wk	9089
Myotis californicus	1st fly: 2 mo	1871a
Myotis lucifugus	wean: 3 wk	3216
	wean: 26 d	6122
	wean: 4 wk	5210
Myotis myotis	1st fly: 5-6 wk	3571
	wean: ~6 wk	4663
Myotis nattereri	wean: 2 mo	6248
Myotis nigricans	1st fly: 3 wk	11752
	nurse: 1 mo	7695
	wean: 5-6 wk	11752, 11755, 11759
Myotis tricolor	lact: 6 wk	825
Myotis thysanodes	1st fly: 17 d	1871a
	wean: est 3 wk	8005
Myotis velifer	1st forage: 4th wk	6106
	wean: est ~43 d	6106
Nyctalus noctula	solid food: shortly after birth	4645
	1st fly: 5 wk	3571
	wean: 2 mo	10071
Nycticeius humeralis	1st fly: 20 d	11470
	wean: 1 mo; nurse own yg 2 wk then nurse any yg	11472
	solid food: 4 wk	5386
Nyctophilus gouldi	wean: 6 wk	8492
Pipistrellus ceylonicus	wean: 25-30 d; max 7 wk	6715
Pipistrellus kuhlii	wean: est ~2 mo	6418
Pipistrellus mimus	wean: 2 wk	6026
	wean: 3 mo	9844
Pipistrellus pipistrellus	wean: ~6 wk	10603
	independent: 2 mo	3571
Pipistrellus subflavus	1st fly: 3 wk	3549
	1st fly: 4 wk	1474, 5210
Plecotus auritus	wean: 40-45 d	10248
	independent: 6 wk	3571
Plecotus rafinesquii	female/young association: 3 wk	5397
Plecotus townsendii	1st fly: 3 wk	1474
	wean: 6 wk	11870
Scotophilus heathi	nurse: 2-3 wk	6028
	nurse: 24-28 d	6717
Scotophilus kuhli	nurse: 2 mo	3907
Tylonycteris pachypus	roost emergence: 6 wk	7161, 7164
Vespertilio murinus	solid food: 50 d	9240
	wean: 3 mo	9240
Vespertilio superans	wean: 30-40 d	3563
LITTERS/YEAR		
Barbastella barbastellus	1	3571
	1	8049
Chalinolobus morio	1	5833
	1	12042
Eptesicus furinalis	2	7695
Eptesicus fuscus	1	1474

Eptesicus nilssoni	1	3571
Eptesicus serotinus	1	3571
Histiotus velatus	1	8368
Lasionycteris noctivagans	1	1474
	1	7235
Lasiurus borealis	1	1474
Lasiurus cinereus	1	1474
Lasiurus ega	1	7695
Lasiurus seminolus	1	1474
Miniopterus australis	1	7163
Miniopterus mullatus?	1	1351
Miniopterus schreibersi	1	1344
	1	3571
	1	11128, 11135
Myotis adversus	2; max 3	2833
Myotis albescens	1-3	7695
Myotis austroriparius	1	1474
Myotis bechsteini	1	3571
Myotis californicus	1	1474
Myotis daubentoni	1	3571
Myotis emarginatus	1	3571
	1	9240
Myotis evotis	1	1474
Myotis grisescens	1	10978
	1	1474
Myotis keenii	1	1474
Myotis leibii	1	1474
Myotis lucifugus	1	577
	1	4059
	1	1474
Myotis myotis	1	4663
	1	9220
	1	3571
Myotis mystacinus	1	3571
Myotis nattereri	1	2045
Myotis nigricans	3; r1-4	11755, 11759
Myotis thysanodes	1	577
	1	1474
Myotis velifer	1	1474
Myotis volans	1	1474
Myotis yumanensis	1	1474
Nyctalus leisleri	1	3571
Nyctalus noctula	1	3571
	1	8049
	1	10071
Nycticeius humeralis	1	1474
Pipistrellus hesperus	1	1474
Pipistrellus mimus	3	6025
Pipistrellus nathusii	1	3571
Pipistrellus pipistrellus	1	2420
	1	3571
	1	8049
Pipistrellus savii	1	3571
Pipistrellus subflavus	1	1474
Plecotus auritus	1	3571
	1	2045
Plecotus rafinesquii	1	1474
Plecotus townsendii	1	1474
Scotophilus heathi	1	6028
Scotophilus kuhli	1	9175
	1	3905, 3907
Tylonycteris pachypus	1	7161
Vespertilio murinus	1	9240
	1	8049

SEASONALITY

Antrozous dubiaquercus	preg f: Apr; n1; Honduras	8554
Antrozous pallidus	testes: enlarge end Aug, large Sept-Oct; CA	8127
	mate: Oct-Nov, Feb; CA	8127
	mate: Oct-Feb; WY	1871a
	mate: fall; sw US	8175
	ovulate: early Apr; n3/4 f; CA	8127
	conceive: spring; sw US	8175
	implant: mid Apr; CA	8127
	preg f: Apr; n2; CA	11228a
	preg f: May; n1; Coahuila, Mexico	522
	preg f: May; n1; Chihuahua, Mexico	1196a
	preg f: May-June; Chihuahua, Mexico	280a
	preg f: Aug-Oct; CA	8127
	birth: Apr-June; sw US	1474
	birth: May-June; CA	8127
	birth: est July; Sonora, Mexico	1471a
Barbastella barbastellus	birth: May-early Aug; Germany	3571
	birth: end July; Germany	7777
	birth: early in the yr; USSR	8049
Chalinolobus argentatus	birth: Mar; Cameroun, Kenya	252
	birth: Oct; Tanzania	252
	preg f: Jan; n2; Cameroun	71
Chalinolobus beatrix	birth: prob Mar; central Africa	252
Chalinolobus gouldii	mate: early winter; sperm stored; WA, Australia	5830
	conceive: end winter; WA, Australia	5830
	preg f: Dec; n22/25 f; VIC, Australia	9989
	birth: end Sept-early Oct; n Australia	5830
	birth: end Nov-early Dec; WA, Australia	5830
Chalinolobus morio	preg f: Aug-Oct; sw Australia	5833
Chalinolobus tuberculatus	birth: Jan-Feb; New Zealand	2259
Chalinolobus variegatus	sexually active m: July, Oct; Uganda	5759
	preg f: Mar; n3/10; Zaire	191
	birth: Mar; central Africa	252
Eptesicus brasiliensis	preg f: June, Aug; n12; Peru	3999a
	lact f: end July-early Aug; n1; Oaxaca, Mexico	523
Eptesicus capensis	preg f: Oct-early Nov; Transvaal, S Africa	8965
	preg f: Nov; Namibia	9945
	birth: end Feb-early Mar; Garamba Nat Park, Zaire	11270
	birth: mid Nov; Tanzania, Zambia	252
	birth: est early-mid summer; Transvaal, S Africa	8965
	birth: summer; S Africa	10158
Eptesicus furinalis	testes: large Apr-June, Sept-Dec; Paraguay	7695
	mate: May-June, Sept-Nov; Paraguay	7695
	preg f: May; n1; Michoacan, Mexico	4671
	preg f: July; n1; Nicaragua	5425
	preg f: end July-Dec; Paraguay	7695
	birth: Oct-Nov, Jan; Paraguay	7695
	lact f: July; n9; Nicaragua	2670a
Eptesicus fuscus	testes: large Aug-Sept, small Oct-June; IN	11627
	mate: fall; KS	8485
	mate: fall; IN	11627
	ovulate: early Apr; PA, MD	1829
	ovulate: spring; PA	11794
	preg f: Mar; n8; Guatemala	5405e
	preg f: mid Apr-May; IN	11627
	preg f: May; n1; NC	1290
	preg f: May; n1; Jalisco, Mexico	11471a
	preg f: June; n8; s central BC, Canada	3217
	preg f: June; n2; MT	5424_2
	preg f: June; n4/8; ND	3703
	preg f: June; n1; CO	1649a
	preg f: June; n11; KS	950a
	preg f: June-July; n2; w SD	10954a

	preg f: June-July; n13; NE	5405b$_1$
	birth: mid Mar-June; PA, MD	1829
	birth: May; IL	6298
	birth: May-mid June; SD	10954
	birth: end May-mid June, peak early June; KS	6107
	birth: end May-early June; CA	6059
	birth: early June; KY	2354
	birth: June-mid July; ALB, Canada	9643
	birth: June-mid July, peak end June; ALB, Canada	9643
	birth: July; BC, Canada	2088
	1st yg: June; IN	11627
	lact f: July; n21; MT	5424$_2$
	lact f: July; n3; ND	3703
	lact f: July; n1; SD	261a
	lact f: July; n8; w SD	10954a
	lact f: July; Baja CA, Mexico	5424e
	lact f: July; n1; Durango, Mexico	523a
	lact f: Sept; n1; NE	5405b$_1$
Eptesicus guineensis	birth: June; central Africa	252
Eptesicus melckorum	preg f: Oct; n2; Cape Provice, S Africa	10158
Eptesicus nilssoni	birth: June; Germany	3571
Eptesicus pumilus	testes: large Nov; TAS, Australia	4038
	birth: end Nov-mid Dec; central Australia	6713
	birth: end Nov-mid Dec; TAS, Australia	4038
Eptesicus regulus	mate: autumn; sw Australia	5834
	conceive: end winter; store sperm; sw Australia	5834
	birth: end Nov-Dec; sw Australia	5834
Eptesicus rendalli	birth: end Mar-early Apr; Garamba Nat Park, Zaire	252, 11270
Eptesicus serotinus	spermatogenesis: May-Sept; Europe	2075
	mate: Apr; n1; Pakistan	9175
	mate: mid Sept; Germany	2945
	mate: Sept, spring; Germany	3571
	mate: after mid Nov; Portugal	436
	mate: fall; Iraq	110
	ovulate: Mar; Portugal	436
	preg f: spring; Iraq	110
	birth: end May; Portugal	436
	birth: June; Germany	7777
	birth: June; Germany	3571
	birth: spring-early summer; Pakistan	9175
Eptesicus tenuipinnis	mate: Jan; central Africa	252
	birth: Mar; n1; Cameroun	2948
	birth: Mar; Cameroun	252
	birth: Oct; Albert Nat Park, Zaire	252
	lact f: July; n1; Liberia	184
Euderma maculatum	lact f: June; n3; NM	5396
Hesperoptenus tickelli	preg f: May (many), Dec (n1); Sri Lanka	8501
	birth: May; Sri Lanka	8493
	birth: early June; Sri Lanka	8494, 8501
Histiotus macrotus	preg f: Oct; n3; Peru	3999a
Histiotus montanus	lact f: Dec; n4; Chile	4054
Histiotus velatus	repro: mid Sept; Rio de Janeiro, Brazil	8368
Idionycteris phyllotis	preg f: June; n1; NM	5396
	preg f: June; AZ	2196
	preg f: June; Durango, Mexico	2196
	preg f: June; n1; Durango, Mexico	3607$_1$
	birth: June-July; US	577
	birth: est June-July; AZ, NV, CO	1474
	lact f: June; n4; Durango, Mexico	3607$_1$
	lact f: June-July; n9; NM	5396
	lact f: July; AZ	1939a
	lact f: Aug; n2; AZ	3254a
Kerivoula cuprosa	birth: est Jan; Cameroun, Kenya	252
Kerivoula lanosa	birth: Jan; central Africa	252

Kerivoula papillosa	preg f: June, Oct, Dec; n4; Malaysia	6446, 6447
Kerivoula smithi	birth: Sept; central Africa	252
Lasionycteris noctivagans	spermatogenesis: May-Oct, peak end Oct; NM	2733
	mate: July; BC, Canada	2088
	mate: end Sept-spring; NM	2733
	ovulate: Apr-May; NM	2733
	preg f: May-June; n4; MN	495
	preg f: June; n1; SD	261a
	preg f: June; n1; w SD	10954a
	preg f: June; n2; Long I, NY	7881a
	preg f: June-July; n9; IA	2868
	birth: end May-early July; SD	10954
	birth: end May-early July; NE	$5405b_1$
	birth: June; n1; MI	6122a
	birth: end June-early July; US	577
	birth: end June-early July; WI	5210
	birth: June-July; US	1474
	birth: early July; NY	7235
	birth: July; BC, Canada	2088
	lact f: June-July; n5; IA	2868
	lact f: end June-July; IA	6105a
	lact f: July; n10; MT	5424_2
	lact f: July; n1; MN	495
Lasiurus borealis	mate: Aug; NY	7651
	mate: Aug-Sept; WY	1871a
	preg f: Apr-May; n12; NC	1290
	preg f: May; n1; WY	1871a
	preg f: May; n1; NE	$5405b_1$
	preg f: May; n1; NM	5396
	preg f: May; n2; AL	589a
	preg f: May-early June; LA	6263
	preg f: end Apr-mid June; IN	7608, 11627
	preg f: May-June; n18; KS, IN	1937
	preg f: June; n1; SC	1973c
	preg f: June; n1; Chihuahua, Mexico	1196a
	birth: May-early June; IL	6298
	birth: end May-early June; KS	4222
	birth: end May-June; KS	5414a
	birth: June; US	1474
	lact f: June; n1; KS	1812a
	lact f: June; n3; NM	5396
	lact f: June; n1; Coahuila, Mexico	522
	lact f: June-mid July; LA	6263
	lact f: July; n<12; MT	5424_2
	lact f: July; n2; MN	495
	lact f: July; n2; SC	1973c
	lact f: Dec; n1; Chile	4054
Lasiurus cinereus	spermatogenesis: May-end Oct, peak early Sept; NM	2733
	mate: Aug; NY	7235
	mate: Sept-spring; NM	2733
	preg f: end May-early June; n1; Chihuahua, Mexico	1196a
	preg f: June; n1; w SD	10954a
	preg f: June; NE	$5405b_1$
	preg f: June; n1; IA	6105a
	birth: May-July; HI	10809
	birth: mid May-July; WY	1871a
	birth: end May-early June; IN	11627
	birth: peak end May-early June, May-July; SD	10954
	birth: June; N America	1474
	birth: June; n1; IL	7619
	birth: June; n6; captive, NM	1043
	birth: June; n1; SAS, Canada	7847
	birth: June-early July; BC, Canada	2088
	birth: mid June; KS	5414a

	birth: end spring-early summer?; NY	7235
	lact f: June; n1; w SD	10954a
	lact f: end June-July; IA	6105a
	lact f: July; n1; se MT	6166
Lasiurus ega	spermatogenesis: Dec-July; Paraguay	7695
	testes: large Mar-May; Paraguay	7695
	epididymal sperm: Mar-Sept; Paraguay	7695
	mate: May; Paraguay	7695
	ovulate: mid Aug; Asuncion, Paraguay	7695
	ovulate: mid Sept; Chaco, Paraguay	7695
	birth: June; NM	577
	birth: end Nov-early Dec; Paraguay	7695
	preg/lact f: July; n7; Baja CA, Mexico	5424e
	lact f: May; n1; Honduras	4048_1
	lact f: May; n1; Costa Rica	3608
Lasiurus intermedius	preg f: June; n1; Sinaloa, Mexico	5413a
	birth: est end May-early June; FL	9896a
	lact f: July; n2; Hidalgo, Mexico	1640a
	lact f: July; n4; Veracruz, Mexico	522a
Lasiurus seminolus	preg f: May; n2; AL	589a
	preg f: May; n1; SC	1973c
	preg f: June; n2; FL	7498a
	birth: June; se US	1474
	lact f: June; n1; FL	7498a
	lact f: July; n1; NC	588a
	lact f: July; n1; SC	1973c
Mimetillus moloneyi	preg f: Feb; n1; tropical Africa	1346
	birth: Feb-Mar, Aug; Africa	5759
	birth: Mar; central Africa	252
	birth: bi-annual, end dry spells; e Africa	5759
Miniopterus australis	testes: large end May; NSW, Australia	2831
	testes: increase June-Aug; New Hebrides	518
	testes: large Nov; e Malaysia	7163
	mate: June-July; NSW, Australia	2827, 2832
	mate: Sept; New Hebrides	518
	mate: Nov-Dec; e Malaysia	7163
	conceive: mid Aug; Australia	9102
	implant: Sept; NSW, Australia	2827, 2832
	implant: mid Sept; Australia	9102
	preg f: Sept-Dec; New Hebrides	518
	birth: end Apr-early May; e Malaysia	7163
	birth: Dec; Australia	9102
	birth: Dec; NSW, Australia	2827, 2832
	birth: Dec; New Hebrides	518
Miniopterus fraterculus	mate: mid May; Natal, S Africa	822
	conceive: mid May; Natal, S Africa	822
Miniopterus inflatus	preg f: Oct; n3; Zimbabwe	10158
	birth: Oct; Cameroun, Gabon, se Zaire	252, 255
	birth: Oct; ne Gabon	1349
Miniopterus minor	birth: end Oct-early Nov; central Africa	252
	birth: Dec; e Africa	6966
	lact: ?-Dec; central Africa	252
Miniopterus mullatus?	birth: Oct; Gabon	1351
Miniopterus schreibersi	spermatogenesis: peak end Apr-mid Sept; Australia	9102
	spermatogenesis: Aug-Sept; s France	2075
	spermatogenesis: Sept; s France	8574
	testes: large Jan-Mar; S Africa	11134
	testes: large May-June, small Oct-early Nov; NSW, Australia	2828
	testes: small Dec-Jan; Germany	546
	follicular growth: Mar-Apr, Aug-Dec; Natal, S Africa	821
	mate: 1st 3 wk Feb; India	3945
	mate: mid Feb-early May, peak Mar-Apr; Transvaal, S Africa	11131, 11134, 11135
	mate: mid Apr; Natal, S Africa	822
	mate: May; 30°S; Aug; 15°S, Australia	2827, 2832

	mate: end May-early June; NSW, Australia	11418
	mate: end May-early June; Australia	9102
	mate: July?; few data; SW Africa	6966
	mate: mid Oct; Japan	11015
	mate: Sept; Pyrenees, France	8463
	mate: fall before hibernation; s France	2075
	mate: fall, spring; nw Sardinia, Italy	3517
	ovulate: Jan-Feb; Portugal	436
	ovulate: Feb-Mar; S Africa	11134
	ovulate: Mar-early Apr; s Transvaal, S Africa	11129
	ovulate: Mar-Apr; S Africa	11128
	ovulate: Mar-Apr, peak end Mar; s Transvaal, S Africa	11129
	ovulate: Apr; Natal, S Africa	821, 822
	ovulate: May; NSW, Australia	11418
	ovulate: Aug-Sept; Pyrenees, France	8464
	ovulate: mid Oct; diapause mid Oct-mid Dec; Japan	5727
	ovulate: mid Oct; Japan	11015
	ovulate: fall before hibernation; s France	2075
	implant: Feb; s France	8574
	implant: Mar; Pyrenees, France	8464
	implant: Mar-Apr; Pyrenees, France	8463
	implant: before mid Mar; Sardinia, Italy	3517
	implant: July-Aug, peak end July; Transvaal, S Africa	11129, 11131, 11133, 11135
	implant: Aug; Australia	9102
	implant: Aug-Sept; NSW, Australia	11418
	implant: mid Aug; Natal, S Africa	821
	implant: mid Sept; s Australia	2832
	implant: Aug-Sept; n Australia	2832
	implant: mid Dec-early Jan; Japan	5727, 5729
	preg f: Mar-Apr; Sardinia, Italy	3517
	preg f: May-Dec; NSW, Australia	2828
	preg f: Sept-Feb; s France	8574
	preg f: Oct, Nov; Transvaal, S Africa	8965
	preg f: mid Oct-early July; Japan	5729
	preg f: Nov; n11; Natal, S Africa	4402
	birth: mid Apr-end June; Algeria	5988
	birth: May; s France	8574
	birth: May-July; Germany	3571
	birth: June; Pyrenees, France	8463, 8464
	birth: end June-early July; Japan	3562, 5727
	birth: Oct-Nov; se Zaire	255
	birth: mid Oct; se Zaire	252
	birth: early Nov; Zambia	252
	birth: end Oct-mid Dec, peak mid-end Nov; s Transvaal, S Africa	11129, 11131, 11135
	birth: Dec; NSW, Australia	2827, 2828, 2832
	birth: Dec; Australia	9102
	birth: Dec; Natal, S Africa	821, 822
	birth: mid Oct-Jan; Australia	2840
	birth: early summer; Transvaal, S Africa	8965
	lact f: June-July; Pyrenees, Europe	8464
	lact f: July-Aug; Japan	3562
	lact f: Dec-Mar; NSW, Australia	2828
	wean: after July ?; s France	8574
Murina cyclotis	preg f: Feb; n1; Pahang, Malaysia	7161
Murina tubinaris	preg f: May; n5; Mishmi Hills, Assam, India	4813b
Myotis adversus	preg f: Mar; n2; Papua, New Guinea	7073
	birth: Jan, Oct; n19; QLD, Australia	2833
	lact f: Mar; n4; Papua, New Guinea	7073
Myotis albescens	spermatogenesis: Mar-Dec; Paraguay	7695
	testes: small Jan-Mar; Paraguay	7695
	mate: May; Oct; Paraguay	7695
	ovulate: July, Oct; Paraguay	7695
	preg f: Jan; n3; Costa Rica	6260

	preg f: June, Aug; n11; Peru	3999a
	preg f: July; n1; Honduras	2670a
	preg f: Aug; n10; Peru	10974
	birth: Oct, Jan; Paraguay	7695
Myotis auriculus	preg f: June; n3; Chihuahua, Mexico	280a
	birth: June; US	577
Myotis austroriparius	epididymis: large Nov-mid Apr; FL, captive	9089
	birth: end Apr-May; FL	577
	birth: end Apr-mid May; FL, captive	9089
	birth: May-June; US	1474
Myotis bechsteini	birth: June-July; Germany	3571
Myotis blythii	testes large: July; Lebanon	6418
	spermatogenesis: Aug-Sept, peak Sept; Yugoslavia	5902
	mate: July-Oct; Algeria, Kirghizia	4980
	mate: Aug-Sept; Czechoslovakia, Bulgaria	4980
	birth: Apr-May; Algeria	5988
Myotis bocagei	preg f: Mar; n1; Niger	8611
	birth: Mar; Gabon	252
	birth: Apr; Rwanda	252
	birth: Jan, June, Aug, Oct; Zaire	252
	birth: yr rd; central Africa	253
Myotis californicus	preg f: Apr; n1; Sinaloa, Mexico	5413a
	preg f: end Apr-mid June; CA	6060
	preg f: May-June; CA	11228a
	preg f: June; n3; Durango, Mexico	3607_1
	birth: May-June; US	1474
	birth: end May-mid June; US	577
	birth: July; BC, Canada	2088
	birth: July; n1; UT	12104a
Myotis capaccinii	spermatogenesis: none Nov-Apr; Europe	2075
Myotis chiloensis	preg f: Feb, July; n3; Costa Rica	3608
Myotis dasycneme	mate: 40% before mid Nov, rest mid Nov-mid Apr; St Petersburg, Russia	10495
	ovulate: Jan-mid Apr; Netherlands	11095
Myotis daubentoni	mate: Feb; Netherlands	11183
	mate: Sept-Jan; Poland	4074
	mate: 17% before mid Nov, rest mid Nov-mid Apr; St Petersburg, Russia	10495
	ovulate: Mar, Dec; Netherlands	11095
	birth: end June-early July; Germany	3571
	birth: early July; Germany	7777
Myotis elegans	preg f: Feb; n1; Campeche, Mexico	5424f
	preg f: Apr; Costa Rica	6260
	preg f: June; n2; Veracruz, Mexico	6148a
	lact f: June; n2; Caribbean, Guatemala	7024
Myotis emarginatus	mate: Aug; France	9240
	mate: Sept-Oct; Netherlands	10070
	birth: June; Netherlands	10070
	birth: end June-mid July; Germany	5175
	birth: end June-July; Germany	3571
Myotis evotis	preg f: May-June; n2; SD	6780
	preg f: May-June; n≥3; SD	261a
	preg f: May-June; n2; CA	11228a
	preg f: June; n2; ID	11646a
	preg f: June; n2; NV	6780
	preg f: mid June-mid July; WY	1871a
	preg f: July; n2; BC, Canada	2088
	preg f: July; n2; MT	5424_2
	birth: end June-July; w US	1474
	birth: July; n5; WA	6780
	lact f: July; n3; se MT	6166
	lact f: July; NM	6780
Myotis fortidens	preg f: June; n4; Sinaloa, Mexico	5413a
Myotis frater	birth: mid June-end July; Japan	6723

Myotis grisescens	spermatogenesis: June-Aug; MO	7330
	estrus: Oct-Nov; MO	7330, 10107
	mate: end fall; US	577
	birth: May-June; central US	1474
	birth: June; MO	4174
	birth: June; US	577
	lact f: June; MO	4173a
Myotis hasseltii	lact f: Jan; Malaysia	7161
Myotis keaysi	preg f: May-June, Aug, Nov; Costa Rica	6261
	preg f: July; n7; Peru	3999a
Myotis keenii	preg f: mid May-mid June; IA	6105a
	preg f: June; n7; Durango, Mexico	3607_1
	birth: early-mid June; IN	2043
	birth: June; n1; MO	2864
	birth: end June-July; e US	1474
	lact f: end June-July; IA	6105a
	lact f: June; n2; Durango, Mexico	3607_1
Myotis leibii	mate: Apr; n1; IN	4202
	preg f: May; n1; NM	502
	preg f: June; n1; MT	5424_2
	preg f: June; n3; ND	3703
	preg f: June; n1; w SD	10954a
	preg f: June-July; SD	10954
	preg f: July; n1; NE	8807a
	preg/lact f: June; s central BC, Canada	3217
	birth: May-July; US	1474
	lact f: July; n1; se MT	6166
Myotis lucifugus	spermatogenesis: Apr-Aug; ne US	343, 4161
	spermatogenesis: June-Aug; MO	7330
	uterus epithelium: high Oct-Feb; MO	9002
	mate: Aug-Sept; ONT, Canada	8843
	mate: end Aug-?; ONT, Canada	10779
	mate: Sept; AZ	6070
	mate: Sept-Oct, Apr; NY	2135
	mate: Sept-Mar; ne US	343
	mate: Oct; IN	4202
	mate: early fall; NY	7651
	mate: fall; MO	4174, 9002
	ovulate: Feb-Apr, peak Apr; MO	4174, 4175
	ovulate: end Apr-May; ONT, Canada	1414
	ovulate: spring; PA	11794
	preg f: May; n14; SD	261a
	preg f: May-June; MN	495
	preg f: June; IA	6105a
	preg f: mid June; 9/21; BC, Canada	3217
	preg f: end June-early July; n40; CO	2353a
	preg f: July; n10; NM	2353a
	birth: May-July; US	1474
	birth: May-July; NE	$5405b_1$
	birth: peak; end May; NY	784
	birth: mid May-mid July; IL	1526
	birth: end May-?; WI	5210
	birth: June-early July; Canada, NY	3214
	birth: June-early July; ONT, Canada	3214
	birth: June-mid July; ALB, Canada	9645
	birth: June-mid July; ONT, Canada	2842
	birth: June-July; ne US	2355
	birth: June-July; ONT, Canada	2842
	birth: mid June-mid July; ALB, Canada	9645
	birth: July; BC, Canada	2088
	birth: July; WY	3254
	birth: mid summer; NY	7651
	lact f: end June-July; n38; CO	2353a
	lact f: July; IA	6105a

	lact f: July; n9; NM	2353a
	lact: ends mid July; ne US	343
Myotis muricola	birth: May, June?; Fukien, Yunnan, China	179
Myotis myotis	mate: Aug; Terek Region, Russia	8049
	mate: Sept; AZ	6070
	mate: Sept-Oct; Netherlands	10070
	mate: mid Sept; Germany	2945
	mate: spring?; Germany	9679
	mate: spring, fall; Japan	4819
	mate: fall, also captive Jan; Germany	9679
	mate: fall; USSR	1524
	mate: est fall; Sardinia, Italy	3517
	ovulate: end Dec-mid Apr; Netherlands	11095
	ovulate: Jan; Netherlands	10069
	ovulate: Feb; Portugal	436
	ovulate: Mar-Apr; Germany	2947
	implant: before mid Mar; Sardinia, Italy	3517
	preg f: Mar-Apr; Sardinia, Italy	3517
	birth: May-June; Portugal	436
	birth: end May; Pyrenees, France	1657
	birth: end May-mid June; Eifel mts, Germany	9220
	birth: end May-mid June; Bohemia, Czechoslovakia	6001
	birth: end May-June/July; Germany	3571
	birth: end May-early July; Germany	5938
	birth: June; Belgium	4663
	birth: June; France	9596
	birth: June; Germany	2945
	birth: June; Netherlands	10070
	birth: June; Transcaucasus, Georgia	8049
	birth: end June; Netherlands	754
	birth: end June-early July; Guadalajara, Spain	2402
Myotis mystacinus	mate: Jan; Netherlands	11183
	mate: mid Nov-mid Apr; St Petersburg, Russia	10495
	ovulate: Dec-Apr; Netherlands	11095
	preg f: Feb, Apr, June, Aug, Oct; n7; Malaysia	6446
	preg f: yr rd, peak Apr-May; Malaysia	7161
	birth: June; Germany	7777
	birth: June-July; Germany	3571
Myotis nattereri	birth: end June-early July; Ireland, UK	2045
	lact f: mid June-mid Aug; Germany	6248
Myotis nigricans	spermatogenesis: Dec-Aug; Panama	11756
	spermatogenesis: yr rd; Paraguay	7695
	testes: small Feb; Banos, Ecuador	10659
	mate: Jan-mid Sept; Panama	11755
	mate: peak May-June; Paraguay	7695
	mate: est end Dec-early Jan; Banos, Ecuador	10659
	preg f: peak Jan/Feb, Apr/May, Aug; Panama	11755
	preg f: Jan-Nov; Costa Rica	3322
	preg f: Feb, Apr, June-Aug, Oct-Nov; n58; Peru	3999a
	preg f: June; n2; Peru	10974
	preg/lact f: 42% Aug, 84% Sept; Guatemala	2552
	preg/lact f: yr rd; Paraguay	7695
	birth: end Feb-Nov; Panama	11755
Myotis oxyotus	preg f: June; n1; Peru	3999a
Myotis riparius	preg f: Apr-June; Costa Rica	6261
	preg f: Aug; n1; Peru	3999a
Myotis simus	preg f: July-Aug; n3; Peru	3999a
	preg f: Oct; Paraguay	7699a
Myotis sodalis	mate: fall; MO	4174
	mate: Oct; midwest US	4231
	preg f: May-June; MO	2867
	preg f: June; n1; IN	7609
	birth: end June; midwest US	4231
	birth: end June-early July; IN	5096c

	lact f: June-Aug; n6; MI	6119a
Myotis thysanodes	preg f: Feb; n1; Chiapas, Mexico	1641
	preg f: May; n2; Jalisco, Mexico	11471a
	preg f: June; n2; Chihuahua, Mexico	280a
	preg f: June; n3; SD	10954
	birth: June-July; w US	1474
	lact f: June; n1; San Luis Potosí, Mexico	2228a
	lact f: Aug; n2; Hidalgo, Mexico	1640a
Myotis tricolor	mate: Apr?; Natal, S Africa	825
	ovulate: end Sept; Natal, S Africa	825
	conceive: Sept; Natal, S Africa	825
	implant: end Sept-Nov; Natal, S Africa	825
	birth: end Oct-mid Nov; Cape Province, S Africa	10158
	birth: mid Nov-mid Dec; Natal, S Africa	825
	birth: 1st 3 wk Dec; Natal, S Africa	11818
	birth: Dec-Jan; Zaire, S Africa	255
Myotis velifer	mate: Apr; n3; KS	3288
	mate: Sept-Oct; sw KS	6106
	mate: peak Sept-Oct, Dec-Mar; AZ	2135, 6070
	conceive: mid Apr; sw KS	6106
	preg f: Mar, Dec; few data; Veracruz, Mexico	4228
	preg f: May; n1; Jalisco, Mexico	11471a
	preg f: June; n1; Chihuahua, Mexico	1196a
	preg f: June; n9; Durango, Mexico	5405b
	preg f: June-July; n19+; Sinaloa, Mexico	5409, 5413a
	birth: June-July; sw US	1474
	birth: est mid June; sw KS	6106
	birth: end June-mid July; KS, OK	577
Myotis vivesi	preg f: May; n27; Pescadora, I, Mexico	9002a
	preg f: June; n2; Partida I, Mexico	1469a
	birth: May-June; CA to Sonora, Mexico	1346
	lact f: June; n25; Partida I, Mexico	1469a
Myotis volans	ovulate: Mar-May; NM	2733
	preg f: May; n1; CA	11228a
	preg f: May; n2; Jalisco, Mexico	11471a
	preg f: June; n2; SD	261a
	preg f: June; n100; CA	2231a
	preg f: June or July; n4+; NE	8807a
	preg f: mid June-mid July; NV	11466
	preg f: July; n1; se MT	6166
	preg f: July; n2; MT	5424_2
	preg f: July; NM	2353a
	preg f: July; NM	11466
	preg f: July; Baja CA, Mexico	5424e
	preg f: July-Aug; WY	3254
	birth: May-early Aug; NM	2733
	birth: June; n many; CA	2231a
	birth: June; w N America	1474
	birth: end June-mid Aug; WY	1871a
	lact f: May-Aug; NM	2733
	lact f: June; n1; Durango, Mexico	3607_1
	lact f: July; n1; se MT	6166
	lact f: July; n3; MT	5424_2
	lact f: Aug; n1; SD	261a
Myotis yumanensis	preg f: June; n7; OK	3803a
	preg f: June; n~34 f; NM	1996a
	preg f: June; n1; Sinaloa, Mexico	5413a
	birth: May-June; w US	1474
	birth: May-July; WY	1871a
	birth: May-July; US	577
	lact f: June; 26/59; sw BC, Canada	3217
	lact f: July; n1; CO	2353a
	lact f: Aug; n2; Jalisco, Mexico	11471a
Nyctalus aviator	birth: May-July; Japan	3033

Nyctalus lasiopterus	birth: mid June-mid July; Japan	6722
Nyctalus leisleri	preg f: June-early Aug, Jan; India	8883
	birth: May-July; Germany	3571
Nyctalus noctula	epididymal sperm: Nov-Mar; Germany	546
	testes: increase mid May-July; lab	5860
	testes: large July-Aug; Germany	4102
	testes: large Sept; n1; small Jan; n1; Germany	3631
	testosterone: peak Sept; lab	8829
	mate: July; Germany	4102
	mate: Aug; Voronesh, Russua	8137
	mate: end Aug; Germany	2945
	mate: Sept-?; Netherlands	10071
	mate: Sept-Oct; captive	5860
	mate: fall from end Aug, spring; Germany	3571
	mate: fall; Europe	11318
	mate: fall; Belgium	11147
	mate: fall; Germany	9679
	ovulate: May; Belgium	11148
	conceive: 5-10 d after hibemation; Germany	3571
	implant: Mar-Apr; Belgium	11147
	preg f: May-June; Belgium	11148
	birth: May-June; Russia	4645
	birth: May-June; Netherlands	754
	birth: end May-early June?; Russia	8049
	birth: end May-July; Germany	3571
	birth: early-mid June; Netherlands	10071
	birth: June; few data; UK	11622
	birth: June; Voronesh, Russia	8137
	birth: June or later; Belgium	11148
	birth: July; Germany	4102
	lact f: Aug; Netherlands	10071
Nycticeius humeralis	preg f: Apr-May; n5; NC	1290
	preg f: May; n1; KS	5414a
	preg f: May; n3; AL	589a
	preg f: June; n2; NE	5405b$_1$
	preg f: June; n1; KS	950a
	preg f: June; n1; IA	6105a
	preg f: June-July; n4; IA	2868
	birth: mid May-mid June; se US	577
	birth: end May-early June; MS	5386
	birth: mid June; MO	11472
	lact f: May; FL	7498a
	lact f: June; n1; KS	5414a
	lact f: June-July; n9; IA	2868
Nycticeius schlieffeni	mate: June; Transvaal, S Africa	11138
	ovulate: Aug; Transvaal, S Africa	11138
	preg f: Sept-Nov; n11; Transvaal, S Africa	11137
	birth: Nov; central Africa	252
	birth: Nov-mid Dec; Transvaal, S Africa	11135, 11138
	birth: est early rainy season; few data; Burkina Faso, w Africa	5903
Nyctophilus geoffroyi	birth: Nov; TAS, Australia	4038, 4039
	lact f: Nov; n1; VIC, Australia	7074
Nyctophilus gouldi	spermatogenesis: summer; ACT, Australia	8492
	sperm storage: Jan-Sept; ACT, Australia	8492
	mate: early Apr; ACT, Australia	8492
	ovulate: Sept-early Oct; ACT, Australia	8492
	birth: Oct-Nov; ACT, Australia	8492
Philetor brachypterus	preg f: June; n1; Malaysia	6447
Pipistrellus spp	repro: yr rd; Papua, New Guinea	200
	preg f: May; n3; Honduras	9136a
Pipistrellus ceylonicus	mate: June; India	3933, 3940, 6715
	mate: yr rd; Sri Lanka	8493
	ovulate: mid July; India	3933, 3940, 6715
	preg f: July-Sept; India	3915

	preg f: Sept; Sri Lanka	8494, 8501
	birth: end Aug-mid Sept; India	6715
Pipistrellus coromandra	preg f: July; n1; Sri Lanka	8501
	lact f: July; n2; Sri Lanka	8501
	lact f: Aug; central India	5705
	juveniles: May; n2; sub-adult: Sept; n1; central India	5705
Pipistrellus dormeri	spermatogenesis: yr rd; India	6716
	preg f: Mar; n5; central India	5705
	preg f: yr rd; India	6716
	juveniles: Aug-Sept; central India	5705
Pipistrellus hesperus	spermatogenesis: June-Sept; CA	6064
	testes: large July-Oct, small Mar-May; CA	6064
	mate: Sept-Apr; CA	6064
	ovulate: mid Mar-end Apr; CA	6064
	preg f: Mar-Aug; CA	6064
	preg f: Apr; n1; Jalisco, Mexico	11471a
	preg f: May; n1; NM	502
	preg f: June; n1; CA	11228a
	birth: June-Aug; CA	6064
	lact f: June; Coahuila, Mexico	522
Pipistrellus javanicus	sperm: few Sept, many mid Oct; Japan	7759
	mate: Oct; Japan	7759
	mate: end Oct; Japan	11014
	ovulate: Apr; Japan	11013, 11014
	ovulate: May; Japan	7759
	birth: June; Japan	7521
	birth: early July; Japan	11013
Pipistrellus kuhlii	mate: fall; Iraq	110
	preg f: Oct; n2; Transvaal, S Africa	10158
	birth: Apr-May; Pakistan	9175
	birth: est end Apr-May; Lebanon	6418
	birth: est Nov; central Africa	252
	wean: end June; Lebanon	6418
Pipistrellus mimus	spermatogenesis: Nov, Feb-early June; India	6026
	spermatogenesis: yr rd; lab	3944
	testes: large Nov, Feb, small June-Sept; India	6026
	mate: end Nov-early Dec, Feb-June; India	6025, 6026
	mate: yr rd; Sri Lanka	8494, 8501
	conceive: end Dec, Feb-June; India	6025
	preg f: May; n1; India	5705
	preg f: peaks May-July, Sept-Oct; lab	3944
	preg f: Dec-Aug, peak Mar; India	6026
	preg f: yr rd, peaks July, Aug, Feb-Mar; India	70
	birth: Feb, May, Aug; India	6025, 6026
	lact f: Mar, May, Dec; Sri Lanka	8493
	yg observed: Mar-Apr; Thar Desert, India	9844
Pipistrellus nanulus	preg f: Mar; n1; Rio Muni, W Africa	5387
Pipistrellus nanus	mate: May-Sept; Kenya	8016
	preg f: Feb, May, July-Aug, Oct; Zaire	11272
	preg f: Aug, Dec-Jan; Zaire	191
	preg f: Aug-Sept; Zimbabwe	10158
	preg f: Sept-Oct; no data June-Sept; Africa	6262
	preg f: Oct; n1; Namibia	9945
	birth: Mar; Zaire	11270
	birth: Oct; no data other mo; Liberia	6086
	birth: Nov; Kenya	8016
	birth: yr rd; central Africa	252
	lact f: June, Nov; no data June-Sept; Africa	6262
	lact f: Aug; Rio Muni, W Africa	5387
Pipistrellus nathusii	birth: mid June; central Russia	8049
Pipistrellus pipistrellus	spermatogenesis: end May-Sept, none Oct-Apr; France	2074, 2075
	epididymal sperm: Nov-Mar; Germany	546
	testes: large Aug, small Nov; UK	8842
	testes: large Aug; n1; Lebanon	6418

	mate: Dec; n1; Spain	6104
	mate: fall; France	2075
	mate: fall; Europe	11318
	mate: fall; UK	2420
	mate: fall; USSR	8049
	mate: fall, spring; Germany	1524
	ovulate: May; Belgium	11148
	ovulate: May; UK	2420
	ovulate: spring; Germany	1524
	birth: May/June-July; Germany	3571
	birth: June; Germany	7777
	birth: June-early July; Azerbaijan	8865
	birth: June/July; Spain	6104
	birth: end June; UK	2420
	birth: end June; UK	1179
	birth: end June-?; Scotland, UK	10603
	birth: July; n3; UK	11622
	birth: early summer; Pakistan	9175
	wean: mid Aug; Scotland, UK	10603
Pipistrellus rueppelli	birth: Apr-May; Tanzania	252
	birth: Aug; Zambia	252
	juveniles: Sept; Zambia	331
Pipistrellus rusticus	birth: Nov; S Africa	11139
Pipistrellus savii	birth: June; Germany	3571
Pipistrellus stenopterus	preg f: Oct; n1; Singapore, Malaysia	7161
	lact f: Oct; n1; Singapore, Malaysia	7161
Pipistrellus subflavus	spermatogenesis: Aug-Feb; e US	6068
	testes: large Aug-Sept; e US	6068
	mate: Nov; IN	4202
	ovulate: spring; MO	4174
	preg f: Apr-May; n24; NC	1290
	preg f: May-mid June; MO	3549
	birth: May; se US	1474
	birth: end May-early June; n2; GA	4386
	birth: end May-mid June; FL	3549
	birth: June-July; ne US	1474
	birth: mid-end June; MA	3549
	birth: mid June-mid July; PA	6175
	birth: end June-early July; IN	3549
	lact f: June; FL	7498a
	lact f: July; n1; IA	6105a
Pipistrellus tenuis	preg f: Jan-Mar, May-June, Aug, Oct; Papua, New Guinea	7073
Plecotus auritus	epididymal sperm: Aug-Sept; Scotland, UK	10247
	mate: mid Oct-Nov; Ireland, UK	7413
	preg f: June-early July; ne Scotland, UK	10248
	birth: Apr?; May-July; Germany	3571
	birth: June; Germany	2945
	birth: June?; Portugal	436
	birth: July; ne Scotland, UK	10248
	birth: spring; UK	2045
	repro: Apr-mid May; Ireland, UK	7413
Plecotus mexicanus	preg f: Apr; n3; Jalisco, Mexico	11471a
	lact f: May; n1; Jalisco, Mexico	10942a
	lact f: July; San Luis Potosí, Mexico	10942a
Plecotus rafinesquii	preg f: July; WA	9578a
	birth: end May-early June; NC	587
Plecotus townsendii	mate: Oct-Apr; CA	8314
	mate: Oct-Feb; US	1474
	mate: est fall; CA	2228
	ovulate: Feb-Mar; CA	8314
	ovulate: Feb-Apr; US	1474
	ovulate: spring; CA	2228
	preg f: July; n1; SD	10954, 10954a
	birth: Apr-July; US	1474

	birth: June; TX	11870
	birth: mid July; BC, Canada	2088
	lact f: July; n19/22; SD	10954, 10954a
Rhogeessa parvula	preg f: Apr; n1; Michoacan, Mexico	4671
	preg f: May-June; n6; Sinaloa, Mexico	5413a
Rhogeessa tumida	preg f: Mar; n2; Costa Rica	3608
	preg f: Mar; n2; Guatemala	5405e
	preg f: Aug; n2; Guatemala	2552
	lact f: Mar; n2; Costa Rica	3608
Scotoecus hirundo	lact f: May; n1; e Africa	5759
Scotophilus heathi	spermatogenesis: Oct-Mar; 1st sperm: Jan; India	6028
	spermatogenesis: Oct-Nov; s India	3935
	testes: scrotal yr rd, small May-Oct; India	6028
	epididymal sperm: Sept-Oct; n2; central India	5705
	quiescent f: Aug-Nov; India	6028
	mate: Jan-early Feb; India	6027, 6028
	mate: Mar-Apr; Pakistan	9175
	mate: Nov-Dec; s India	3935, 6717
	ovulate: early Mar; India	6027, 6028
	ovulate: end Dec; s India	3935, 6717
	implant: mid Apr; India, Sri Lanka	6027
	preg f: Jan-Apr; India	6717
	birth: mid July; India	6027, 6028
	lact f: June-July; central India	5705
Scotophilus kuhli	epididymal sperm: Oct; n2; central India	5705
	spermatogenesis: Dec-?, peak Feb-Mar; India	7415
	testes: large Jan-Feb, small May-Oct; s India	3906
	testes: large Feb-Mar; s India	3907
	mate: mid Feb-mid Mar; s India	3905, 3906, 3937, 7415
	preg f: Mar-July; s India	3907, 3915
	preg f: Sept-Oct; n3; Sri Lanka	8501
	birth: Mar; India	9175
	birth: June-July; s India	3905, 3907, 7415
	lact f: June; Jaomam/Kimog, Asia	179
	juveniles: June-July; central India	5705
Scotophilus leucogaster	preg f: Feb-early Mar; n1; Africa	5759
	preg f: Mar; n1; Zaire	191
	preg f: Apr-Oct; S Africa	11145b
	preg f: May; n1; Niger	8611
	birth: prob Dec; Tanzania	252
Scotophilus nigrita	mate: end Dec-early Jan; central Africa	252
	preg f: Jan-Mar; boreal central Africa	252
	preg f: Aug-Oct; austral central Africa	252
	preg f: Mar; n2; Zaire	191
	birth: peak Jan-Mar, Aug-Sept; n12; e Africa	5759
	birth: Mar; Angola, Burundi	252
	birth: Oct-Nov; se Zaire	252
	lact f: ?-mid Apr/early May; ne Zaire	252
	lact f: end Dec-Jan; Rwanda, Uganda	252
Tylonycteris pachypus	testes: large Oct, small Jan-May; w Malaysia	7164
	mate: Nov; w Malaysia	7164
	mate: end Nov-mid Dec; Ula Gombak, w Malaysia	8839
	ovulate: Jan; w Malaysia	7164
	implant: Jan; w Malaysia	7164
	preg f: Mar-May; Selangor, Malaysia	7161
	preg f: Apr-May; Malaysia	7164
	birth: Apr-May; w Malaysia	7164
Tylonycteris robustula	testes: large Oct, small May; Malaysia	7164
	mate: mid Nov; Malaysia	7164
	ovulate: Jan; Malaysia	7164
	implant: end Jan; Malaysia	7164
	preg f: Apr; Malaysia	7164
	preg f: Apr-May; Malaysia	7164
	birth: Apr; Malaysia	7161, 7164

Vespertilio murinus	spermatogenesis: May-Sept; Europe	2075
	mate: Sept; s France	9240
	mate: end Sept; s France	9240
	ovulate: Mar-Apr; s France	9238, 9240
	birth: May-June; Russia	8049
	birth: end May-early June; s France	9240
	wean: Aug; s France	9240
Vespertilio superans	birth: summer; Ohzukue I, Jima, Fukuoka Prefecture, Japan	3563

Order Chiroptera, Family Mystacinidae, Genus *Mystacina*

The 12-35 g, New Zealand *Mystacina tuberculata* eats pollen, fruit, and insects (2257) in forested areas. Males form leks (2258). Monoestry is reported (2257). Young are born naked with their eyes open (2257).

NEONATAL MASS		
Mystacina tuberculata	3.2 g; n1 with placenta	2257
LITTER SIZE		
Mystacina tuberculata	1 emb ea; n2 f	2257
SEASONALITY		
Mystacina tuberculata	mate: Mar; New Zealand	2257, 2258
	birth: Dec; New Zealand	2257

Order Chiroptera, Family Molossidae

Free-tailed bats (*Chaerephon* 10-30 g; *Cheiromeles* 165-200 g; *Eumops* 30-70 g; *Molossops* 16-22 g; *Molossus* 8-35 g; *Mops* 7-65 g; *Mormopterus* 10-15 g; *Myopterus* 56-66 mm; *Nyctinomops* 10-30 g *Otomops* 25-50 g; *Promops* 15-20 g; *Tadarida* 10-40 g) are insectivores from the tropics and subtropics of the world. *Molossus* males may be slightly larger than females and possess a gular gland (4575, 10663).

Often only the right ovary is functional and the right uterine horn is used preferentially (191, 2133, 2337, 5305, 5306, 5548, 5603, 5835, 6067, 7120, 7682, 7683, 8872, 8951, 8952, 10375, 11136, 11140, 11804; cf 10154). The hemochorial or endotheliochorial, labyrinthine, diffuse or discoidal placenta has a superficial, mesometrial or antimesometrial implantation into a bicornuate or simplex uterus (181, 2133, 4289, 5306, 5835, 6067, 7563, 7683, 8952, 8952b, 9178, 10310, 10374, 10375, 10376, 10377, 11136, 11140, 11799). Placentophagia (4508, 11270; cf 9896), menstruation (8952a), monoestry (5546, 5835), postpartum estrus (252, 4400, 6067, 7120, 11136, 11735), delayed fertilization and sperm storage (2133, 6069, 8950), and communal nursing (7035) occur.

The naked neonates generally have open eyes except those of *Tadarida* who open their eyes at 1-2 wk (331, 577, 1938, 2337, 3973, 4508, 9933, 10158, 11270; cf 8197). Females have 2 mammae (2337, 6067, 6720). Males have seminal vesicles, ampullary glands, and a baculum (6065, 6069, 6071). *Mormopterus* has a horizontally bifid penis (6069) and *Mops* a penial scent gland (5759). Details of endocrinology (5306), female anatomy (5306, 5603, 9482, 11136), male anatomy (5604), sperm (1273), and milk composition (5087, 5296, 10524) are available.

NEONATAL MASS		
Chaerephon pumila	3.2 g	11140
	3.5 g; n1; term emb	7683
Molossus molossus	3.3 g ea; n2 small yg	3900a
	f: 3.3 g; se0.2; n12	4508
	m: 3.9 g; se0.3; n8	4508
Mops condylurus	10 g	7683
Mops midas	~9-10 g; term emb	10154, 10158
Tadarida brasiliensis	f: 2.40 g; n5	9933
	m: 2.66 g; n3	9933
	2.7 g; n1	4676
	3-4 g	502
	3-4 g; from graph	8197

NEONATAL SIZE		
Molossus molossus	CR: ~30 mm; term emb	6067
	57.5 mm; se3; n7	4508
Mops thersites	31.67 mm; r30-34; n3	71
Tadarida brasiliensis	f: FA: 17.6 mm; n5	9933
	m: FA: 18.6 mm; n3	9933
	CR: 23-27 mm	2337
	CR: 25.8 mm	11702
LITTER SIZE		
Chaerephon plicata	1 emb	5548
	1 emb; n1 f	10872
Chaerephon pumila	1	10158
	1	252
	1	331
	1 yg	7683
	1 emb; n1 f	8611
	1 emb ea; n5 f	187
	1 emb; n13 f	8965
	1 emb ea; n54 f	11140
	1.1 emb; mode 1; r1-2; n10 f	7120
Chaerephon russata	1	252
	1 emb ea; n20 f	1207
Eumops auripendulus	1 emb; n1 f	3999a
Eumops bonariensis	1 emb ea; n5 f	950, 1163
Eumops glaucinus	1 emb ea; n2 f	3608
	1.05 emb; r1-2; n18 f	950, 1163
Eumops perotis	1	6061
	1 yg	1938
	1 yg	577
	mode 1; occasionally 2	1474
	1 emb; n>1 f	5023
	1 emb ea; n3 f	4079
Eumops underwoodi	1 yg	577
	1	1474
Molossops abrasus	1 emb; n1 f	3999a
Molossops greenhalli	1 emb; n1 f	11055
	1 yg ea; n4 f	3900a
Molossops planirostris	1; n1	452
Molossops temminckii	1 emb ea; n2 f	10974
Molossus ater	1	1937
	1 emb; n1 f	217
	1 emb; n1 f	1937
	1 emb; n1 f	3208
	1 emb; n1 f	3999a
	1 emb; n1 f	5405e
	1 emb; n1 f	5409
	1 emb; n1 f	11507a
	1 emb ea; n2 f	950, 1162
	1 emb ea; n3 f	5413a
	1 emb ea; n7 f	1163
	1 emb ea; n8 f	5425
	1 emb ea; n30 f	5424f
Molossus bondae	1	1078
	1 emb; n1 f	3608
	1 emb ea; n8 f	2670a
Molossus molossus	1 emb; n1 f	6307
	1 cl; occasionally 2	6067
	1 emb ea; n6 f	1635
	1 emb ea; n7 f	5411a
	1 ea; n11 lt	452
	1 emb ea; n18 f	527a
	1 emb ea; n27 f	11471a
	1 emb ea; n34 f	3999a
Molossus pretiosus	1 emb ea; n2 f	2670a

Molossus sinaloae	1 emb ea; n2 f	9457a
	1 *emb ea*; n15 f	1162
	1.05 emb; r1-2; n20 f	950, 1163
Mops condylurus	1	11270
	1	5759
	1 emb; n1 f	8611
	1 yg	7683
	1 ea; n2 lt	331
	1 emb ea; n3 f	10154
	2 emb; n1 f	252
Mops congicus	1	252
	1 emb *ea*; n5 f	191
	1 emb ea; n5 f	1207
Mops demonstrator	1; n1 f	252
Mops leonis	1	252
	1 emb *ea*; n12 f	1207
Mops midas	1 ea; n14 lt	10154, 10158
	1-2	252
Mops nanulus	1	252
Mops niveiventer	1; a few 2	252
Mops thersites	1	252
Mops trevori	1 emb; n1 f	1207
	1 *emb*; n1 f	191
Mormopterus norfolkensis	1 ea; n2 lt	7075
Mormopterus planiceps	1	9100
	1 yg	2133
	1 ea; n18 lt	7075
Mormopterus setiger	1 yg ea; n3 f	252
Nyctinomops aurispinosus	est 1 yg	5125a, 5411
Nyctinomops femorosaccus	1 emb; n1 f	1937
	1 emb; n1 f	2865b
	1 emb ea; n2 f	781
	1 emb ea; n2 f	6099
Nyctinomops laticaudatus	1 emb; n1 f	1162
	1 *emb* ea; n9 f	1163
	1 emb *ea*; n16 f	5424f
	1 *emb ea*; n27 f	217
Nyctinomops macrotis	1 yg	577, 1474
	1 emb ea; n2 f	1937
	1 emb ea; n8 f	3607,
	1 yg ea; n36 f	7354
Otomops martiensseni	1 emb; n many	7682
Otomops wroughtoni	1 yg	1340, 1344
Promops centralis	1 *emb* ea; n2 f	3999a
	1 *emb* ea; n4 f	1163
Tadarida aegyptiaca	1 *emb*; n1 f	10158
	1 emb; n1 f	8501
	1 yg *ea*; n6 f	1340, 1344
Tadarida australis	1	5835
	1; n1 lt	7075
Tadarida brasiliensis	1	7035
	1 yg	10590
	1 yg	6062
	1 emb; 2 rare	577
	1; 2 occasional	4443
	1 emb; n1 f	527a
	1 yg; n1 lt	2197, 5405b,
	1 emb; n1 f	6639
	1 emb ea; n2 f	4079
	1 emb ea; n3 f	5873
	1 emb ea; n4 f	6259
	1 emb ea; n6 f	3999a
	1 yg ea; n6 f	485
	1 emb *ea*; n>50 f	178

	1 emb ea; n300 f	9896
	1 ea; n2000 f	2337
	1.25; r1-2; n4 f	7611
	1.58 emb; r1-3; n26 f	2539
Tadarida fulminans	1 ea; n6 f	10158
Tadarida teniotis	1 emb ea; n2 f	6418
SEXUAL MATURITY		
Chaerephon pumila	f: 1st preg: 3 mo	7120
	f: 1st preg: FA: >37.5 mm; shortly after weaning	6848
	m: 5 mo	7120
Molossus molossus	f: 1st mate: 3 mo?	6067
Mormopterus planiceps	f: 1st yr	2133
	m: est not 1st yr	6069
Otomops martiensseni	f/m: ~1 yr; ~25 g	7682
Tadarida aegyptiaca	f/m: 1st mate: 1 yr	5547
Tadarida brasiliensis	f: 9 mo	9896
	m: 18 mo	9932
GESTATION		
Chaerephon ansorgei	2-2.5 mo	255
Chaerephon pumila	60 d	11135, 11140
Molossus molossus	~3.5 mo	6067
Mops condylurus	est ~2 mo	7683
	3 mo	11270
Mormopterus planiceps	3-5 mo	2133
Otomops martiensseni	~3 mo	7682
Tadarida aegyptiaca	77-90 d	5546, 5547
Tadarida brasiliensis	77 d	8798
	11-12 wk	9896
	90+ d	2337
	< 100 d	6062
	~3 mo	4678
	~4 mo	10377
LACTATION		
Chaerephon ansorgei	nurse: est 2-3 mo; field data	252, 255
Chaerephon pumila	wean: < 21 d	11140
Molossus molossus	wean: ~6 wk	6067
	nurse: 65 d	4508
	wean: 4 mo	11735
Molossus sinaloae	nurse: 6-8 wk	4575
Tadarida aegyptiaca	wean: est 2 mo	5546
LITTERS/YEAR		
Chaerephon pumila	1-2	252
	max 3 lt	11135
	max 5	7120
Eumops perotis	1	1474
Eumops underwoodi	1	1474
Molossus molossus	1; sometimes 2	4508
	2	6067
Mops condylurus	1	11270
	2	7683
Mops congicus	1	252
Mops nanulus	1	252
Mormopterus planiceps	1	2133
Otomops martiensseni	1	7682
Tadarida aegyptiaca	1	5547
Tadarida brasiliensis	1	2337
	1	6259
SEASONALITY		
Chaerephon ansorgei	mate: Feb, Oct; Zaire	252, 255
	birth: Jan, May; Zaire	252, 255
	lact f: Jan-Mar; se Zaire	252
Chaerephon bivittata	preg f: Mar; n1; e Africa	5759
Chaerephon chapini	mate: est Oct-Jan; ne Zaire	252
	mate: Aug; Zambia	252

Chaerephon jobensis	yg found: Dec-Jan; Australia	10698
Chaerephon nigeriae	birth: Dec; se Zaire	252
	birth: early Jan; Zambia	252
Chaerephon pumila	spermatogenesis: Oct-Jan, Apr; no location given	252
	spermatogenesis: yr rd; Uganda	252
	epididymal sperm: yr rd; Victoria Nyanza, Uganda	6848
	mate: peaks Sept, Nov, Feb; S Africa	11135
	mate: yr rd; Victoria Nyanza, Uganda	6848
	conceive: Aug-Nov; Transvaal, S Africa	11140
	preg f: Jan; n1; Albert Park, Zaire	11272
	preg f: Apr-Oct; n Ghana	7120
	preg f: May; n1; Niger	8611
	preg f: Nov; n1; central Africa	187
	preg f: Nov; n13; Transvaal, S Africa	8965
	preg f: peak Dec; S Africa	10158
	preg f: Dec-Jan; few data; n Zimbabwe	331
	preg f: yr rd; Uganda	7683
	birth: early-mid Apr; Zaire	252
	birth: Aug; Uganda	252
	birth: Aug, Nov; few data; W Africa	5903
	birth: Aug; n Ghana	7120
	birth: Oct-Nov, Jan-Apr; Transvaal, S Africa	11135, 11140
	lact f: Dec; n9; Transvaal, S Africa	8965
Chaerephon russata	preg f: Sept; Africa	1207
	birth: Oct; central Africa	252
Eumops auripendulus	preg f: Feb; French Guiana	1350a
	preg f: Sept; n1; Peru	3999a
Eumops bonariensis	preg f: Mar, May-June; n33; Yucatan, Mexico	1163
	lact f: June-July; n76; Yucatan, Mexico	1163
Eumops glaucinus	preg f: mid Apr-June; n64; Yucatan, Mexico	1163
	preg f: May, Dec; n2; Costa Rica	3608
	birth: June-July; US	577
	lact f: Apr-May, Aug; n8; Costa Rica	3608
	lact f: mid June-July; n63; Yucatan, Mexico	1163
Eumops perotis	m repro: spring; se US, nw Mexico	1938
	preg f: Apr; n1; Coahuila, Mexico	522
	birth: May-June; CA, AZ, TX	1474
	birth: most by early July; CA	6061
Eumops underwoodi	birth: end June-July; US	577
	birth: July; AZ	1474
Molossops abrasus	preg f: July; n1; Peru	3999a
	preg f: Aug-Oct, Dec; Brazil	10618a
	preg f: Oct-Dec; se Brazil	7983
Molossops greenhalli	birth: June; n4; Trinidad	3900a
	lact f: May; n1; Costa Rica	3608
	lact f: July or Aug; n1; Honduras	11055
Molossops mattogrossensis	preg f; Aug-Dec; Brazil	11734
Molossops temminckii	preg f: Aug; n1; Peru	10974
Molossus ater	preg f: end Feb-early Apr; Veracruz, Mexico	4228
	preg f: Mar; n1; Guatemala	5405e
	preg f: Mar; n2; Trinidad	3900a
	preg f: Mar-June; n44; Yucatan, Mexico	1163
	preg f: 33% Mar; 0% July; n26; Nicaragua	5425
	preg f: Apr-May, July-Aug; Yucatan peninsula, Mexico	5424f
	preg f: June; n1; Durango, Mexico	5405b
	preg f: June-July; n≥3; Sinaloa, Mexico	5409, 5413a
	preg f: July; n1; Veracruz, Mexico	11507a
	preg f: Sept-Oct; French Guiana	1350a
	preg f: Nov; n1; El Salvador	3208
	preg f: Nov; n1; Peru	3999a
	birth: mid Sept; Trinidad, lab NY	8951
	lact f: June-mid Aug; n46; Yucatan, Mexico	1163
	lact f: July; n4; Hidalgo, Mexico	1640a
	lact f: July-Sept; Trinidad	3900a

Molossus bondae	preg f: Jan, Aug; n2; Costa Rica	6260
	preg f: Mar; n1; Costa Rica	3608
	preg f: July; n8; Nicaragua	2670a
	lact f: Sept-Dec, Mar-May; sw Colombia	10785
	lact f: July; n3; Nicaragua	2670a
	repro: Aug-Dec, Feb-June, none Jan, July; sw Colombia	10785
Molossus molossus	estrus: end Feb-Mar, May-June; Puerto Rico	6067
	preg f: Mar-Oct; Puerto Rico	6067
	preg f: Apr, July-Dec; n34; Peru	3999a
	preg f: May; Ecuador	1345a
	preg f: June; n1; Chihuahua, Mexico	1196a
	preg f: July; n3; El Salvador	2670a
	preg f: July; n7; Lesser Antilles	5411a
	preg f: July; n18; Guadeloupe, Lesser Antilles	527a
	preg f: mid July-mid Aug; Trinidad	1635
	preg f: Aug; n27; Jalisco, Mexico	11471a
	preg f: Sept-Feb; ne Brazil	11735
	preg f: Oct-Dec; French Guiana	1350a
	birth: Apr, end Sept-Oct; El Colegio, Colombia	4508
	birth: June, Sept; Puerto Rico	6067
	lact f: Jan-May; ne Brazil	11735
	lact f: July; n11; Lesser Antilles	5411a
	lact f: Sept; n10; Trinidad	3900a
	repro f: Mar-early July; Nicaragua	5425
Molossus pretiosus	preg f: Feb 0/18; Apr 3/4; July 1/1; Boaco, Nicaragua	5425
	preg f: Mar 2/3, mid Aug 3/18; nw Diriamba, Nicaragua	5425
	preg f: May, July, Oct; Costa Rica	6260
	preg f: Aug; n2; Costa Rica	2670a
Molossus sinaloae	preg f: Mar-June; n100; Yucatan, Mexico	1163
	preg f: min Mar-mid May; Yucatan peninsula, Mexico	5424f
	preg f: Apr; n2; Quintana Roo, Mexico	9457a
	preg f: July; n12; Honduras	2670a
	lact f: June-Aug; n518; Yucatan, Mexico	1163
	lact f: July; n12; Honduras	2670a
Mops condylurus	mate: Feb; se Africa	11270
	mate: Apr-May, Nov-Dec; Uganda	5759
	preg f: Jan; n2; n Zimbabwe	331
	preg f: Feb; n1; Niger	8611
	preg f: Dec-Mar, June-Sept; Uganda	7678, 7683
	preg f: yr rd, peak Aug-Oct, Nov-Feb; se Zaire	252
	birth: Feb-Mar, July-Aug; Uganda	5759
	birth: Mar, Sept; Uganda	7678
	birth: mid May; se Africa	11270
	juveniles seen: Feb; n Zimbabwe	331
Mops congicus	preg f: Aug-Sept: n5; Africa	1207
	preg f: Sept; n5; Zaire	191
	birth: Oct; ne Zaire	252
	lact: early Mar; Cameroun, e Zaire, w Uganda	5759
Mops demonstrator	preg f: Apr, June, Sept; ne Zaire, Uganda	252
	preg f: Apr, June, Sept; central Africa	11270
	birth: Apr, June; ne Zaire, Uganda	252
Mops leonis	preg f: Apr; Medge, Zaire	1207
	birth: Apr; ne Zaire	252
Mops midas	mate: Feb; central Africa	252
	preg f: Dec-Feb; Botswana	10154
	birth: est Feb-Mar; S Africa	10158
	birth: est Feb-Mar; Botswana	10154
	birth: May, Oct; central Africa	252
Mops mops	birth: Jan-Feb, July; Kuala Lumpur, Malaysia	3974
Mops nanulus	conceive: Jan; Uganda, w Kenya, Zaire	5759
	birth: end Mar-Apr, July-Sept; Uganda, w Kenya, Zaire	5759
	birth: end Aug-early Sept; central Africa	252
Mops niveiventer	birth: peak Nov-Feb/Mar; se Zaire, Zambia	252

Mops thersites	birth: end Apr; Cameroun	252
	birth: est end Apr; Cameroun	71
	birth: June; central Africa	252
	birth: Aug; ne Zaire	252
	no repro: Feb; Rwanda	252
Mops trevori	preg f: Sept; n1; Zaire	191
	preg f: Sept; n1; Africa	1207
Mormopterus planiceps	testes: large Feb-Mar; small June-Oct; SA, Australia	6069
	spermatogenesis: Sept-June; SA, Australia	6069
	epididymal sperm: Mar-Sept; SA, Australia	6069
	mate: end July-Aug; SA, Australia	2133
	ovulate: Aug; SA, Australia	2133
	implant: Sept; SA, Australia	2133
	birth: Dec-Jan; SA, Australia	2133
Mormopterus setiger	birth: Jan; se Kenya	252
	lact f: Feb; n2; central Africa	252
Nyctinomops aurispinosus	preg f: Sept; n3; Bolivia	5125a
Nyctinomops femorosaccus	preg f: July; n1; AZ	6099
	preg f: July; n1; NM	2865b
	preg f: June; n1; Sinaloa, Mexico	5413a
	preg f: June-July; TX	2864a
	lact f: June-July; TX	2864a
	lact f: July-early Aug; TX	6099
Nyctinomops laticaudatus	preg f: Apr-May; n16; Yucatan peninsula, Mexico	5424f
	preg f: June; n15; Yucatan, Mexico	1163
	preg f: June; n27; Tamaulipas, Mexico	217
	lact f: July; n1; Yucatan, Mexico	1163
	lact f: Aug; n many; Yucatan peninsula, Mexico	5424f
	no preg f: Mar-May; n11; Belize	7624
Nyctinomops macrotis	preg f: May-June; n4; Cuba	7354
	preg f: June; UT	8612
	preg f: June; n1; NM	7354
	preg f: June; n1; Chihuahua, Mexico	7354
	preg f: June; n8; Durango, Mexico	3607,
	preg f: June-July; n36; TX	7354
	birth: end May-early June; sw US	1474
	lact f: June-July; n10; Cuba	7354
	lact f: end June-Aug; TX	7354
	lact f: July-mid Sept; UT	8612
	lact f: July-Sept; NM	7354
	lact f: Aug-Sept; n27; NM	2026b
Otomops martiensseni	testes: large June-Oct, peak Aug; Kenya	5604, 7682
	preg f: Oct-Jan, occasionally Feb; Kenya	5603, 7682
	birth: Jan-Feb; se Zaire, Rwanda	255
	birth: Mar-Apr; Kenya	252
Promops centralis	preg f: Apr, June; n8; Yucatan, Mexico	1163
	preg f: Apr, Oct; n2; Peru	3999a
	preg f: May; n1; Michoacan, Mexico	4671
	lact f: Apr; n2; Trinidad	3900a
	lact f: mid June-mid Aug; n16; Yucatan, Mexico	1163
	lact f: Aug; n1; Suriname	3705c
Tadarida aegyptiaca	quiescence: July-Mar; India	5547
	mate: end May-June; India	5547
	ovulate: 2-3rd wk June; India	5547
	preg f: Mar; n1; Sri Lanka	8501
	preg f: end June-3rd wk Sept; India	5546
	preg f: Sept-Oct, Dec; Transvaal, S Africa	10158
	preg f: Nov; n1; Zimbabwe	10158
	birth: Jan; central Africa	252
	birth: 1-3rd wk Sept; India	5547
	birth: summer; S Africa	10158
	lact f: Dec; n Botswana	10158

Tadarida australis	conceive: end Aug; Australia	5835
	birth: Dec-Jan; Australia	5835
	wean: early May; Australia	5835
Tadarida brasiliensis	mate: July-Aug; NM	502
	mate: winter; TX	2337
	ovulate: early spring; sw US	5305
	conceive: end Feb; FL	10377
	preg f: May; n3; Querétaro, Mexico	9603a
	preg f: May-June; n3; Haiti	5873
	preg f: June; Puerto Rico	344a
	preg f: June; Chihuahua, Mexico	280a
	preg f: June; n1; Chihuahua, Mexico	1196a
	preg f: June; n25; Sinaloa, Mexico	5413a
	preg f: July; Baja CA, Mexico	5424e
	preg f: July; n1; Guadeloupe, Lesser Antilles	527a
	preg f: Aug, Oct; n6; Peru	3999a
	birth: end Jan-early July; CO	10590
	birth: May-June; NM	502
	birth: end May-June; se US	9896
	birth: early-mid June 86%; TX	9933
	birth: June; FL	10377
	birth: June; n1; NB, Canada	2197
	birth: mid June; TX	2337
	birth: end June; TX	2853
	birth: early July; sw US	5305
	birth: July; n1; NE	$5405b_1$
	birth: most by mid July; CA	6062
Tadarida teniotis	preg f: June; n2; Lebanon	6418
	lact f: Oct; n1; Lebanon	6418